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Dongguan SMARTWIN Intelligent Equipment Co.,Ltd.

DGSMARTWIN

AKS系列伺服驱动器用户手册

AKS Series Servo Drives User Manual

▪ Revision History

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▪ Unpacking

If you need assistance with the installation and configuration of AKS drives, please contact the technical support department of Dongguan SMARTWIN Intelligent Equipment Co., Ltd.

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AKS SERIES

1. Preface

1.1 Product Overview

AKS is a full-featured and high-performance servo drive developed with innovative technologies, delivering industry-leading power density. For detailed specifications of all AKS models, refer to the part number description in the Ordering Information section.



SERVO DRIVES

2. Safety & Standard

2.1 Safety Signal

For safe operation, strictly abide by all safety labels attached to the equipment. Do not damage or remove them. The specifications of safety labels are stated as follows:

Safety Signal	Content Description	Description	ISO7000/ IEC60417
	Warning	Remind operators to operate with caution, fully grasp on-site conditions and take proper measures to prevent adverse consequences.	0434
	Dangerous Voltage	Do not touch the terminals within 15 minutes after power off, otherwise electric shock may occur.	5036
	Earthing Protection	To prevent electric shock, securely connect the grounding terminal and strictly follow the instructions in the user manual.	5019
	Warning: Hot Surface	Do not touch the heat sink after power-on to avoid scalding.	5041

2.2 Safety Level Definition

The AKS servo drive is designed as an internal component of machinery systems. Machine manufacturers and system integrators shall guarantee personal safety and complete machine system performance. They must take the intended application of AKS drives into account during risk assessment, and implement corresponding safety measures based on assessment results. The AKS servo drive shall be used in compliance with all applicable safety regulations, directives and relevant technical specifications and requirements.

	Indicates improper operation may result in death or serious personal injury.
	Indicates that improper operation may cause death or serious personal injury.
	Indicates improper operation may cause minor personal injury or equipment damage.

Only qualified personnel are allowed to perform installation, operation, inspection and maintenance work. Such personnel shall receive sufficient professional training and possess adequate expertise to predict and identify potential hazards arising from product application, parameter adjustment and operation of mechanical, electrical and electronic components of the complete machine system.

All relevant staff must be fully familiar with all applicable standards, directives and accident prevention regulations for such operations.

2.3 Installation Safety

Note: Improper handling of AKS servo drives may cause personal injury and/or equipment damage.



When connecting the AKS servo drive to other control devices, follow the two basic rules below to avoid drive damage:
The AKS servo drive must be grounded via the earthing wire of AC main power supply.All connected motion controllers, PLCs or PCs shall share the same grounding wire as the AKS servo drive.

■Review all relevant product documents before installing or commissioning the AKS servo drive.
■Install the drive strictly in accordance with product specifications and installation instructions.
■All system components must be grounded. Ensure electrical safety via low-resistance grounding connection (Class 1 protection per EN/IEC 61800-5-1). Connect the motor to protective earth with a separate grounding wire whose cross-section is no less than that of motor power cables.
■As part of machinery design, manufacturers shall complete hazard analysis and adopt proper measures to prevent personal injury and equipment damage caused by unintended movements.
■The drive complies with IP20 (IEC 60529) and Class 1 (UL50). Manufacturers must equip it with a suitable enclosure meeting at least IP54 (IEC 60529) and Class 2 (UL50), made of metal or 5VA flame-retardant material with no openings at the bottom.
■Protective earth leakage current exceeds 3.5mA. To meet IEC 61800-5-1 and UL508C requirements, adopt dual PE grounding (one via mains earth wire, the other via heat sink to grounded equipment base) or use copper cables with cross-section over 10mm². Fasten with mounting screws and PE terminal screws.
■Do not use yellow wires with or without green stripes for purposes other than protective earthing.
■Power cables shall be rated for minimum 600V and 75℃.

2.4 Operation Safety

Note: Machinery manufacturers shall be responsible for the implementation, testing and certifica -tion of equipment safety. Operation & maintenance conditions as well as safety precautions mu st be clearly specified in the machine manual.

■Perform all equipment operations strictly in accordance with product specifications and installation guidelines.
■Equipment manufacturers shall provide power disconnection devices in compliance with local regulations.
■Keep all protective covers and cabinet doors closed during operation.
■Live parts and high-temperature components exist inside the equipment during operation; the heat sink temperature of AKS servo drive can reach up to 90℃.
■High voltage may remain in control cables and power cables even when the motor is stationary.
■Machine axes bearing suspended or unbalanced loads shall be fitted with additional mechanical safety locks such as motor holding brakes to prevent uncontrolled load dropping. The AKS servo drive cannot hold suspended loads when STO function is activated. Severe injuries may occur without proper load protection measures.

2.5 Maintenance Safety

Note: Improper operation of AKS servo drive may cause personal injury and/or equipment damage.

■Review all relevant product documents before maintaining the AKS servo drive or its driven machinery.
■Carry out maintenance procedures strictly in line with official maintenance requirements and instructions.
■Never connect or disconnect the product while powered on to avoid electric arcs, personal injury and contact damage.
■After power cut, wait at least 5 minutes before touching or disconnecting normally live componen ts such as capacitors, switch contacts and screw terminals.
■Measure the electrical contacts with a multimeter before handling. Only start maintenance when the DC voltage drops below 30V.

2.6 Safety Guidelines

The product illustrations in this manual may show the device without outer covers or safety guards to display internal details. Always install all covers and guards completely during actual use and operate in strict accordance with user manual instructions.

All figures are for reference only and may differ slightly from the actual purchased products, please refer to the real product.

Upon Unpacking Inspection	
	■Do not install the product if damages, rust or used traces are found on the product and accessories during unpacking. ■Do not install it if water ingress, missing parts or damaged components are found inside. ■Check strictly against the packing list; stop installation if the item name does not match.
	■Check whether the outer package is intact for damage, dampness, moisture and deformation before unpacking. ■Unpack in order layer by layer; violent knocking is strictly prohibited. ■Inspect the equipment and accessories for surface damage, rust and collision marks while unpacking. ■Check carefully against the packing list after unpacking to confirm complete quantity of goods and documents.
During Storage and Transportation	
	■Use professional lifting equipment and assign qualified operators to carry large or heavy products, otherwise personal injury or product damage may occur. ■Ensure front cover, terminal blocks and other components are firmly fastened with screws before vertical hoisting to prevent parts falling off and causing injury or damage. ■No personnel shall stand or stay beneath the product during hoisting. ■Hoist steadily and evenly with steel wire ropes. Avoid vibration, impact, overturning and prolonged suspension, so as to prevent personal injury and equipment damage.

	■Handle the product gently during transportation and watch your surroundings to avoid tripping or falling, which may cause personal injury or product damage. ■Hold the product shell firmly when carrying it manually to prevent parts from falling and causing injury. ■Store and transport the product in full compliance with specified requirements to avoid damage. ■Keep away from water splash, rain, direct sunlight, strong electric field, strong magnetic field and severe vibration. ■Avoid storage period exceeding 3 months. Strengthen protection and conduct necessary inspection for long-term storage. ■Pack the product securely before vehicle transportation, and use sealed containers for long-distance delivery. ■It is forbidden to ship this product together with any goods that may cause adverse effects or damage to it.

When installing

	■Only professionally trained personnel with electrical expertise are permitted to operate this device. Unauthorized personnel operation is strictly prohibited.
	■Thoroughly read the user manual and safety precautions before installation. ■Do not install the product in areas with strong electric field or electromagnetic interference. ■Confirm the mounting position has sufficient mechanical strength to bear equipment weight to avoid mechanical hazards before installation. ■Avoid wearing loose clothes or ornaments during installation to prevent electric shock risks. ■When installed in enclosed spaces such as cabinets, equip adequate cooling devices like cooling fans or air conditioners to meet ambient requirements and prevent overheating or fire hazards. ■Product modification is strictly prohibited. ■Do not loosen fastening bolts of internal parts, components or bolts marked in red. ■When built into cabinets or end equipment, the host device shall be equipped with qualified fireproof, electrical and mechanical protective enclosures whose protection level complies with relevant IEC standards and local regulations.

	■Install shielding devices when mounting equipment generating strong electromagnetic interference such as transformers to prevent product malfunction. ■Mount the product on flame-retardant materials like metal. Keep flammables away and avoid attaching them to the product to eliminate fire hazards.
	■Cover the top of the product with cloth or paper during installation to prevent metal scraps, oil, water and other foreign matters from entering inside and causing faults. Remove the cover after work to avoid blocking vents which may lead to poor heat dissipation and overheating. ■Resonance may occur when changing the speed of machinery running at constant speed. Install anti-vibration rubber under the motor base or activate vibration suppression function to effectively reduce resonance.
	■Non-professionals are strictly prohibited from performing equipment installation, wiring, maintenance, inspection and parts replacement. ■Cut off all power supplies before wiring. Residual voltage remains in internal capacitors after power-off. Wait for the time specified on the warning label before operation. Measure DC voltage of main circuit and confirm it is within safe range to prevent electric shock. ■Conduct wiring, remove outer covers or touch circuit boards only when power is fully cut off to avoid electric shock. ■Ensure reliable grounding of all equipment to eliminate electric shock hazards.
	■Never connect input power supply to the equipment output terminal, otherwise it will cause equipment damage and even fire hazards. ■Ensure accurate phase sequence consistency between drive unit and motor terminals during connection to avoid reverse motor rotation. ■All cables shall meet specified wire gauge and shielding requirements; the shield layer of shielded cables shall be reliably grounded at single end only. ■Tighten terminal screws in accordance with specified torque. Insufficient or excessive torque may lead to overheating, connection damage and fire risks. ■After wiring, verify all connections are correct, and ensure no loose screws, gaskets or exposed wires remain inside the product to prevent electric shock and equipment failure.
	■Follow specified ESD anti-static procedures and wear anti-static wristbands during wiring operations to prevent damage to internal circuits. ■Use twisted pair shielded cables for control circuit wiring, and connect the shield layer to the product grounding terminal to avoid abnormal operation.

Before Power-on

	■Confirm complete installation, firm wiring and motor restart permission before power-on. ■Ensure power supply meets product specifications to prevent equipment damage and fire hazards. ■Do not open cabinet doors or protective covers, touch terminals or disassemble any parts when powered on to avoid electric shock.
	■Complete wiring and parameter setting, then conduct trial run to confirm safe operation, preventing personal injury and equipment damage. ■Make sure rated voltage matches supply voltage before powering on; wrong voltage may cause fire. ■Keep all personnel away from the product, motor and mechanical structure before energization to avoid injury or fatal danger.

During Operation

	■Non-professionals are forbidden to operate the product to avoid personal injury or fatal danger. ■Do not touch wiring terminals or disassemble any parts during operation to prevent electric shock.
	■Never touch the casing, fan or resistors to check temperature to avoid scalds. ■Keep foreign objects and metal debris away from running equipment to prevent fire and damage.

During Maintenance

	■Only qualified personnel are allowed to conduct installation, wiring, maintenance, inspection and parts replacement. ■Equipment maintenance is prohibited while powered on to avoid electric shock. ■After power-off, wait for the time specified on warning labels before maintenance work. ■For PM motors, induced voltage still exists at motor terminals during rotation even after power is cut off. Do not touch the terminals to prevent electric shock.
	■Carry out daily and regular inspection & maintenance in accordance with official requirements, and keep complete maintenance records properly.

During Maintenance	
	■Non-professionals are strictly prohibited from equipment installation, wiring, maintenance, inspection and parts replacement. ■Equipment maintenance while powered on is forbidden to avoid electric shock. ■Cut off all power supplies and wait for the time specified on the warning label before inspection and maintenance.
	■Apply for equipment maintenance in accordance with the warranty agreement. ■After fuse blowout, circuit breaker or ELCB tripping, wait for the specified time before restoring power and operation to prevent casualties and equipment damage. ■All troubleshooting and repairs shall be performed by professionals per relevant guidelines with complete maintenance records kept. ■Replace vulnerable parts strictly in accordance with replacement instructions. ■Stop using damaged equipment immediately to avoid personal injury and further equipment damage. ■Recheck wiring and reset parameters completely after equipment replacement.
When Scrapping	
	■Dispose of equipment and products in accordance with relevant national regulations and standards to prevent property loss and personal injury. ■Recycle scrapped equipment and products per industrial waste treatment standards to avoid environmental pollution.

2.6.1 Other Notes

Precautions for Dynamic Brake：

■Dynamic brake is only used for emergency stop under fault condition or sudden power failure. Frequent artificial fault triggering and power cut-off are forbidden.

Safety Notes：

■Ensure an action interval of more than 5 minutes for dynamic brake at high speed, otherwise the internal dynamic brake circuit may be damaged.

■It commonly occurs on rotary mechanical structures: after dynamic brake stops the motor, the shaft load may drive the motor to rotate reversely. At this time the motor runs in generating state, generating short-circuit current inside the dynamic brake. Continuous external driving will cause drive smoking, fire hazard and motor burnout.

2.7 Material Safety Data Sheet

Marked in accordance with SJ/T 11364 standard, which applies to electronic and electrical products sold within the territory of the People's Republic of China.The hazardous substance data listed in the table complies with China RoHS 2.0 issued on January 21, 2016, namely *Order on Restricting the Use of Certain Hazardous Substances* in Electrical and Electronic Equipment.The product contains specified hazardous substances and can be safely used for 20 years. It shall be sent into recycling system after the service life expires.

▪ Hazardous Substances List：

Component Name	Lead (Pb)	Mercury (Hg)	Cadmium (Cd)	Hexavalent Chromium (Cr VI))	Polybrominated Biphenyls (PBB)	Polybrominated Diphenyl Ethers (PBDE)
Golden Components	×	○	○	○	○	○
Plastic Components	○	○	○	○	○	○
Electronic Components	×	○	○	○	○	○
Contact	○	○	○	○	○	○
Cables and cable accessories	○	○	○	○	○	○

O: The content of this hazardous substance in all homogeneous materials of the part is below the limit specified in GB/T26572.
X: The content of this hazardous substance exceeds the limit specified in GB/T26572 in at least one homogeneous material of the part.

2.8 Certification

- Conduct tests and certifications in accordance with the following standards

Standard	Command/Instrucment	Certification Signal
IEC61800-5-1	Low Voltage Directive 2014/35/EU — Speed Controlled Electric Drive Systems	CE
IEC61800-5-2:2017	Machinery Directive 2006/42/EC Adjustable Speed Power Drive Systems - Functional Safety Requirements	CE
IEC61800-3	Electromagnetic Compatibility (EMC) Directive 2004/108/EC Adjustable Speed Electrical Power Drive Systems	CE
IEC61800-3	Electromagnetic Compatibility (EMC) Directive 2014/30/EU Adjustable Speed Electrical Power Drive Systems (PDS) Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs	CE
EN50581	European Directive 2011/65/EU (RoHS, Restriction of Hazardous Substances) Declare conformity and provide technical documents required for compliance with substance restriction requirements.	CE
SJ/T11364	Marking for Restricted Use of Hazardous Substances in Electrical and Electronic Products (China RoHS 2.0) Hazardous substances in electrical and electronic products; Environmental Protection Use Period (EPUP) and recyclability	20
EUREACH	Comply with Regulation (EC) 1907/2006 (REACH), governing the manufacture, application of chemical substances and their potential risks to human health and ecological environment.	REACH COMPLIANT

IEC International Electrotechnical Commission

EN European Standards

CSA Canadian Standards Association

EU European Parliament and Council of the European Union

3. Order Information

The Electrical Specification Model of 220V Servo Drive					
Drive Model	Input Voltage Level	Output Current	Peak Current	Control Method	Supported Position Feedback / Encoder Types
AKS-4D52AAPDB	Single/Three Phase 220VAC±10%	4.5A rms	16A rms	PULSE	Digital Type
AKS-4D52AECDB				EtherCAT/ PULSE	Digital Type
AKS-4D52AECAB					Digital Type Analog Type
AKS-0062AAPDB	Single/Three Phase 220VAC±10%	6A rms	18A rms	PULSE	Digital Type
AKS-0062AECDB				EtherCAT/ PULSE	Digital Type
AKS-0062AECAB					Digital Type Analog Type
AKS-0082AAPDB	Single/Three Phase 220VAC±10 %	8A rms	24A rms	PULSE	Digital Type
AKS-0082AECDB				EtherCAT/ PULSE	Digital Type
AKS-0082AECAB					Digital Type Analog Type
AKS-0102AAPDB	Single/Three Phase 220VAC±10%	10A rms	28A rms	PULSE	Digital Type
AKS-0102AECDB				EtherCAT/ PULSE	Digital Type
AKS-0102AECAB					Digital Type Analog Type
AKS-0132AAPDB	Single/Three Phase 220VAC±10 %	13A rms	28A rms	PULSE	Digital Type
AKS-0132AECDB				EtherCAT/ PULSE	Digital Type
AKS-0132AECAB					Digital Type Analog Type

The Electrical Specification Model of 380V Servo Drive					
Drive Model	Input Voltage Level	Output Current	Peak Current	Control Method	Supported Position Feedback / Encoder Types
AKS-0123DAPDS	Three Phase 380VAC±10%	12A rms	36A rms	PULSE	Digital Type
AKS-0123DECDS				EtherCAT/ PULSE	Digital Type
AKS-0123DECAD					Digital Type Analog Type
AKS-0243DAPDS	Three Phase 380VAC±10%	24A rms	72A rms	PULSE	Digital Type
AKS-0243DECDS				EtherCAT/ PULSE	Digital Type
AKS-0243DECAD					Digital Type Analog Type
AKS-0303DAPDS	Three Phase 380VAC±10%	30A rms	90A rms	PULSE	Digital Type
AKS-0303DECDS				EtherCAT/ PULSE	Digital Type
AKS-0303DECAD					Digital Type Analog Type
AKS-0403DAPDS	Three Phase 380VAC±10%	40A rms	110A rms	PULSE	Digital Type
AKS-0403DECDS				EtherCAT/ PULSE	Digital Type
AKS-0403DECAD					Digital Type Analog Type
AKS-0503DAPDS	Three Phase 380VAC±10%	50A rms	140A rms	PULSE	Digital Type
AKS-0503DECDS				EtherCAT/ PULSE	Digital Type
AKS-0503DECAD					Digital Type Analog Type

4. Specification Overview

4.1 Specification & Nameplate

- Model Description

AKS - 006 2A EC AB

AKS Series

Input Voltage	
2A	220VAC±10%
3D	380VAC±10%

Communication Interface	
AP	Only supporting pulse mode
EC	Supporting Ethercat bus
PN	Supporting PROFINET bus

	2A 220VAC			3D 380VAC	
	Continuous current Arms	Peak current Arms		Continuous current Arms	Peak current Arms
4D5	4.5	16	012	12	36
006	6	18	024	24	72
008	8	26	030	30	90
010	10	26	040	40	110
013	13	26	050	50	140

Function	
DB（Medium-voltage）/DS（High-voltage）	Supporting digital encoder, Tamagawa absolute encoder
AB（Medium-voltage）/AD（High-voltage）	Supporting digital, analog encoder, rotary encoder, BiSS-C encoder, Endat, Tamagawa absolute, Nikon absolute encoder

⋆380V drives require an external 24V UPS power supply for dynamic braking function.

*Only DB and DS functions are optional for AP communication.

- Nameplate Description

DGSMARTWIN®

400-807-1566

www.dgsmartwin.com

Mode No: AKS-4D52AECDB

SN码：AAMO003717 SN241114167

Rating	INput	Output
Voltage	120/240AC	0~240VAC
Frequency	50/60Hz	0~1000Hz
Phase	1Ph/3Ph	3Ph
FL Current	8.5/4A	4.5A
Power@240VAC		1.1KVA
Ambient:45℃	Enclosure:IP20	

RoHS

CE

Made in China

4.2 Product Dimension

- 220VAC Servo Drive Electrical Specifications & Models

Model	Height H	Width W	Depth D	Height H1	Width W1	Mounting Screws	Recommended Tightening Torque	Overall Weight
	Unit: mm					Specification /Quantity	Unit: N·m	Unit: kg
AKS-4D52A	173	57.5	160.5	164	/	M4/2pls	1.35	0.97
AKS-0062A	173	57.5	160.5	164	/	M4/2pls	1.35	0.97
AKS-0082A	194	60	174.5	185	/	M4/2pls	1.35	1.15
AKS-0102A	194	60	174.5	185	/	M4/2pls	1.35	1.15
AKS-0132A	194	60	174.5	185	/	M4/2pls	1.35	1.15

- **380VAC**Servo Drive Electrical Specifications & Models

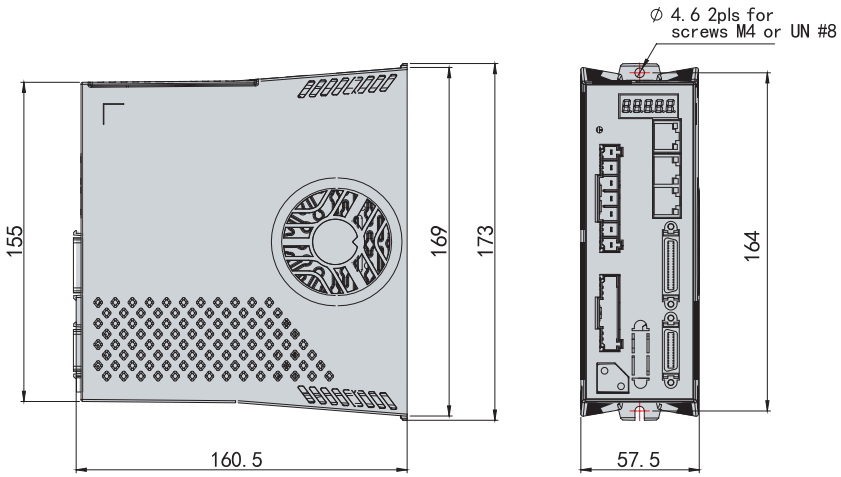
Model	Height H	Width W	Depth D	Height H1	Width W1	Mounting Screws	Recommended Tightening Torque	Overall Weight
	Unit: mm					Specification /Quantity	Unit: N·m	Unit: kg
AKS-0123D	245	106	188.5	236	60	M4/2pls	1.35	3.2
AKS-0243D	310	124	208.5	300	100	M4/2pls	1.35	5.3
AKS-0303D	350	140	208.5	340	116	M4/2pls	1.35	10.05
AKS-0403D	350	150	208.2	340	116	M4/2pls	1.35	10.05
AKS-0503D	350	150	208.2	340	116	M4/2pls	1.35	10.05

- The Outline Dimension&Parameter of Medium-Voltage of AKS-4D52A/AKS-0062A

AKS-4D52A/AKS-0062A



Outline Dimension

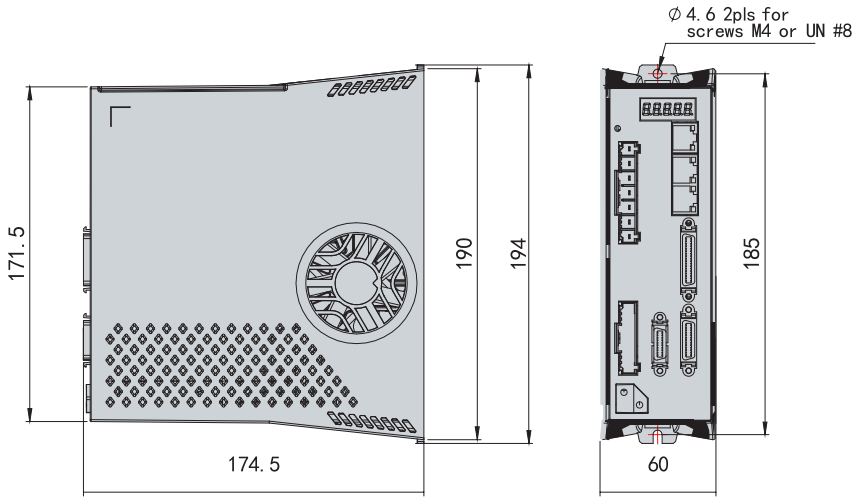


- The Outline Dimension&Parameter of Medium-Voltage of AKS-0082A/AKS-0102A/AKS-0132

AKS-0082A/AKS-0102A/ AKS-0132A

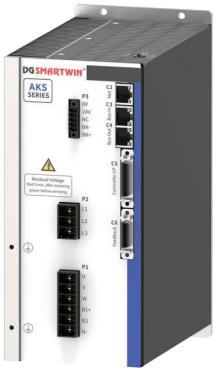


External Dimensions

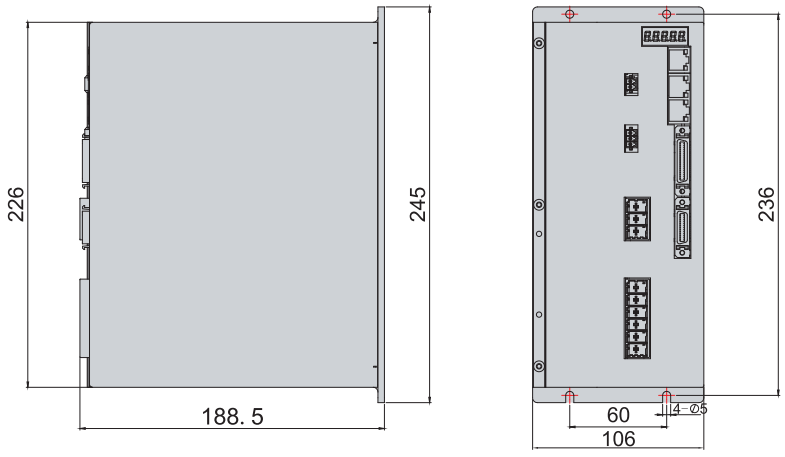


- The Outline Dimension&Parameter of High-Voltage of AKS-0123D

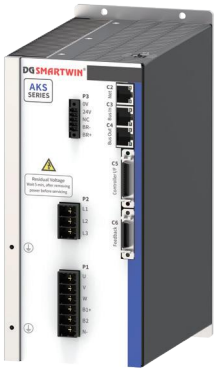
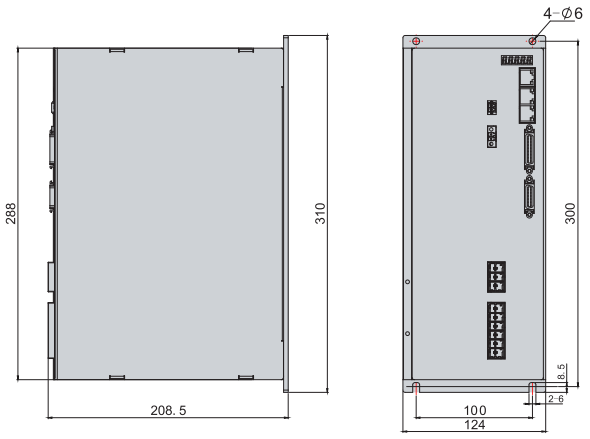
AKS-0123D



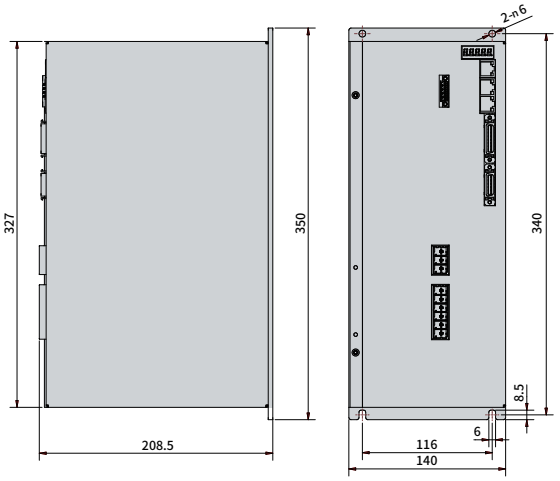
External Dimensions



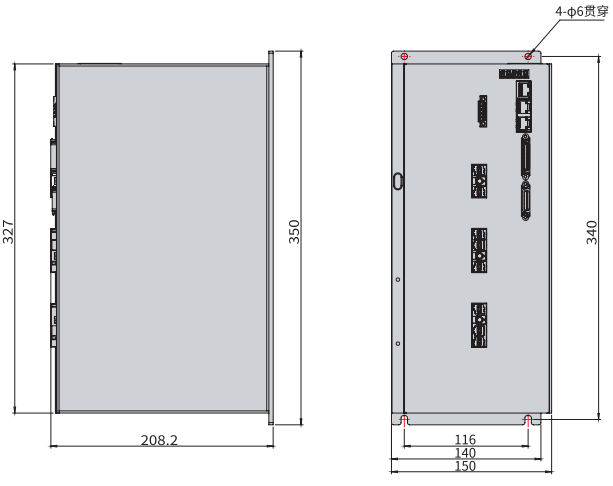
- The Outline Dimension&Parameter of High-Voltage of AKS-0243D

AKS-0243D	Outline Dimension
	

- The Outline Dimension&Parameter of AKS-0303D

AKS-0303D	Outline Dimension
	

- The Outline Dimension&Parameter of AKS-0403D/AKS-0503D

AKS-0403D/AKS-0503D	Outline Dimension
	

4.3 Operating Environment Requirements

- Applied Environment Requirements

Environment Type	Detailed Desciption
Operating Temperature/Humdity	Below 0-45℃，90%RH(Non-condensing)
Storing Temperature/Humdity	Below 0-70℃，90%RH(Non-condensing)
Altitude Height	⩽1000m No derating required； Above 1000m, derate by 1% of rated value per 100m elevation increase.
Ingress Protection Rating	IP20，Note：Except terminals (IP00)
Pollution Level	PD2 Level（According to IEC 60664-1） Not for use in places with corrosive or flammable gas, chemicals, oil water, iron dust or salt-containing dust.
Vibration resistance strength	Operating： 5Hz~8.4Hz：3.5mm displacement 8.4Hz~200Hz：1g Product Package： 5Hz~100Hz：0.01g²/Hz 200Hz：0.001g²/Hz Grms=1.14g
Shock resistance	19.6m/s²

4.4 Mechanical Installation

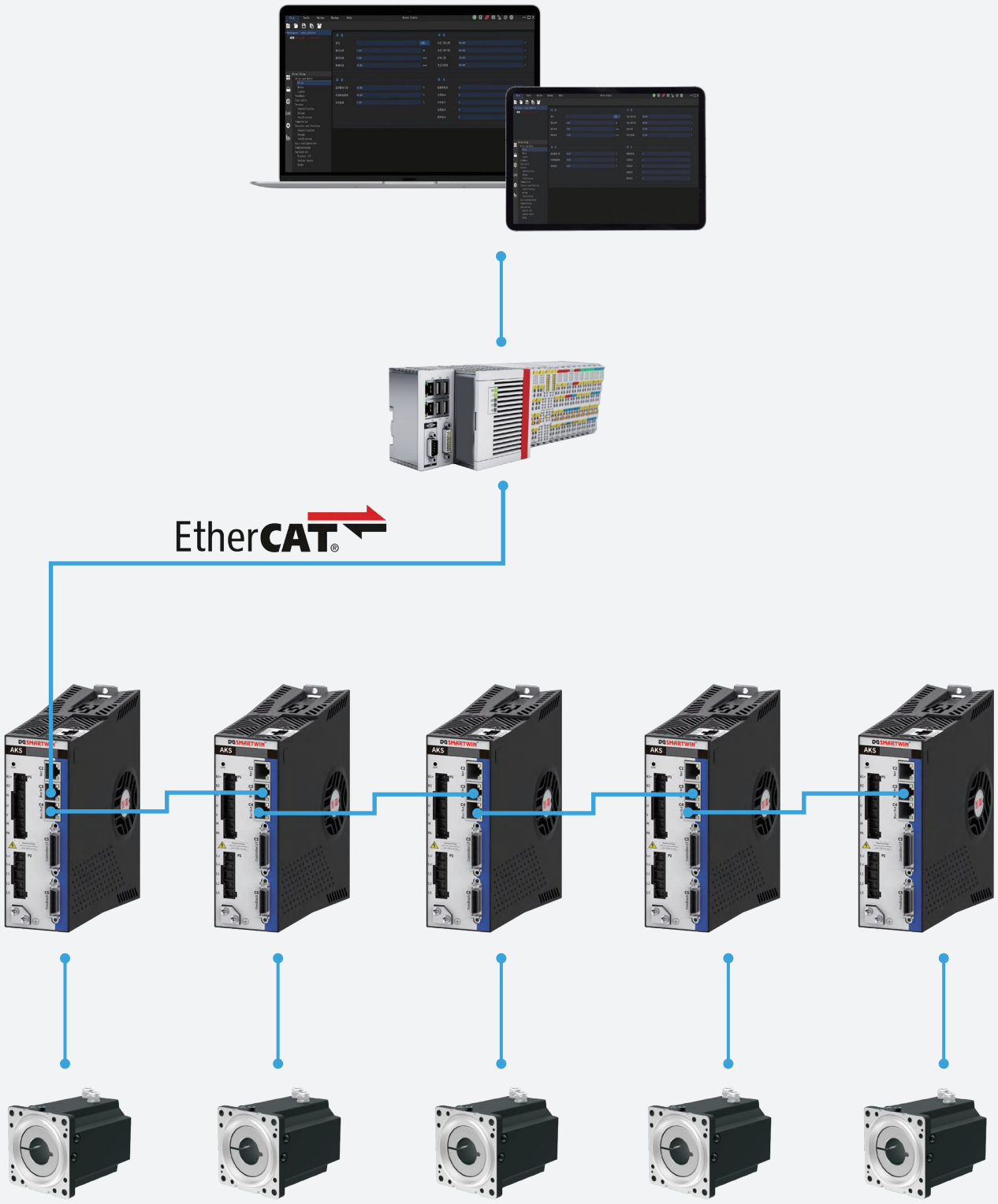
4.4.1 Installing AKS

Mount the AKS on grounded conductive metal panel using the bracket on its rear side. The metal panel shall be sufficiently rigid. Refer to Section 4.2 Product Dimensions for installation dimensions.

4.4.2 Installing Multiple Units

When multiple AKS units are installed side by side in a cabinet or enclosure, the minimum unit spacing is recommended to be 10mm. For all AKS models, the recommended minimum top and bottom clearance is 50mm. It is critical to keep the ambient temperature inside the enclosure below 45℃. The maximum allowable temperature varies by model, refer to Operating Temperature in Section 3.4 for details. If AKS units are mounted on a backplate, ensure the backplate temperature does not exceed 45℃. It is recommended to install a cooling fan at the cabinet bottom for optimal air circulation.

- Schematic Diagram for Multiple AKS Units Installation in Cabinet



4.5 General Specification

Basic Specification	Detailed Description
Control Method	IGBT PWM control, sinusoidal current driving mode 220V/380V: Single-phase or three-phase full-bridge rectification
Supported Motor Type	Rotary servo motor, DDL, DDR, DC motor, Voice wire motor
Built-in Function	Motor parameter writing Innertia identification & auto-tunning Error compensation function Gantry function (rigid gantry&compatible gantry) PCOM function MB function
Control Algorithm	PDC&PID control algorithm Linear control algorithm
Control Mode	PC current mode Analog current mode PC speed mode Analog speed mode PC position mode Pulse position mode
Protective Function	Including but not limit to：overcurrent, overvoltage, undervoltage, overload, main circuit detection abnormality, heatsink overheating, phase loss, overspeed, encoder abnomality, CPU abnormality, parameter abnormality...
LED Display Function	Main power CHARGE, 5-digit nixie tube display
Debugging & Detection	Status display, user parameter setting, monitoring display, alarm tracking display, JOG operation and auto-tuning operation, mapping function for speed/torque command signals, etc.

4.6 Electrical & Mechanical Specification

4.6.1 The Electrical & Mechanical Specification of medium-voltage AKS-4D52A/ AKS-0062A

AKS SIZE-A		AKS-4D52A	AKS-0062A
Electrical Specification			
Power Circuit Input Power (L1, L2, L3)	Rated Input Voltage	220VAC±10%	220VAC±10%
	Phase	Single/Three-Phase	Single/Three-Phase
	Line Frequency	50/60Hz	50/60Hz
	Continuous Current	4Arms	5.8Arms
	Line fuse (FRN-R, LPN or equivalent) (A)	10A	10A
	Withstand voltage (L1/L2/L3对PE)	1500VAC(2121VDC)	1500VAC(2121VDC)
Control Circuit Input Power (L1C, L2C)	Rated Input Voltrage	220VAC±10%	220VAC±10%
	Phase	Single-Phase	Singl- phase
	Control Circuit Power Consumption	25W	25W
	Control Circuit Power Fuse (Delay)	0.5A	0.5A
STO (Safety Torque Off)	STO Signal Voltage	24VDC±10%	24VDC±10%
Motor Input (U, V, W)	Rated Continuous Input Current	4.5Arms	6Arms
	Rated Continuous Input Peak Current	6.3Apeak	8.4Apeak
	Overload Output Current within 2s	16Arms	18Arms
	Peak overload output current within 2s	22.6Apeak	25.4Apeak
	Output Power	0.55KW	0.75KW
	PWM Carrier Frequency	10KHz	10KHz
Soft Start	Max. Surge Current	7A	7A
	Max. Charging Time	350mS	350mS
Physical Specification			
Weight	Net Weight	0.75kg	0.75kg
Cable dimension	Control Circuit Power Wire Gauge (wire length ≤ 3m)	18-28AWG	18-28AWG
	Power Circuit Power Input Output Wire Gauge	18AWG	18AWG
Power Input PE Grounding	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm
Protective Grounding	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm
Mounting Clearance	Side	15mm	15mm
	Top/Bottle	50mm	50mm

AKS SIZE-A		AKS-4D52A	AKS-0062A
Overvoltage & Undervoltage Protection			
Trip Voltage	Bus Undervoltage Threshold	120VDC	120VDC
	Bus Undervoltage Threshold	400VDC	400VDC
Over-Temperature Protection			
Power Board Temperature	Fan Start Temperature Threshold	45°C	45°C
	Power Board Over-temperature Fault Threshold	85°C	85°C
External Regenerative Resistor (B1+,B2)			
Regenerative Resistor	Peak Current	15A	15A
	Minimum External Resistance	50Ω	50Ω
	Rated Power	200W	200W
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	385VDC	385VDC
	Bus Discharge Disable Voltage Threshold	340VDC	340VDC

4.6.2 The Mechanical&Electrical Specification of Medium-Voltage AKS-0082A/AKS-0102A /AKS-0132A

AKS SIZE-B		AKS-0082A	AKS-0102A	AKS-0132A
Electrical Specification				
Power Circuit Input Power (L1,L2,L3)	Rated Input Voltage	220VAC±10%	220VAC±10%	220VAC±10%
	Phase	Single/Three-Phase	Single/Three-Phase	Single/Three-Phase
	Line Frequency	50/60Hz	50/60Hz	50/60Hz
	Continuous Current	5Arms	8Arms	10Arms
	Rated current of fuse (FRN-R, LPN or equivalent)	10A	10A	15A
	Withstand Voltage (L1/L2/L3 on PE)	1500VAC (2121VDC)	1500VAC (2121VDC)	1500VAC (2121VDC)
Control Circuit Input Power (L1C, L2C)	Rated Input Voltage	220VAC±10%	220VAC±10%	220VAC±10%
	Phase	Single-Phase	Single-Phase	Single-Phase
	Control Circuit Power Consumption	25W	25W	25W
	Control Circuit Power Fuse (Delay)	0.5A	0.5A	0.5A
STO (Safety Torque Off)	STO Signal Voltage	24VDC±10%	24VDC±10%	24VDC±10%
Motor Output (U, V, W)	Rated Continuous Input Current	8 Arms	10 Arms	13 Arms
	Rated Continuous Input Peak Current	11.3Apeak	14.1Apeak	18.3Apeak
	Overload Output Current within 2s	26Arms	26Arms	26Arms
	Peak Overload Output Current within 2s	36.7Apeak	36.7Apeak	36.7Apeak
	Output Power	1 KW	1.3 KW	1.5 KW
	PWM Carrier Frequency	10KHz	10KHz	10KHz
Soft Start	Max. Surge Current	7A	7A	7A
	Max. Charging Time	350mS	350mS	350mS
Physical Specification				
Weight	Net Weight	1.15kg	1.15kg	1.15kg
Cable Dimension	Control Circuit Power Wire Gauge (wire length ≤ 3m)	18-28AWG	18-28AWG	18-28AWG
	Power Circuit Power Input Output Wire Gauge	18AWG	18AWG	18AWG
Power Input PE Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm	M4/1.35Nm

AKS SIZE-B		AKS-0082A	AKS-0102A	AKS-0132A
Physical Specification				
Protective Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm	M4/1.35Nm
Mounting Clearance	Side	15mm	15mm	15mm
	Top/Bottle	50mm	50mm	50mm
Overvoltage & Undervoltage Protection				
Trip Voltage	DC bus undervoltage threshold	120VDC	120VDC	120VDC
	DC bus overvoltage threshold	400VDC	400VDC	400VDC
Over-temperature Protection				
Power Board Temperature	Fan start temperature threshold	45°C	45°C	45°C
	Power board over temperature fault threshold	85°C	85°C	85°C
External Regenerative Resistor（B1+,B2）				
Regenerative Resistor	Peak Current	15A	15A	15A
	Minimum External Resistance	45Ω	45Ω	45Ω
	Rated Power	500W	500W	500W
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	385VDC	385VDC	385VDC
	Bus Discharge Disable Voltage Threshold	340VDC	340VDC	340VDC

4.6.3 The Mechanical&Electrical Specification of High-Voltage AKS-0123D

AKS SIZE-C		AKS-0123D
Electrical Specification		
Power Circuit Input Power (L1,L2,L3)	Rated Input Voltage	380VAC±10%
	Phase	Three-phase
	Line Frequency	50/60Hz
	Continuous Current	8Arms
	Rated current of fuse (FRN-R, LPN or equivalent)	15A
	Withstand Voltage (L1/L2/L3 on PE)	1800VAC(2520VDC)
Control Circuit Input Power (P3)	Rated Input Voltage	24VDC±10%
	Rated Consumed Current	2A
	Control Circuit Power Consumption	50W
STO (Safety Torque Off)	STO Signal Voltage	24VDC±10%
Motor Output (U, V, W)	Rated Continuous Input Current	12Arms
	Rated Continuous Input Peak Current	16.9Apeak
	Overload Output Current within 2s	36Arms
	Peak Overload Output Current within 2s	50.9Apeak
	Output Power	6KW
	PWM Carrier Frequency	10KHz
Soft Start	Max. Surge Current	9A
	Max. Charging Time	1300mS
Physical Specification		
Weight	Net Weight	3.2kg
Cable Dimension	Control Circuit Power Wire Gauge (wire length ≤ 3m)	24-28AWG
	Power Circuit Power Input Output Wire Gauge	10-14AWG
Power Input PE Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Protective Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Mounting Clearance	Side	25mm
	Top/ Bottle	50mm
Overvoltage & Undervoltage Protection		
Trip Voltage	DC bus undervoltage threshold	300VDC
	DC bus overvoltage threshold	780VDC

AKS SIZE-C		AKS-0123D
Over-temperature Protection		
Power Board Temperature	Fan start temperature threshold	45°C
	Power board over temperature fault threshold	85°C
External Regenerative Resistor（B1+,B2）		
Regenerative Resistor	Peak Current	26A
	Minimum External Resistance	30Ω
	Rated Power	1000W
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	720VDC
	Bus Discharge Disable Voltage Threshold	700VDC

4.6.4 The Mechanical & Electrical Specification of High-Voltage AKS-0243D

AKS SIZE-D		AKS-0243D
Electrical Specification		
Power Circuit Input Power (L1,L2,L3)	Rated Input Voltage	380VAC±10%
	Phase	Three-Phase
	Line Frequency	50/60Hz
	Continuous Current	20Arms
	Rated current of fuse (FRN-R, LPN or equivalent)	30A
	Withstand Voltage (L1/L2/L3 on PE)	1800VAC(2520VDC)
Control Circuit Input Power (P3)	Rated Input Voltage	24VDC±10%
	Rated Consumed Current	2A
	Control Circuit Power Consumption	50W
STO (Safety Torque Off)	STO Signal Voltage	24VDC±10%
Motor Output(U, V, W)	Rated Continuous Input Current	24Arms
	Rated Continuous Input Peak Current	33.9Apeak
	Overload Output Current within 2s	72Arms
	Peak Overload Output Current within 2s	101.8Apeak
	Output Power	12KW
	PWM Carrier Frequency	10KHz
Soft Start	Max. Surge Current	13A
	Max. Charging Time	1500mS
Physical -Specification		
Weight	Net Weight	5.3kg
Cable Dimension	Control Circuit Power Wire Gauge (wire length ⩽ 3m)	24-28AWG
	Power Circuit Power Input Output Wire Gauge	8-12AWG
Power Input PE Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Protective Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Mounting Clearance	Side	25mm
	Top/Bottle	50mm
Overvoltage & Undervoltage Protection		
Trip Voltage	DC bus undervoltage threshold	300VDC
	DC bus overvoltage threshold	780VDC

AKS SIZE-D		AKS-0243D
Over-temperature Protection		
Power Board Temperature	Fan start temperature threshold	45℃
	Power board over temperature fault threshold	85℃
External Regenerative Resistor（B1+,B2）		
Regenerative Resistor	Peak Current	40A
	Minimum External Resistance	20Ω
	Rated Power	1500W
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	720VDC
	Bus Discharge Disable Voltage Threshold	700VDC

4.6.5 The Mechanical & Electrical Specification of High-Voltage AKS-0303D

AKS SIZE-E		AKS-0303D
Electrical Specification		
Power Circuit Input Power (L1,L2,L3)	Rated Input Voltage	380VAC±10%
	Phase	Three-Phase
	Line Frequency	50/60Hz
	Continuous Current	24Arms
	Rated current of fuse (FRN-R, LPN or equivalent)	30A
	Withstand Voltage (L1/L2/L3 on PE)	1800VAC(2520VDC)
Control Circuit Input Power (P3)	Rated Input Voltage	24VDC±10%
	Rated Consumed Current	2A
	Control Circuit Power Consumption	50W
STO (Safety Torque Off)	STO Signal Voltage	24VDC±10%
Motor Output (U, V, W)	Rated Continuous Input Current	30Arms
	Rated Continuous Input Peak Current	42.4Apeak
	Overload Output Current within 2s	90Arms
	Peak Overload Output Current within 2s	127.2Apeak
	Output Power	15KW
	PWM Carrier Frequency	10KHz
Soft Start	Max. Surge Current	13A
	Max. Charging Time	1500mS
Physical Specification		
Weight	Net Weight	6.4kg
Cable Dimension	Control Circuit Power Wire Gauge (wire length ⩽ 3m)	24-28AWG
	Power Circuit Power Input Output Wire Gauge	8-12AWG
Power Input PE Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Protective Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm
Mounting Clearance	Side	25mm
	Top/Bottle	50mm
Over -voltage&Under-voltage Protection		
Trip Voltage	DC bus undervoltage threshold	300VDC
	DC bus overvoltage threshold	780VDC

AKS SIZE-E		AKS-0303D
Over-temperature Protection		
Power Board Temperature	Fan start temperature threshold	45°C
	Power board over temperature fault threshold	85°C
External Regenerative Resistor（B1+,B2）		
Regenerative Resistor	Peak Current	40A
	Minimum External Resistance	20Ω
	Rated Power	1500W
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	800VDC
	Bus Discharge Disable Voltage Threshold	760VDC

4.6.6 The mechanical & Electrical Specification of High-Voltage AKS-0403D/AKS-0503D

AKS SIZE-F		AKS-0403D	AKS-0503D
Electrical Specification			
Power Circuit Input Power (L1,L2,L3)	Rated Input Voltage	380VAC±10%	380VAC±10%
	Phase	Three-phase	Three-phase
	Line Frequency	50/60Hz	50/60Hz
	Continuous Current	32Arms	40Arms
	Rated current of fuse (FRN-R, LPN or equivalent)	50A	50A
	Withstand Voltage (L1/L2/L3 on PE)	1800VAC(2520VDC)	1800VAC(2520VDC)
Control Circuit Input Power (P3)	Rated Input Voltage	24VDC±10%	24VDC±10%
	Rated Consumed Current	2A	2A
	Control Circuit Power Consumption	50W	50W
STO (Safety Torque ff)	STO Signal Voltage	24VDC±10%	24VDC±10%
Motor Output (U, V, W)	Rated Continuous Input Current	40Arms	50Arms
	Rated Continuous Input Peak Current	56.56Apeak	70.7Apeak
	Overload Output Current within 2s	110Arms	140Arms
	Peak Overload Output Current within 2s	155.5Apeak	197.9Apeak
	Output Power	20KW	25KW
	PWM Carrier Frequency	10KHz	10KHz
Soft Start	Max. Surge Current	16A	16A
	Max. Charging Time	1500mS	1500mS
Mechanical Specification			
Weight	Net Weight	10.5kg	10.5kg
Cable Dimension	Control Circuit Power Wire Gauge (wire length ⩽3m)	24-28AWG	24-28AWG
	Power Circuit Power Input Output Wire Gauge	8-14AWG	8-14AWG
Power Input PE Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm
Protective Earthing	PE Grounding Screw Size/Torque	M4/1.35Nm	M4/1.35Nm
Mounting Clearance	Side	25mm	25mm
	Top/Bottle	50mm	50mm
Over -voltage&Under-voltage Protection			
Trip Voltage	DC bus undervoltage threshold	300VDC	300VDC
	DC bus overvoltage threshold	780VDC	780VDC

AKS SIZE-F		AKS-0403D	AKS-0503D
Over-temperature Protection			
Power Board Temperature	Fan start temperature threshold	45℃	45℃
	Power board over temperature fault threshold	85℃	85℃
External Regenerative Resistor（B1+,B2）			
Regenerative Resistor	Peak Current	50A	100A
	Minimum External Resistance	15Ω	8.4Ω
	Rated Power	Determining by System	Determining by System
Bus Voltage Discharge	Bus Discharge Enable Voltage Threshold	800VDC	800VDC
	Bus Discharge Disable Voltage Threshold	760VDC	760VDC

4.7 STO/ Functional Safety Specification

4.7.1 The Overview of Safety Torque Off

Functional safety of AKS servo drive is implemented by Safe Torque Off (STO). Activation of STO cuts off power supply for motor motion control and torque output, preventing accidental startup and motor rotation while the drive stays powered on.

The integrated STO adopts standard electromechanical safety technology, removing separatecomponents and simplifying wiring and maintenance.

Besides, its switching response time is shorter than that of conventional electromechanical safetyparts.

Note: Once STO activates, its circuit power cuts off and motor power supply is disabled.

STO is defined in EN/IEC61800-5-2, matching Category 0 uncontrolled stop per IEC60204-1.EN/IEC61800-5-2 specifies functional safety requirements for adjustable speed electric drive systems.

No driving power will be supplied to the motor when STO is activated. STO can be applied to cut power and avoid unexpected startup.

Warning: Drives carrying suspended loads must be fitted with additional mechanical safety devices such as motor brakes. The drive cannot control the load when STO is activated. Insufficient protection may result in serious personal injury.

The STO function of AKS cuts off power supply to power modules and suppresses PWM pulses generated by drive IGBTs.

4.7.2 The STO Module of the Servo Drive

The STO function blocks PWM signal output to the drive power stage via external redundant hardware terminals STO1 and STO2, so as to stop motor movement.

Both STO1 and STO2 terminal input signals shall stay at active high level to enable normal drive operation.

- The STO function sheet are as following:

STO1 Input	STO2 Input	PWM Signal
H	H	Normal
L	H	Prohibit
H	L	Prohibit
L	L	Prohibit

STO（Safety Torque Off）	
Definition	Cutting the power input of motor makes that it can't generate torque.
Description	STO function brings the machine into safe torque-free state and prevents unexpected startup. Once activated, the running motor coasts to stop.
Safety State	Disable PWM gate control signals of the drive
Operation Mode	High needs or continuous mode

4.7.3 The Safety Specification of the STO Function

Safety Standard	Executive Standard	
Functional Safety	IEC 61508: 2010 ISO 13849-1: 2015 ISO 13849-2: 2012 IEC 62061: 2021 IEC 61800-3: 2017	EN 61508: 2010 EN ISO 13849-1: 2015 EN ISO 13849-2: 2012 EN IEC 62061: 2021 EN IEC 61800-3: 2018
EMC	IEC 61000-6-7:2014 IEC 61326-3-1:2017 IEC 61800-5-2: 2016 IEC 61800-3:2017	EN 61000-6-7:2015 EN 61326-3-1:2017 EN 61800-5-2: 2017 EN IEC 61800-5-3: 2018
LVD	IEC 61800-5-1:2007/AMD1:2016 EN 61800-5-1:2007/A1:2017	

Safety Parameter	
SIL	SIL3, IEC61508 Maximum SIL3, EN IEC62061
PFH_D	PFH D ⩽0.16×10-7[1/h] (16% of SIL3)
Cat.	3, EN ISO 13849–1
PL	e, EN ISO 13849–1
MTTFd	441.3 years（High）
DCavg	⩾90%（Medium）
T_1	10 years
HFT	1
SC	SC3
Applied Mode	High Needs or Continuous Mode

4.7.4 STO Symstematic Requirements

- STO Power

Avoid excessive voltage level at STO input; low voltage shall be adopted for safety-related compenents.

Only 24V power supply complying with PELV/SELV requirements of EN60204-1 can be used.

Note: The built-in 24VDC power of AKS is not approved for safety use and forbidden for STO power supply.

- STO Cable

According to the standard requirenents of EN61326-3-1, The STO cable length must be less than 30 meters.

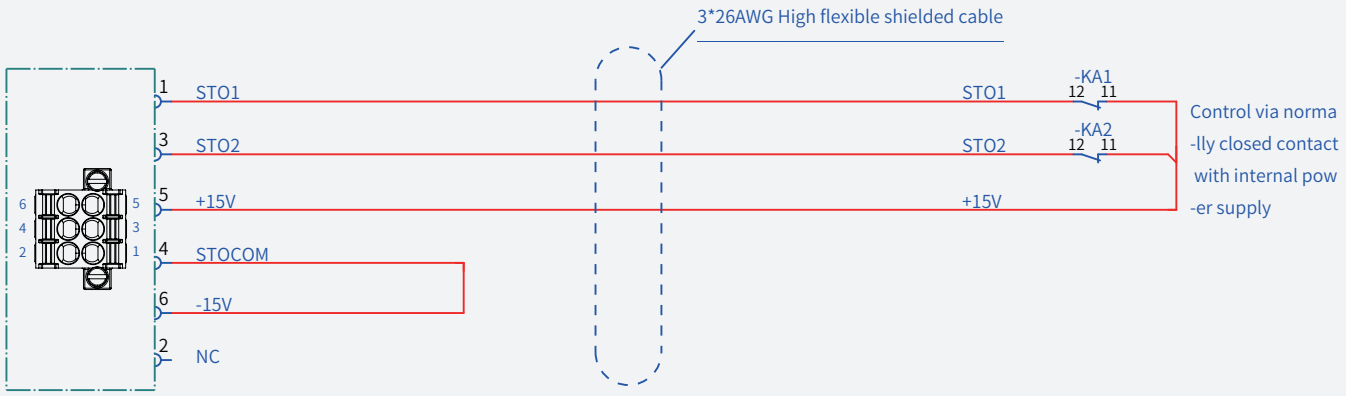
Shielded double-insulated cables must be used. Double insulation requires cable conduit installation, and cable sheath shall be connected to power ground.

Optimal cable size is 22AWG (0.34mm2). 24AWG (0.25mm2) is also applicable as STO circuit current less than 0.45A.

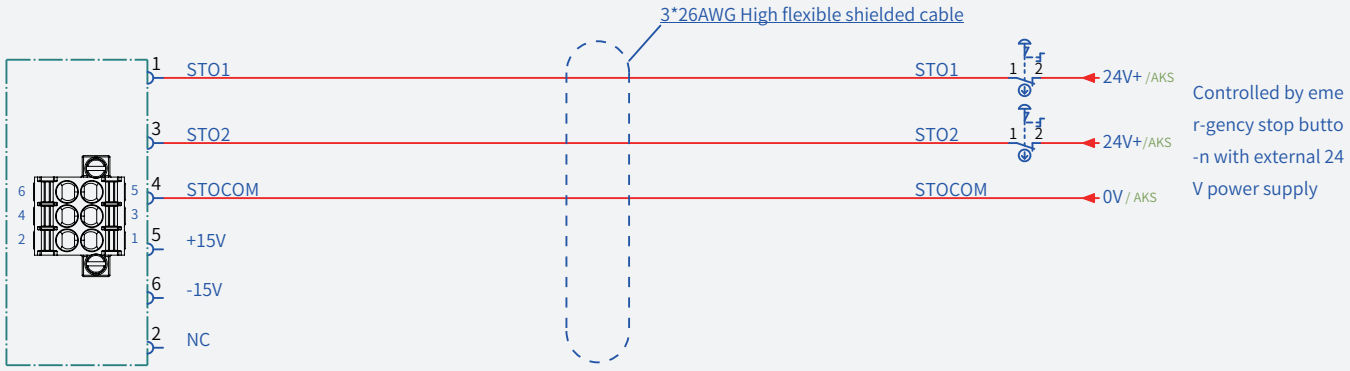
STO cables shall be spatially isolated from all mechanical, electrical, thermal and chemical interference sources.

4.7.5 Definition of STO Wiring Diagram

- Definition of Wiring Diagram for STO Port Using Internal Power Supply



- Definition of Wiring Diagram for STO Port Using External Power Supply



4.7.6 STO Maintenance

Users shall conduct manual STO function test at least every three months.

For diagnostic test, cut off STO supply voltage, confirm drive enters STO fault state and motion is locked.

Maintenance procedures are implemented as follows:

1. Set the system to supply rated STO voltage and enable the drive.
   - The drive enters fault state and displays STO fault alarm.
   - The drive cannot be enabled.
2. Restore STO voltage and reset faults. Verify drive enable availability.

4.8 Communication Interface Specification

Characteristics	Specification
EtherCAT	CiA301 (CANopen Application Layer and Communication Profile) and CiA402 (CANopen Device Profile for Drives and Motion Control) for drives and motion control.
	Communication cycle: 0.125ms~16ms. Contact manufacturer for parameter customization if longer than 16ms.

4.9 Control I/O Interface Specifications

Project	Specification	
Analog Input（3 lines）	Voltage Range	Analog ±10 VDC differential
	Input Resolution	16-bit
	Input Impedence	8 kΩ
	Zero Attenuation	200 Hz
	Bandwidth (-3dB)	1 KHz
Pulse & Direction	Signal	RS422 line receiver
	Max. Input Frequency	4 MHz
Equivalent Encoder Output	Signal	AB quadrature signal / zero differential signal, RS422 line transmitter
	Max. Output Frequency	4 MHz
Ripid Digital Input (6 line)	Signal	Configurable, optically isolated, sinking input. EtherCAT: Sinking or sourcing input optional by user definition
	Voltage	24 V
	Max. Input Current	10 mA
	Transmission Delay Time	1 μs
Common Digital Input (2 line)	Signal	Configurable, open collector, optically isolated, sinking output. EtherCAT: Sinking or sourcing output customizable by user
	Voltage	24 V
	Max. Current	40 mA
	Transmission Delay Time	1 ms
Ripid Digital Input (1 line)	Signal	Configurable, open collector, optically isolated sinking output. EtherCAT: Sinking or sourcing output user configurable.
	Voltage	24 V
	Max. Current	10 mA
	Transmission Delay Time	1 μs
The Second Encoder	Signal	AB quadrature signal / index differential signal, RS422 line receiver
	Max. Input Frequency	4 MHz (prior to AB quadrature)
	Max. Zero Pulse Width	1 μs
	Function	Dual-loop system, master/slave control or handwheel control

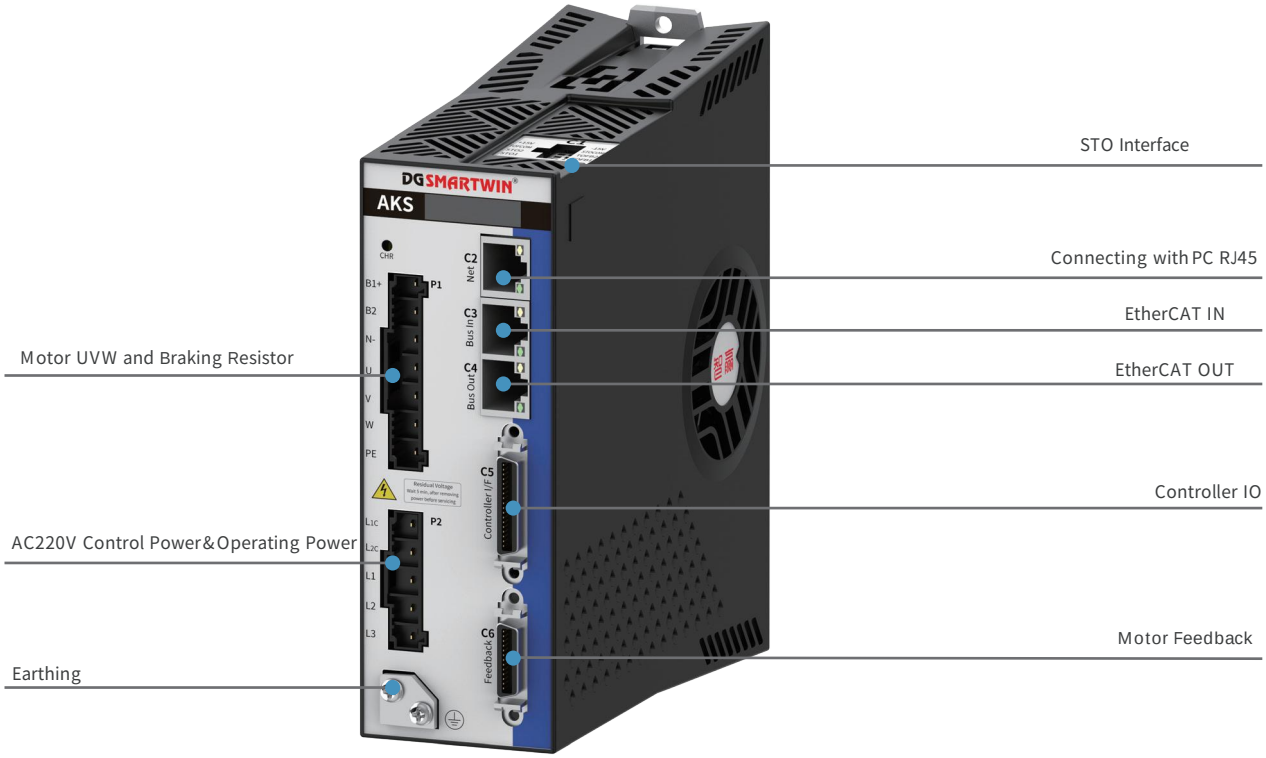
4.10 The Specification of Motor Feedback Interface

Motor Feedback	Specification	
Overview	Drive Output Voltage	5 VDC (7 VDC*)
	Drive Max. Output Current	250 mA
	Max. Cable Length	AWG 28-3 m;AWG 24-10 m
Incremental Encoder	Signal	Differential output of AB quadrature encoder signal with/without index/Hall signal, Tamagawa line-saving type, RS422 or RS485 transmission
	AB Quadrature Maximum Input Frequency	4 MHz (Before quadrature)
	Max. Index Pulse Width	1 μs
Hall Sensor	Signal	Single-ended, Open collector
Resolver	Signal	Sine and cosine differential
	Transformation Ratio	0.45-1.6
	Excitation frequency	8 kHz
	Drive Input Voltage	6-22 Vpp
	Max. DC Resistance	120 Ω (Stator)
	Max. Drive Current	55 mArms
	Output voltage to driver	10 Vpp
Sine Encoder	Signal	Sine-cosine differential, with or without Hall signal
	Signal Level	1 Vpp @ 2.5 V
	Max. Input Frequency	270 kHz
	Protocol	EnDat® 2.1, HIPERFACE ®
	Input Impedence	120 Ω
	Interpolation	Max. to 65536 (16-bit)
	Effective Interpolation	Max. to 16384 (14-bit)
SSI Encoder	Signal	Synchronous encoder: differential data and clock signals Asynchronous encoder: data signal only
	Protocol	EnDat 2.2, BiSS-C, Tamagawa,Nikon and other SSI
Motor Temperature	Signal	PTC or NTC thermistor, user-defined fault threshold

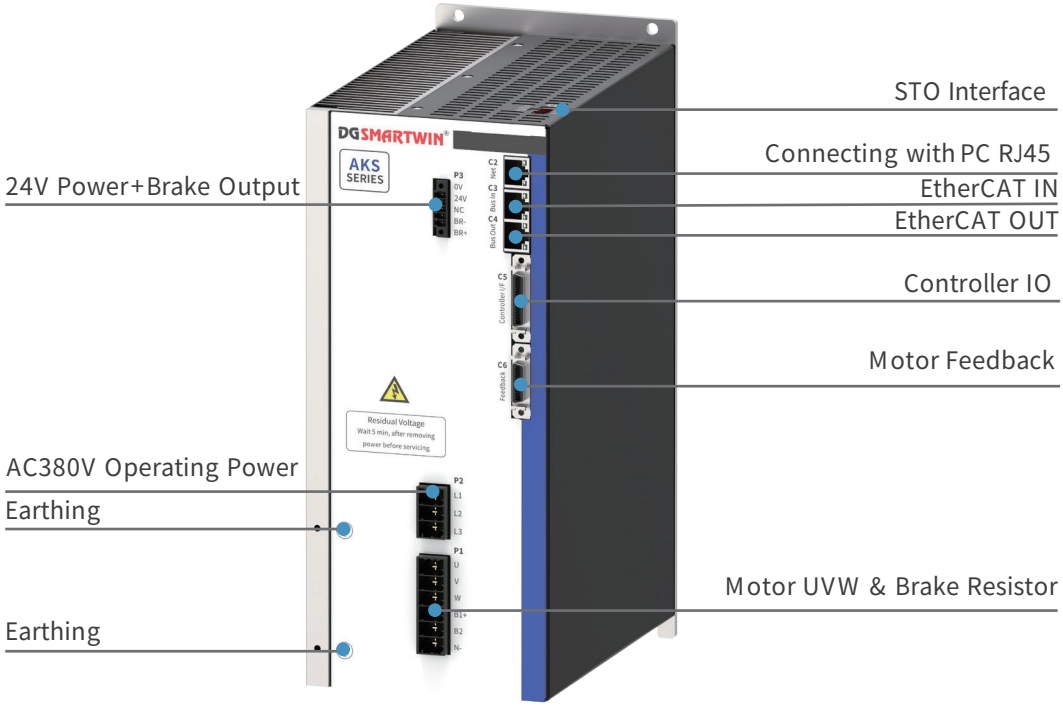
5. Drive Hardware

5.1 System Wiring

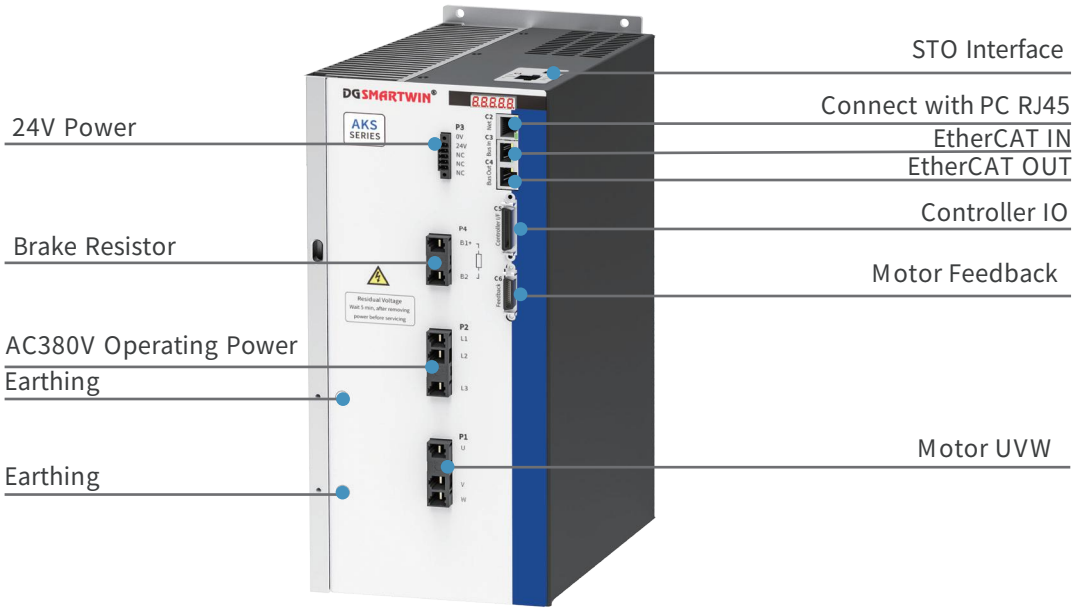
5.1.1 Wiring Diagram of Medium Voltage AKS-4D52A/AKS-0062A/AKS-0082A/AKS-0102A/AKS-0132A Servo System



5.1.2 Wiring Diagram of High Voltage AKS-0123D/AKS-0243D/AKS-0303D Servo System



5.1.3 Wiring Diagram of High Voltage AKS-0403D/AKS-0503D Servo System



5.2 Control Board Wiring

5.2.1 NET Debugging Communication Wiring



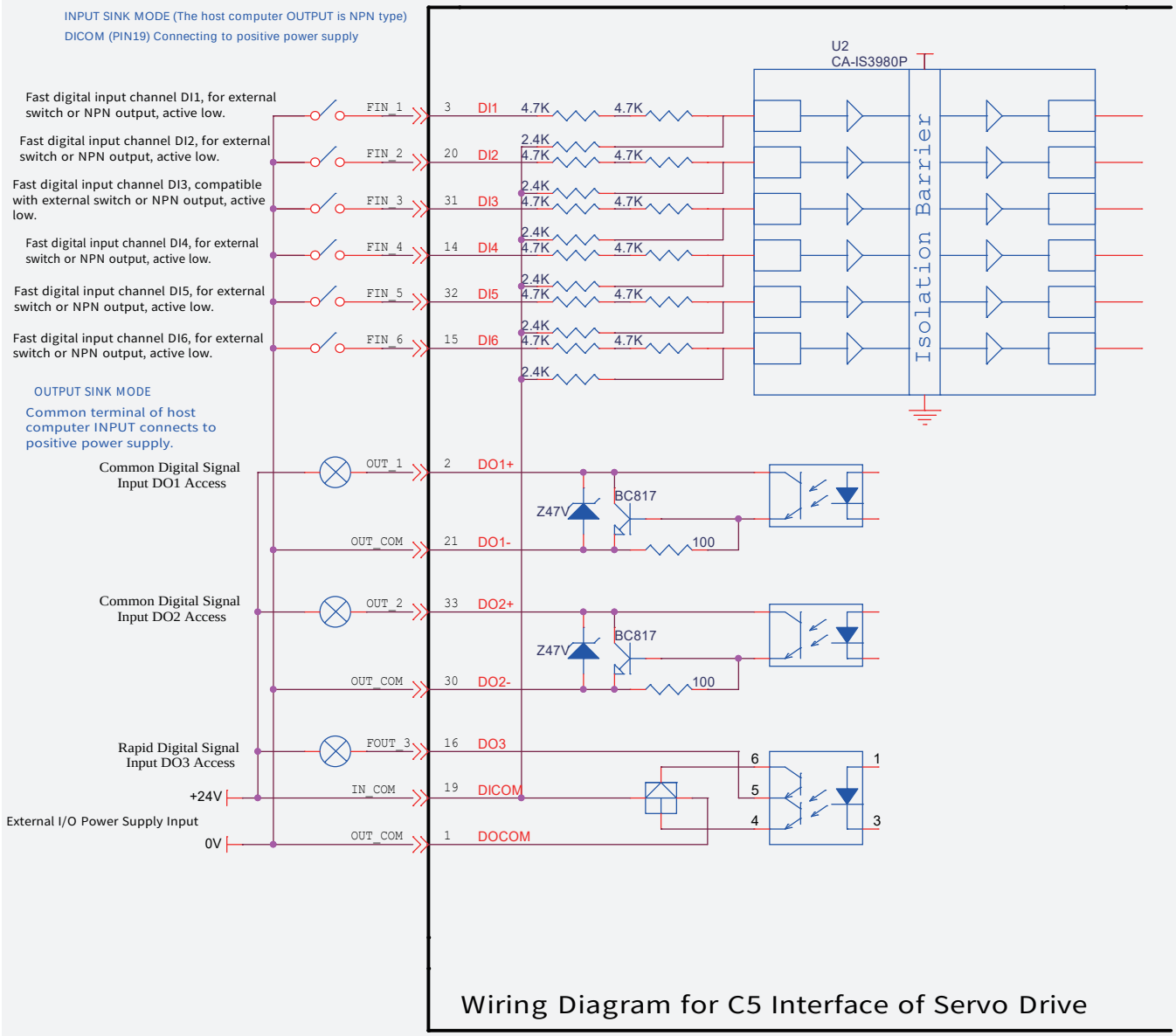
Figure 5-1 NET Debugging Communication Wiring

5.2.2 EtherCAT Bus Communication Wiring

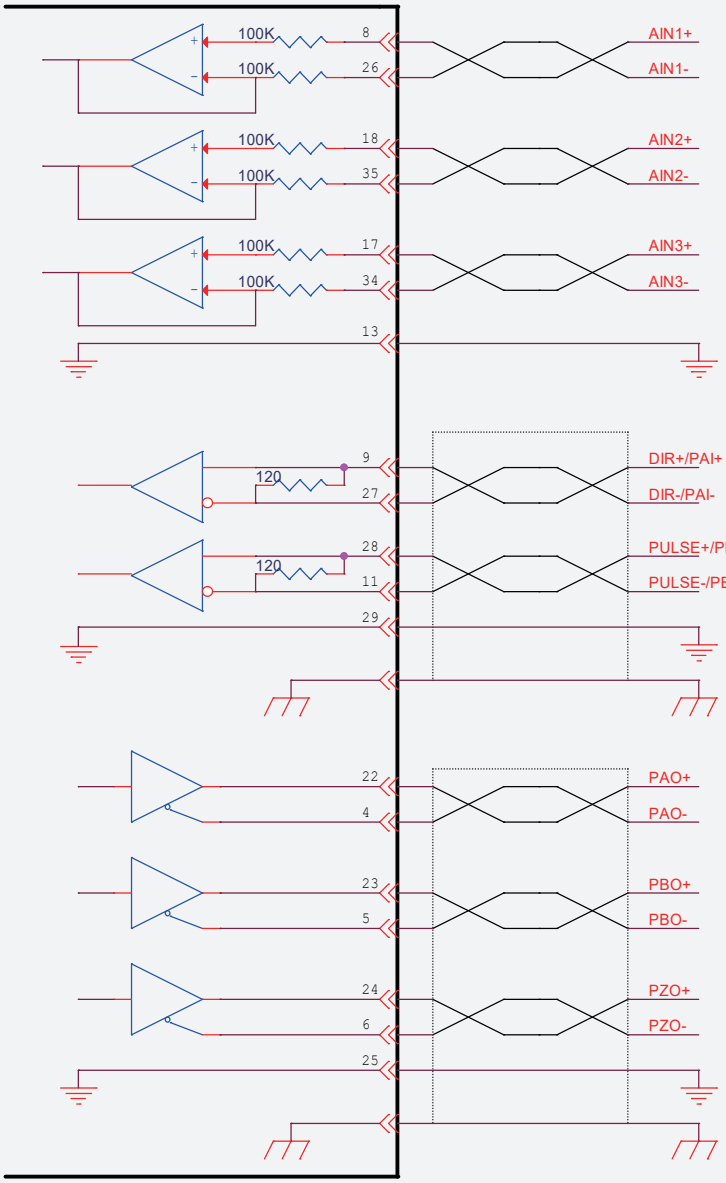


Figure 5-2 EtherCAT Bus Communication Wiring

5.2.3 C5-I/O Interface NPN Type Wiring Diagram



Wiring Diagram for C5 Interface of Servo Drive



Analog ±10V input channel AIN1

Analog ±10v input Channel AIN3

Analog ±10v input channel AIN3

1. The analog input sampling interface of the drive adopts a pseudo-differential design. Single-point grounding between AIN- of each channel and the GND of analog signal output from sensors or host devices can reduce noise interference.
2. Do not connect AIN- of each channel to the analog output (AO+) of sensors or host devices, otherwise the analog sampling circuit of the drive will malfunction.

Differential input for direction & pulse commands, or AB quadrature pulse

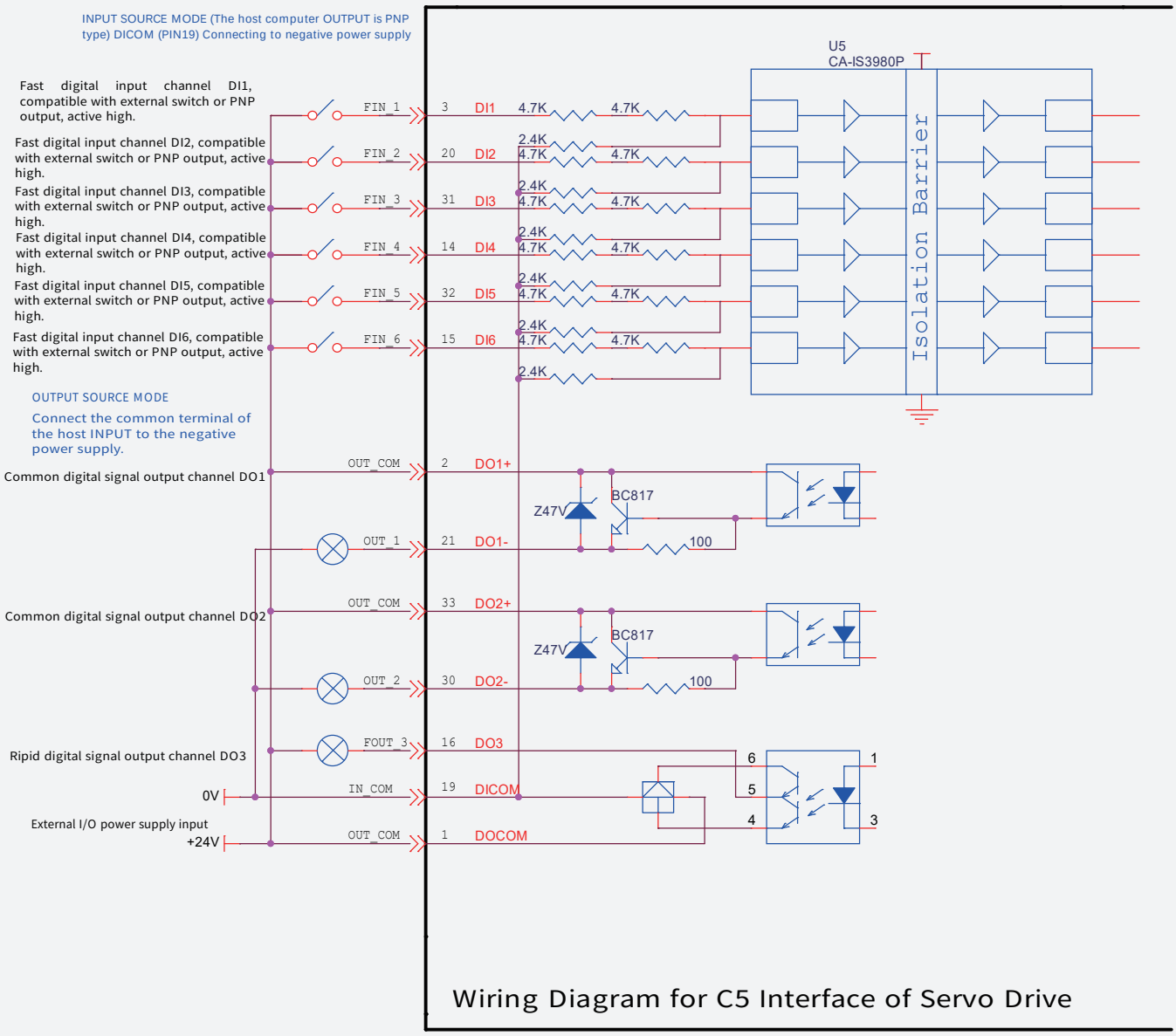
PIN29 is the reference ground for pulse input, connect it to the reference ground of the host pulse output.
Crimp both ends of the cable shield to the metal housing or frame ground (FGND) of the drive and host connector respectively.

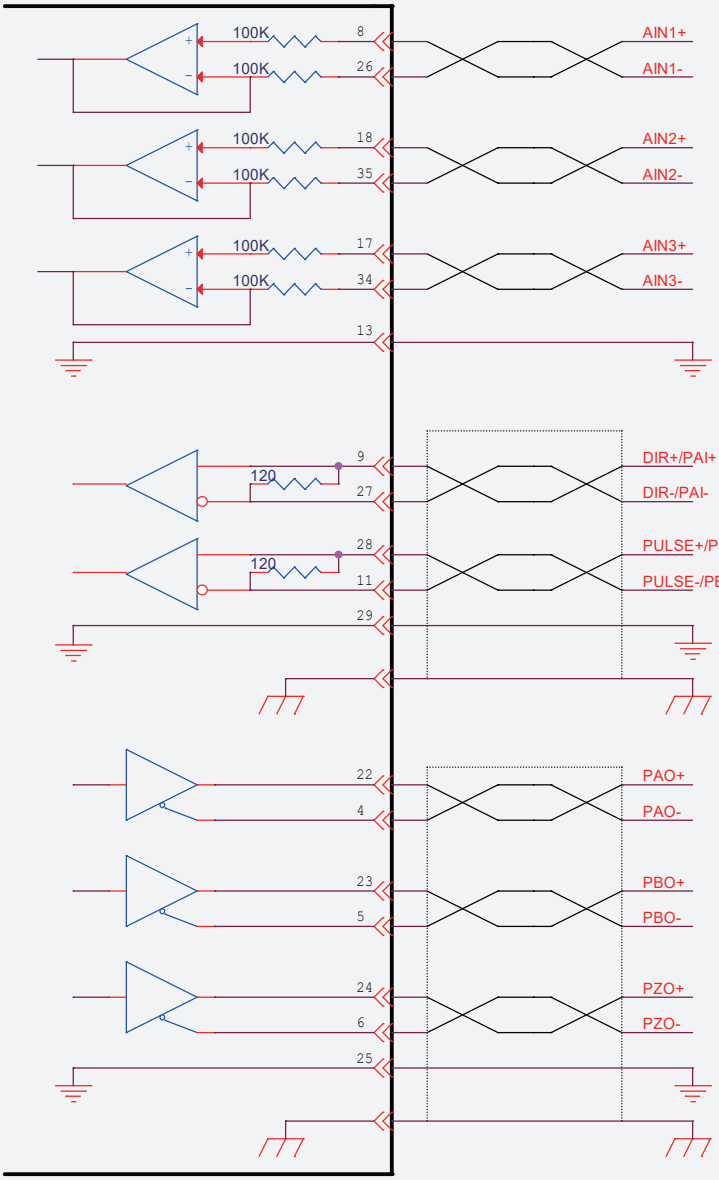
Differential output of divided pulses for equivalent encoder

PIN25 is the reference ground for pulse output, connect it to the reference ground of the host pulse input.
Crimp both ends of the cable shield to the metal housing or frame ground (FGND) of the drive and host connector respectively.

1. Ensure the differential input uses a 5V system; otherwise, the input pulses of the servo drive will be unstable.
2.Connect the drive signal ground (GND) to the 5V reference ground of the host device to reduce noise interference.
3. Firmly crimp both ends of the shielded cable outer shield to the protective earth (PE) of the drive and host device enclosure respectively.

5.2.4 Wiring Diagram for PNP Connection of C5-I/O Interface





Analog ±10v input channel AIN1

Analog ±10v input channel AIN2

Analog ±10v input channel AIN3

1. The analog input sampling interface of the servo drive adopts a pseudo-differential design. Single-point grounding between AIN- of each channel and the GND of analog signal outputs from sensors or host devices helps reduce noise interference.
2. Never connect AIN- of any channel to the analog output (AO+) of sensors or host devices, as this will cause malfunction of the drive's analog sampling circuit.

Differential input for direction and pulse commands, or AB quadrature pulse

PIN29 serves as the reference ground for pulse input and shall be connected to the reference ground of the host pulse output.

Crimp both ends of the cable shield to the metal housing or frame ground (FGND) of the servo drive and host connector respectively

Differential output of frequency-divided pulses for virtual encoder

PIN25 is the reference ground for pulse output, connect it to the reference ground of the host pulse input.

Crimp both ends of the cable shield to the metal housing or frame ground (FGND) of the drive and host connector respectively.

1. Ensure the differential input adopts a 5V system, otherwise the input pulses of the servo drive will be unstable.
2. Connect the drive signal ground (GND) to the 5V reference ground of the host device to reduce noise interference.
3. Securely connect both ends of the shielded cable's outer shield to the protective earth (PE) of the drive and host device enclosure.

Sheet 5-1 Pin Definition of C5 Device I/O Interface

Pin	Function	Instruction
1	Isolated power supply for digital input and output	Shares power supply for digital output or input reference with Pin19. Connect to 24V positive or 24V ground of external IO power according to NPN/PNP type.
2	Digital Output 1+	Opto-isolated programmable digital output, configured via Digital Output 1
3	Ripid Digital Output 1	Opto-isolated programmable digital input, configured via Digital Output 1
4	Equivalent Encoder Output A-	Equvalent Encoder Differential Output Signal A-
5	Equivalent Encoder Output B-	Equvalent Encoder Differential Output B-
6	Equivalent Encoder Output Z-	Equvalent Encoder Differential Output Z-
7	Reserved	Keep Floating
8	Analog Input 1+	Positive terminal of analog differential input 1, read via Analog Input 1
9	Direction Command Input+ /The Second Encoder A+	Positive terminal of direction command differential input or A+ signal of secondary encoder differential input
10	Reserved	Keep Floating
11	Pulse Command Input- /The Second Encoder B-	Negative terminal of pulse command differential input or B- signal of secondary encoder differential input
12	Reserved	Keep Floating
13	Analog Input Reference Grounding	Shall connect to reference ground of upper analog output terminal
14	Ripid Digital Output 4	Isolated programmable digital input, read via Digital Input 4
15	Ripid Digital Output 6	Isolated programmable digital input, read via Digital Input 6
16	Ripid Digital Output 3	High-speed opto-isolated programmable digital output, configured via Digital Output 3
17	Analog Input 3+	Positive terminal of analog differential input 3, read via Analog Input 3
18	Analog Input 2+	Positive terminal of analog differential input 2, read via Analog Input 2
19	Isolated Power Supply for Digital Port	Shares power supply for digital output and input reference with Pin1. Connect to 24V positive or 24V ground of external IO power based on NPN/PNP type.
20	Ripid Digital Input 2	Isolated programmable digital input, read via Digital Input 2
21	Digital Output 1-	Opto-isolated programmable digital input, read via Digital Input 1
22	Equivalent Encoder Output A+	Equvalent Encoder Differential Output Signal A+
23	Equivalent Encoder Output B+	Equvalent Encoder Differential Output Signal B+
24	Equivalent Encoder Output Z+	Equvalent Encoder Differential Output Signal Z+
25	Equivalent Encoder Output Reference Grounding	Connect to reference ground of lower encoder receiving port
26	Analog Input 1-	Negative terminal of analog command input 1, read via Analog Input 1

Pin	Function	Instruction
27	Direction Command Input- /The Second Encoder A-	Negative terminal of direction command differential input or A- signal of secondary encoder differential input
28	Pulse Command Input+ /The Second Encoder B+	Positive terminal of pulse command differential input B+ signal of secondary encoder differential input
29	Pulse & Direction Command Reference Ground Secondary Encoder Input Reference Ground	Connect to reference ground of upper pulse direction command transmitting port Connect to reference ground of the secondary encoder transmitting port
30	Digital Output 2-	Opto-isolated programmable digital output, read via Digital Input 2
31	Ripid Digital Input Output 3	Isolated programmable digital input, read via Digital Input 1
32	Ripid Digital Input Output 5	Isolated Programmable digital input, read via Digital Input 5
33	Digital Output 2+	Opto-isolated programmable digital output, read via Digital Input 2
34	Analog Input 3-	Negative terminal of analog command differential input 3, read via Analog Input 3
35	Analog Input 2-	Negative terminal of analog command differential input 2, read via Analog Input 2
36	Reserved	Reserved

5.2.5 Motor Position Feedback Wiring

Wire the motor feedback interface according to the type of feedback device applied.

Refer to the pin assignment table below.

Pins 1, 2, 14 and 15 have dual functions.

Pins 12 and 25 are for motor temperature sensor input.Leave unused pins unconnected.

Sheet 5-2 C6-Pin Definition of Motor Feedback Input Interface

Pin	Function	Pin	Function
1	Incremental EncoderA+/SSI Encoder Data+	14	Incremental Encoder A-/SSI Encoder Data-
2	Incremental Encoder B+/SSI Encoder Clock+	15	Incremental Encoder B-/SSI Encoder Clock-
3	Incremental Encoder Z+	16	Incremental Encoder Z-
4	Hall U	17	Hall V
5	Hall W	18	8VDC power supply positive output for encoder
6	Resolver sine+	19	Resolver sine-
7	Resolver cosine+	20	Resolver cosine-
8	Resolver reference+	21	Resolver reference-
9	Sine&Cosine Encoder sine+	22	Sine&Cosine sine-
10	Sine&Cosine Encoder cosine+	23	Sine&Cosine cosine-
11	5VDC power supply positive output for encoder	24	8VDC/5VDC power supply negative output for encoder
12	Motor Temperature Sensor Input	25	Motor Temperature Sensor Input
13	5VDC power supply positive output for encoder	26	External Shielding

- Feedback Wiring Guide

The tables below list common feedback types. Contact our technical support if your motor feedback type is not covered.

Pin layouts are specified in tables. Record motor pin numbers marked in the User Motor Pin Label column for future reference.

Set motor feedback type, resolution and other parameters via Motor Setup and Feedback Setting interfaces in AKS Control System. Refer to Motor Setup Wizard and Motor Feedback documents.

Sheet 5-3 Resolver Feedback

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
6	Twisted		Sine+
19			Sine-
7	Twisted		Cosine+
20			Cosine-
8	Twisted		Reference+
21			Reference-
12	Twisted		Motor Temperature Sensor
25			Motor Temperature Sensor
24			Optional: Twisted Inner Shield Ground
26			Cable Shielding

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-4 Incremental Encoder A/B/Z with Single-side Hall Sensor

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		A+
14			A-
2	Twisted		B+
15			B-
3	Twisted		Z+
16			Z-
4			Hall U
17			Hall V
5			Hall W
12	Twisted		Motor Temperature Sensor
25			Motor Temperature Sensor
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1.The Hall sensors specified in this table are single-ended type. Refer to the corresponding section for differential output Hall sensors.

2. Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-5 Single-side Hall Sensor (Hall Sensor as the Only Feedback)

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
4			Hall U
17			Hall V
5			Hall W
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1.The Hall sensors specified in this table are single-ended type. Refer to the corresponding section for differential output Hall sensors.

2. Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-6 Incremental Encoder A/B with Differential Hall Sensor

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		A+
14			A-
2	Twisted		B+
15			B-
9	Twisted		Hall U+
22			Hall U-
10	Twisted		Hall V+
23			Hall V-
3	Twisted		Hall W+
16			Hall W-
12	Twisted		Motor Temperature Sensor
25			Motor Temperature Sensor
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

Leave pins 12/25 vacant if the motor has no temperature sensor.

Connect signal wires to C5 and C6 interfaces correspondingly when using differential Hall sensors and incremental encoder A/B/Z signals, as described below.

Sheet 5-7 Single-side Hall Sensor (Hall Sensor as the Only Feedback)

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
9	Twisted		Hall U+
22			Hall U-
10	Twisted		Hall V+
23			Hall V-
3	Twisted		Hall W+
16			Hall W-
12	Twisted		Motor Temperature Sensor
25			Motor Temperature Sensor
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-8 Tamagawa Incremental Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		A+/Hall U+
14			A-/Hall U-
2	Twisted		B+/Hall V+
15			B-/Hall V-
3	Twisted		Z+/Hall W+
16			Z-/Hall W-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. This encoder shares wiring for ABZ signals and Hall UVW signals. It outputs Hall signals right after power-on, then transmits position signals.
2. Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-9 Incremental Sine/Cosine Encoder-Without Hall Sensor

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. Phase search shall be performed every time the drive or motor is powered on.

2. Leave pins 12/25 vacant if the motor has temperature sensor.

Sheet 5-10 Incremental Sine/Cosine Encoder-Without Hall Sensor

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
4			Hall U
17			Hall V
5			Hall W
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has temperature sensor.

Sheet 5-11 Incremental Sine/Cosine Encoder-With Index Signal

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
3	Twisted		Z+
16			Z-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-12 Incremental Sine/Cosine Encoder-With Index Signal

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
3	Twisted		Z+
16			Z-
4			Hall U
17			Hall V
5			Hall W
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has temperature sensor.

Sheet 5-13 Sick 8V With Sine/Cosine Signal (HIPERFACE Procotol)

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		SSI Data+
14			SSI Data-
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-14 Sick 8V With Sine/Cosine Signal (HIPERFACE Procotol)

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		SSI Data+
14			SSI Data-
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
18	Twisted		8V Power Positive
24			8V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-15 HEIDENHAIN EnDat2.2

Pin	Twisted Cable	User Motor Pin Marking	Signal Function& Description
1	Twisted		SSI Data+
14			SSI Data-
2	Twisted		SSI Clock+
15			SSI Clock-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-16 HEIDEHAIN EnDat2.1 With Sine/Cosine Signal

Pin	Twisted Cable	User Motor Pin Marking	Signal Function &Description
1	Twisted		SSI Data+
14			SSI Data-
2	Twisted		SSI Clock+
15			SSI Clock-
9	Twisted		Sine+
22			Sine-
10	Twisted		Cosine+
23			Cosine-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-17 Nikon 17-bit Single-turn Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function &Description
1	Twisted		Serial Data+
14			Serial Data-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：Leave pins 12/25 vacant if the motor has no temperature sensor.

Sheet 5-18 Nikon 17-bit Single-turn Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function &Description
1	Twisted		Serial Data+
14			Serial Data-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. Backup battery for encoder is not included in AKS products. Connect the battery to the encoder and observe polarity.
2. Battery voltage shall exceed 3.6V. Recommended model: Toshiba ER3V, 3.6V 1000mAh.
3. Connect to pin 12/25 if the motor is equipped with temperature sensor.

Sheet 5-19 Tamagawa 17 Incremental Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function &Description
1	Twisted		Serial Data+
14			Serial Data-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. 17-bit serial digital encoder outputs 131072 pulses per revolution.
2. Connect to pin 12/25 if the motor is equipped with temperature sensor.

Sheet 5-20 Tamagawa 17 Single-turn Absolute Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function &Description
1	Twisted		Serial Data+
14			Serial Data-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. The 17-bit serial digital encoder outputs 131072 pulses per revolution and delivers single-turn absolute position.
2. Connect to pin 12/25 if the motor is equipped with temperature sensor.

Sheet 5-21 Tawagama 17 Multi-turns Absolute Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function & Description
1	Twisted		Serial Data+
14			Serial Data-
11	Twisted		5V Power Positive
24			5V Power Negative
26			Shield Terminal

Note：

1. The 17-bit serial digital encoder outputs 131072 pulses per revolution. It outputs absolute position by recording single-turn position and rotation turns.
2. The encoder backup battery is not supplied with AKS products. Connect the battery to the encoder correctly with proper polarity. The battery voltage shall be higher than 3.6V. Recommended model: Toshiba ER3V, 3.6V 1000mAh.
3. Connect to pin 12/25 if the motor is equipped with temperature sensor.

Sheet 5-22 Feedback Wiring BiSS-C Encoder

Pin	Twisted Cable	User Motor Pin Marking	Signal Function & Description
1	Twisted		Serial Data+ (SLO+)
14			Serial Data- (SLO+)
2	Twisted		Serial Clock+ (MA+)
15			Serial Clock-(MA-)
11	Twisted		Direct Current Voltage+5V
24			Direct Current Voltage 0V
26			Shield

5.2.6 STO Connecting

The STO interface of all 220VAC medium-voltage AKS models is C1. See Figure 5-3.

Warning:

When driving non-horizontal loads, the system shall be equipped with external mechanical safety components such as motor mechanical brakes. The drive cannot hold the load position once the STO function is activated. This may cause severe personal injury or equipment damage, which must be prevented.

Safe Torque Off (STO) is a safety function that cuts off power output from the drive to the motor and stops torque generation. STO enable terminals and STO ground shall be wired for AKS drive operation with 24 VDC supply voltage. Connect the STO interface accordingly.

Note: If STO control is not required in application, connect STO1 and STO2 to +15V, and STOCOM to -15V



Figure 5-3 STO Interface

Sheet 5-23 STO Definition

Pin	Function	Instruction
1	STO 1	STO Input Access 1
2	NC	Unused Pin
3	STO 2	STO Output Access 2
4	STOCOM	STO Input Reference Point
5	+ 15V	Positive 15V Reference Point
6	- 15V	Negative 15V Reference Point

6. Peripheral Device Selection

6.1 Cable & Crimping

Refer to the cable table below for all cables required by AKS system.

Strip 2mm insulation off wire ends before crimping, see Figure 6-1.

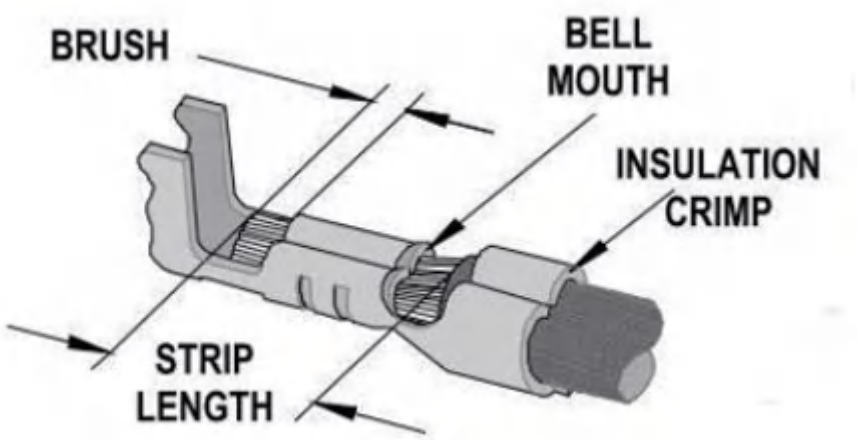


Figure 6-1 Wire Crimping Pin

Sheet 6-1 C1--USB2.0A to Mini-B cable

Project	Specification	Part Properity
Wire Gauge	20-28AWG	
EMI Filter	2 ferrite cores placed near each connector	

Sheet 6-2 C2--Controller interface cable

Project	Specification	Part Properity
Wire Gauge	20-28AWG	
EMI Filter	2 ferrite cores placed near each connector	

Sheet 6-3 C3--Machine interface cable

Project	Specification	Part Properity
C2 flying lead cable	1m flying lead cable	
	2m flying lead cable	
	3m flying lead cable	
	5m flying lead cable	
	10m flying lead cable	

Sheet 6-4 C4——Feedback Cable

Project	Specification	Part Properity
C2 flying lead cable	1m flying lead cable	
	2m flying lead cable	
	3m flying lead cable	
	5m flying lead cable	
	10m flying lead cable	

Sheet 6-5 C5|C6——RJ45 Industry Ethernet/EtherCAT Cat5e Cable

Project	Specification	Part Properity
RJ45 CAT5E Cable	Length: 0.5 m	
	Length: 1 m	
	Length: 2 m	
	Length: 10 m	
Shielding	85% copper braided shielding coverage	
Twisted Cable	Needing	
Max. Length	10m	
Wire Gauge	20-28AWG	

Sheet 6-6C7--RS232 Cable

Project	Specification	Part Properity
RS232 Cable	85% copper braided shielding coverage	
Shielding	10m	
Max. Length	24-28AWG	
Wire Gauge		

Sheet 6-7USB to RS232Adapter Cable

Project	Specification	Part Properity
USB to RS232 cable		
Shielding	85% copper braided shielding coverage	
Max. Length	1.5m	

Sheet 6-8 C8--Daisy Chain

Project	Specification	Part Properity
Cable		
Shielding	85% copper braided shielding coverage	
Twisted Cable	Needing	
Max. Length	0.5m	
Wire Gauge	24-28AWG	

Sheet 6-9 C3<>C3--Gantary Cable

Project	Specification	Part Properity
Cable	0.5m Cable	
	1m Cable	
	2m Cable	
Twisted Cable	Needing	
Wire Gauge	24-28AWG	

6.2 DC Bus Regeneration

6.2.1 Overview of Regenerative Energy

When the motor and load decelerate rapidly, the kinetic energy of the load feeds energy back from the motor shaft to the drive, which is defined as regenerative energy or regeneration.This energy must be dissipated or absorbed.

Undissipated regenerative energy accumulates alongside the electricity stored in the bus module capacitors. The regenerative resistor is unnecessary if the capacitors can accommodate the extra energy.

Excessive energy may be injected into capacitors due to large inertia mismatch between load and motor, pushing voltage over the overvoltage threshold and triggering drive shutdown. To avoid overvoltage and system damage, surplus regenerative energy shall be dissipated via regenerative resistors.

Refer to regenerative resistance calculation to assess system requirements and determine resistor capacity.

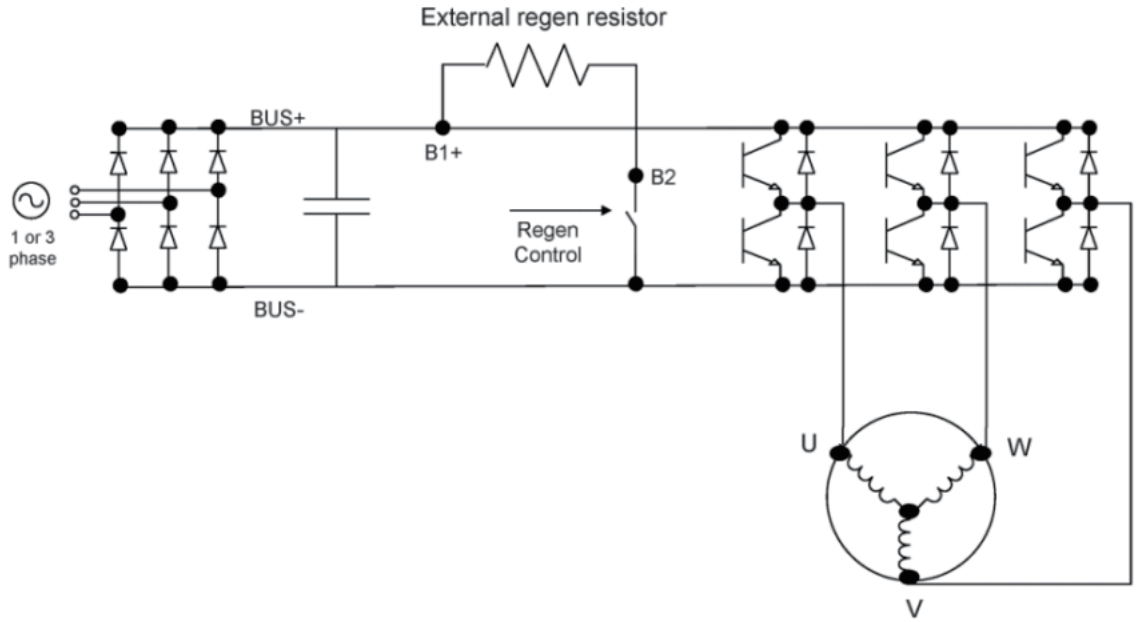


Figure 6-2 Regenerative resistor working circuit

6.2.2 The Calculation of Regenerated Resistor

The resistance value (Ohms, Ω) of the regenerative resistor is determined by the AKS servo drive; the required power depends on the specific application. Therefore, multiple regenerative resistor options may be available for each drive.

Recommended manufacturers and part numbers for regenerative resistors for AKS are listed in the table below.

Sheet 6-10 Recommended Regenerative Resistors for AKS

Specification	Instruction
Below AKS-006	50Ω-200W
Below AKS-008	45Ω-500W
Below AKS-012	50Ω-1000W
Below AKS-030	30Ω-1500W

- Calculate the feedback energy per deceleration:

$$E_{dec} = \frac{1}{2}(J_M + J_L)(W_1^2 + W_2^2)$$

In the formula：

E_{dec}= Feedback energy during deceleration(J or ft · lb)

J_M = Rotor Innertia (kg · m2 or lb · ft · sec2)

J_L= Load Innertia(kg · m2 or lb · ft · sec2)

ω_1 = Speed at the start of deceleration (rad/s)

ω_2 = Speed at the end of deceleration (rad/s)

- Determining motor energy consumption:

Part of the energy is consumed by the motor as current flows through winding resistance：

$$E_{motor} = 3I_M^2 * \frac{R_M}{2} * t_d$$

In the formula：

E_{motor}= motor energy consumption(J)

R_M = motor resistor（ΩL-L)

I_M = current during deceleration(ARMS/phase)

t_d = deceleration time(s)

- Determining friction energy consumption:

$$E_{friction} = \frac{1}{2} T_f (w_1 - w_2) t_d$$

In the formula：

$E_{friction}$= friction energy consumption (J or ft · lb)

T_f = friction torque (N · m or lb · ft

ω_1) = Speed at deceleration start (rad/s)

ω_2 = Speed at deceleration end (rad/s)

t_d = deceleration time (s)

- Determine amplifier energy consumption:

$$E_M = E_{dec} - E_{motor} - E_{friction}$$

In the formula：

E_M= Total energy consumed by amplifier (J or ft · lb)

If the energy is less than the storable energy of the bus module, no regenerative resistor is required.

E_M shall be expressed in joules. Conversion rate: 1ft · lb = 1.356J

- Determine bus absorbable energy:

$$E_{BUS} = \frac{1}{2} C (V_M^2 - V_{NOM}^2)$$

In the formula：

E_{BUS}= Energy absorbable by bus module （J）

C = Bus module capacitance (F); refer to *Bus Specification* section

v_M = Max. bus voltage (V); refer to *Bus Specification* section

V_{NOM} = Bus nominal voltage=$\sqrt{}$2Vmains （V）

Determining whether a regenerative resistor is required：

If $E_{BUS} > E_M$ No regenerative resistor required

If $E_{BUS} < E_M$ Regenerative resistor required

- **Determining the resistor value:**

The calculation procedure for regenerative requirements consists of two parts. Both the resistance value and power rating of the regenerative resistor must be determined.

- Determining resistance:

$$R_{Max} = \frac{V_M^2}{V_B I_M \sqrt{3}}$$

In the formula：

V_B= Motor back EMF minus motor loss

$$V_B = K_B N - \sqrt{3} I_M \frac{R_M}{2}$$

In the formula：

V_M = Max. bus voltage (V); refer to Bus Specification section

K_B = Back EMF constant（VL-L/rpm）

N = Motor speed before deceleration（rpm）

I_M = Current during deceleration

R_M = Motor resistor（ΩL-L）

- Determining average disspation power:

$$P_{AV} = \frac{E_M - \frac{1}{2} C (V_{High}^2 - V_{Low}^2)}{t_{cycle}}$$

In the formula：

E_M= Total energy consumed by amplifier（J）

C = Bus capacitance module (F); refer to *Bus Specification* section

V_{High} = Hysteresis point: Regeneration circuit activation (V)；refer to *Bus Specification* section

V_{Low} = Hysteresis point: Regeneration circuit shutdown; refer to *Bus Specification* section

t_{cycle} = Deceleration interval time+deceleration time (s)

- Determining peak resistor power:

$$P_{PK} = \frac{V_M^2}{R_{Regen}}$$

In the formula：

R_{Regen}= Regenerated resistor resistance（ΩL-L）

V_M = Max. bus voltage (V); refer to *Bus Specification* section

- Motor & Drive Specification:

  AKS drive, single-phase AC 240V input:

  In the formula：

  J_{motor}= 0.000484lb·ft·sec2

  R_M = 1.32Ω

  I_{peak} = 18A

  T_M = 19.8lb · ft

  C = 0.001120F

  V_M = 420V

  K_B = 81.2V/krpm=0.0812V/rpm

  k_t = 1.188lb·ft/A

- System Specification

  T_f= 1.5I_b·f_t

  J_{Load} = 0.0015lb·f_t·sec2

  V_{Max} = 2,500rpm=>ω_M=261rad/s

  V_{NOM} = 240Vrms=340VDC

  Deceleration interval time=5s

- Calculating deceleration time:

$$t_d = \frac{(J_{Motor}+J_{Load})w_M}{T_M+T_f} = (0.000484 + 0.001) * (261)(19.8 + 1.5) = 0.0181s$$

- Current calculation during decelerration:

$$I_M = \frac{(J_{Motor} + J_{Load})}{T_M + T_f} * \frac{w_1 - w_2}{t_d} = \frac{0.001 + 0.000484}{1.188} * \frac{261}{0.0181} = 18A$$

- Calculating Energy:

$$E_{dec} = \frac{1}{2}(J_M + J_L)(W_1^2 + W_2^2) = \frac{1}{2}(0.0015 + 0.000484)(261^2 - 0) = 67.57f_t * I_b = 91.63J$$

$$E_{motor} = 3I_M^2 * \frac{R_M}{2} * t_d = 3 * 18^2 * \frac{1.32}{2} * 0.0181 = 11.61J$$

$$E_{friction} = \frac{1}{2}T_f(w_1 - w_2)t_d = \frac{1}{2} * 1.5 * (261 - 0) * 0.0181 = 3.54f_tI_b = 4.8J$$

$$E_M = E_{dec} - E_{motor} - E_{friction} = 91.633 - 11.6110 - 4.8 = 75.22J$$

$$E_{BUS} = \frac{1}{2}C(V_M^2 - V_{NOM}^2) = \frac{1}{2} * 0.001120 * (420^2 - 340^2) = 34.05J$$

As for E_M>E_{BUS}, the generated resistor is required.

- Calculating resistor specification

$$V_B = K_B N - \sqrt{3} I_M \frac{R_M}{2} = 0.0812 * 2500 - \sqrt{3} * 18 * \frac{1.32}{2} = 182.423V$$

$$R_{Max} = \frac{V_M^2}{V_B I_M \sqrt{3}} = \frac{420^2}{182.423 * 18 * \sqrt{3}} = 31.02\ \Omega$$

$$P_{AV} = \frac{E_M - \frac{1}{2} C (V_{High}^2 - V_{Low}^2)}{t_{cycle}} = \frac{44.42 - \frac{1}{2} * 0.001120 * (400^2 - 380^2)}{5.03} = 7.1\text{W}$$

$$P_{PK} = \frac{V_M^2}{R_{Regen}} = \frac{420^2}{132} = 5686.65\text{W}$$

6.2.3 The protection of regenerated resistor overload

Motor energy regeneration in AKS remains active all the time.
There are two modes for the motor energy regeneration function:

Regeneration with resistor overload protection.

The regenerative resistor circuit requires built-in overload protection to ensure stable operation.
This mode adopts configurable AKS regenerative resistor parameters for protection.

Regeneration without resistor overload protection

	Use regenerative resistor protection with caution. Rising bus voltage generated by the motor may trigger protection and result in overvoltage condition.

6.2.4 Regenerated Resistor Parameter

The capacity of regenerative resistor is defined by several parameters.

To enable regenerative resistor protection of AKS, set REGENRES and/or REGENPOW to values other than -1, which is the default setting.

Sheet 6-11 Regenerated Resistor Parameter

VarCom	Instruction
REGENRES	Regenerative resistor resistance, unit: Ohm
REGENPOW	Regenerative resistor power, unit: Watt
REGENMAXONTIME	Maximum sustainable activation time of regenerative resistor, unit: millisecond
REGENMAXPOW	Maximum power calculation of regenerative resistor, unit: Watt

Note: Set REGENRES=-1 and REGENPOW=-1 if no regenerative resistor equipped.
Either parameter set to -1 disables overload protection algorithm for the resistor.

6.3 Fuse

Circuit protection shall comply with National Electrical Code and relevant national, provincial and local regulations.

- US fuse: CC, J or T type, fast acting. Must be UL and CSA certified; UR certification alone is not acceptable.
- EU fuse：F type, fast acting.
- Fuse holder: Standard type or finger-safe type complying with IEC60529 standard. Examples:
  - Bussmann: CH series modular fuse holder, for fuses below 30A, Class J, 3-pole: CH30J3.
  - Ferraz (Mersen): UltraSafe fuse holder, for fuses ≤30A, Class J, 3-pole, with LED indicator: US3J3I.

6.4 Leakage Current

Leakage current of PE conductor is the combination of leakage current from equipment and cables. It contains multiple frequency components, which are converted into 50Hz equivalent current by residual current circuit breaker. Hence, conventional multimeter cannot measure leakage current accurately.

Empirically, cable leakage current can be estimated based on the PWM frequency of output stage.

- Ileak=nx20mA+Lx1mA/m，Output stage PWM frequency: 8kHz
- Ileak=nx20mA+Lx2mA/m，Output stage PWM frequency: 16kHz

(Ileak=leakage current，n=drive quantity，L=motor cable length)

Since PE leakage current exceeds 3.5mA, double PE connection or cable cross-sectional area over 10mm2 shall be adopted per IEC61800-5-1. This requirement can be fulfilled by PE terminals and con necting screws.

7. Electromagnetic Interference Suppression

7.1 CE Filter Technology

The AKS drive complies with CE standards specified in certification criteria. Proper bonding, grounding and filtering techniques are mandatory to meet the requirements.

There are two typical types of noise current. The first is conducted noise emission flowing through ground loops. The quality of system grounding scheme is inversely proportional to line noise amplitude, featuring line-to-neutral or line-to-ground common-mode characteristics. The second is radiated high-frequency noise emission, mostly generated via line-to-line capacitive coupling with differential-mode properties.

For correct EMI filter installation, unpainted metal surface of the enclosure is required. It ensures sufficient contact area with the filter housing and forms a low-impedance path between the housing and backplate. The backplate is further connected to the enclosure or earth via high-frequency grounding busbar.

7.2 Earthing

When connecting the AKS drive to other control devices, abide by two basic rules to avoid drive damage:

- The AKS drive must be grounded via the AC mains protective earth wire.
- Ground wires connected to drives, PLCs or PCs linked with AKS shall share the same earth connection.

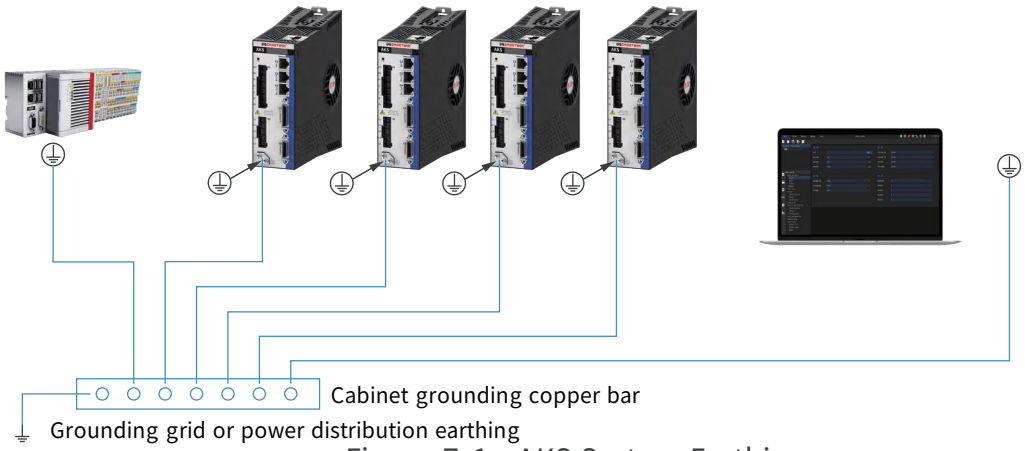


Figure 7-1 AKS System Earthing

System grounding is critical for normal operation of the drive system.

The AC input earth cable must be connected to the PE terminal on the AKS front panel for safety and EMI reduction.

Adopt single-point grounding wiring to eliminate ground loops.

It is highly recommended to mount the AKS on a metal backplate and apply high-frequency earthing. Ensure full electrical contact across the entire rear surface of the drive. Conductive panels such as aluminum or galvanized steel are preferred. Remove all paint and coatings on the mounting area of coated metal panels. This creates an ultra-low impedance path for high-frequency interference signals among filter, drive, power supply and earth. Use flat braided wires or copper busbars for high-frequency grounding, and keep the wiring as short as possible.

Guarantee reliable electrical connection between cabinet components. Connect rear panel and cabinet door to the main cabinet frame with multiple conductive braids. Do not rely on hinges or mounting bolts for earthing. Ensure solid connection between cabinet and qualified grounding system. Ground wire gauge shall be identical or one size smaller than mains cable.

The main controller shall also be properly grounded.

7.3 Input Power Filter

EMI AC line filtering on input power cables is required for AKS electronic components to meet CE standards in industrial environments.

Proper system sizing must be ensured. Filter selection depends on rated voltage, rated current and single/three-phase input type. One line filter is applicable for multi-axis control systems.

Refer to Line Filter document for recommended manufacturers and part numbers. Implementation shall follow below rules:

- ■Keep input and output wiring of the power filter separated.
- ■The filter must be mounted on the same panel as the drive.
- ■Install the filter as close to the drive as possible to avoid noise caused by capacitive coupling with other signal wires and cables.
- ■Remove all paint and coatings when mounting the filter. Unpainted metal backplate is preferred.
- ■The filter is equipped with grounding terminals and must be grounded.
- ■The filter may generate high leakage current. Ensure grounding prior to power connection.
- ■Do not touch the filter within 10 seconds after power cut-off.

7.4 Other EMI suppression suggestions

Separate power and control cables with a minimum spacing of 200mm for better anti-interference. Cross cables at 90 degrees if necessary.

Feedback cables cannot be extended to avoid shield breakage and signal interference.

Use standard wiring methods. Employ metal-shell connectors for cable tapping and fully connect shells along the sheath. Maintain continuous shielding and avoid tapping via terminal blocks.

Use shielded twisted-pair cables for differential analog inputs, with shielding grounded at both ends.

8. PC Commissioning Wiring

8.1 IP Address Setting



Figure 8-1 IP Address Setting

The drive communicates with PC debugging software via 100Mbps Ethernet. Connect to drive port C2 with network cable and configure network parameters accordingly;

NIC IP address: 192.168.X.X. (Arbitrary values allowed except 11.11. Default drive IP: 192.168.11.11)

Subnet mask: 255.255.0.0 (Important note). No gateway or DNS configuration required.

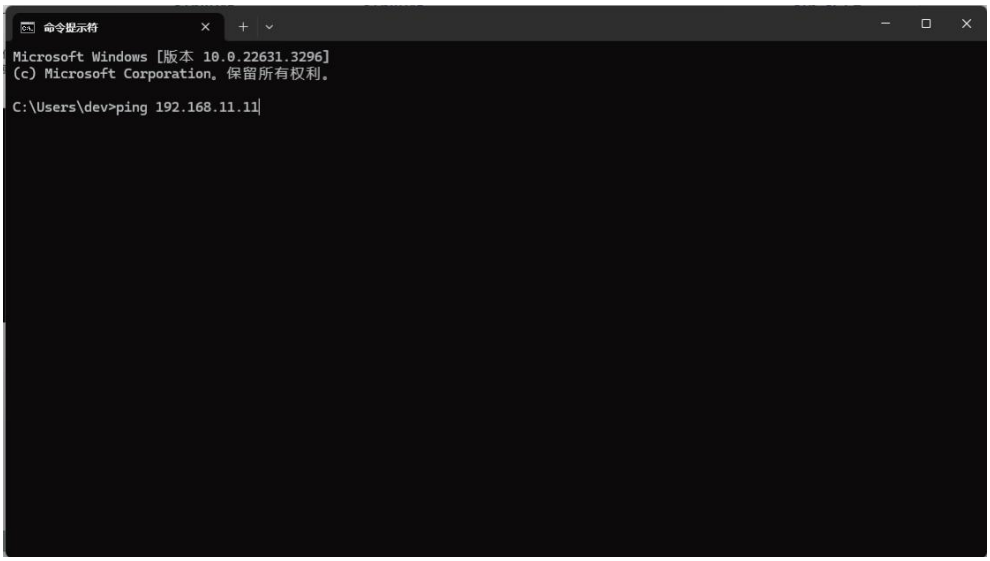


Figure 8-2 IP Address Setting

Ping 192.168.11.11 to verify data transmission and hardware connection. If correct IP and physical connection still fail detection, disable redundant network adapters on this network segment then re-scan.

8.2 New Project Creation and Wiring

First launch the drive debugging software as administrator.



After opening the software, follow the on-screen steps as shown:

- Step 1 ：Left-click "File";
- Step 2 ：Select "New Project"
- Step 3: Click "Modify" in the pop-up window and select the save path for the project.
- Step 4: Enter a custom project name.
- Step 5: Click "OK" to complete project creation.

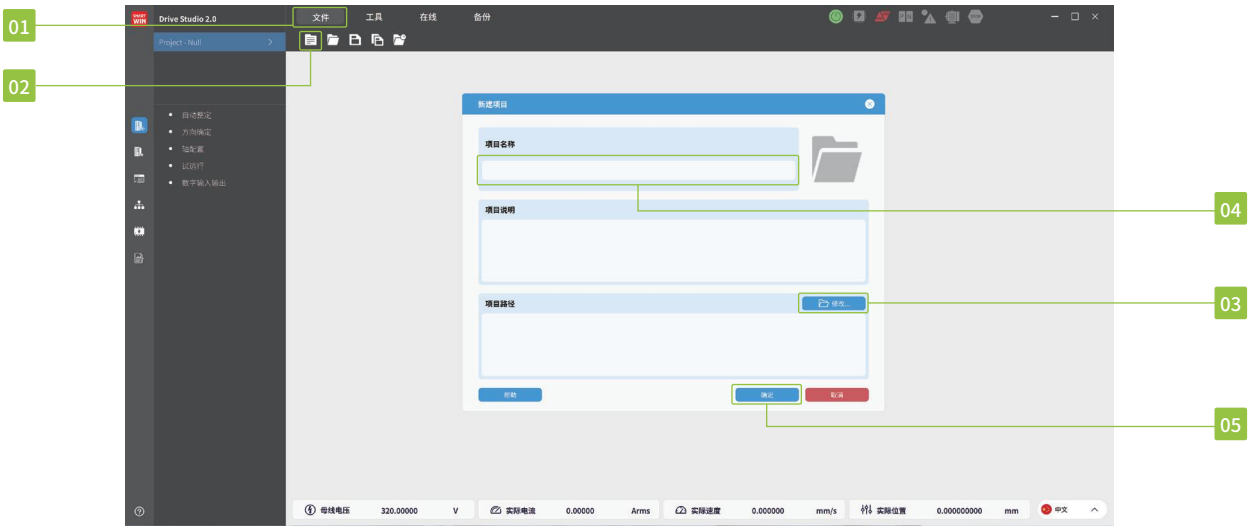


Figure 8-3 New Project Creation and Wiring

After opening the project：

- Step 1: Left-click "Workspace";
- Step 2: Click "New Drive";

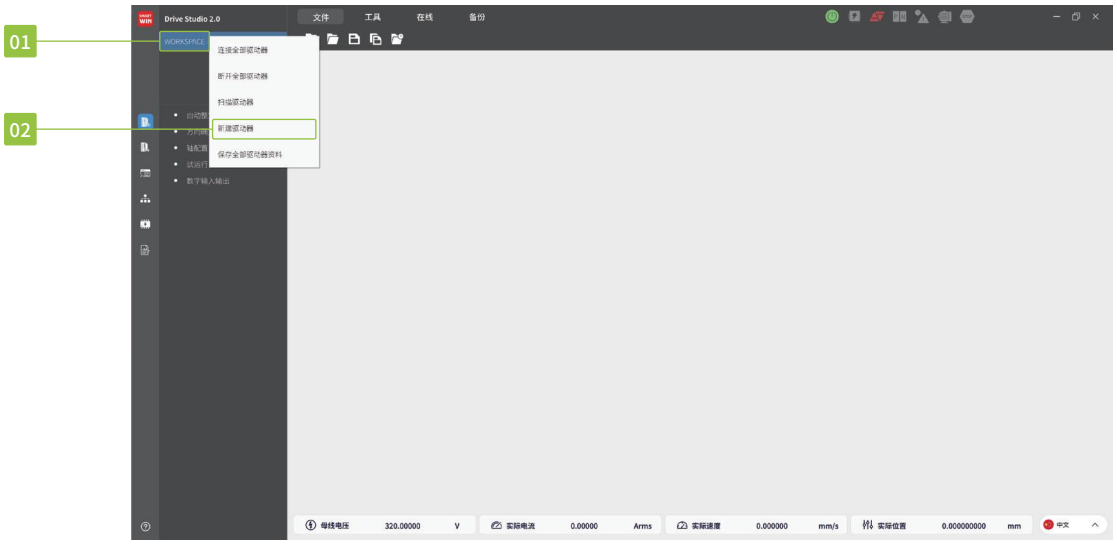


Figure 8-4 New Drive Creation

- Step 3: Name the new drive;
- Step 4: Click "OK" to finish naming. Multiple drives can be created and connected via network switch;



Figure 8-5 Name the newly created drive

- Step 5: Left-click "Workspace", then click Scan Drive;

Drives will be detected if wiring and IP settings are correct. Click Connect to establish connection.

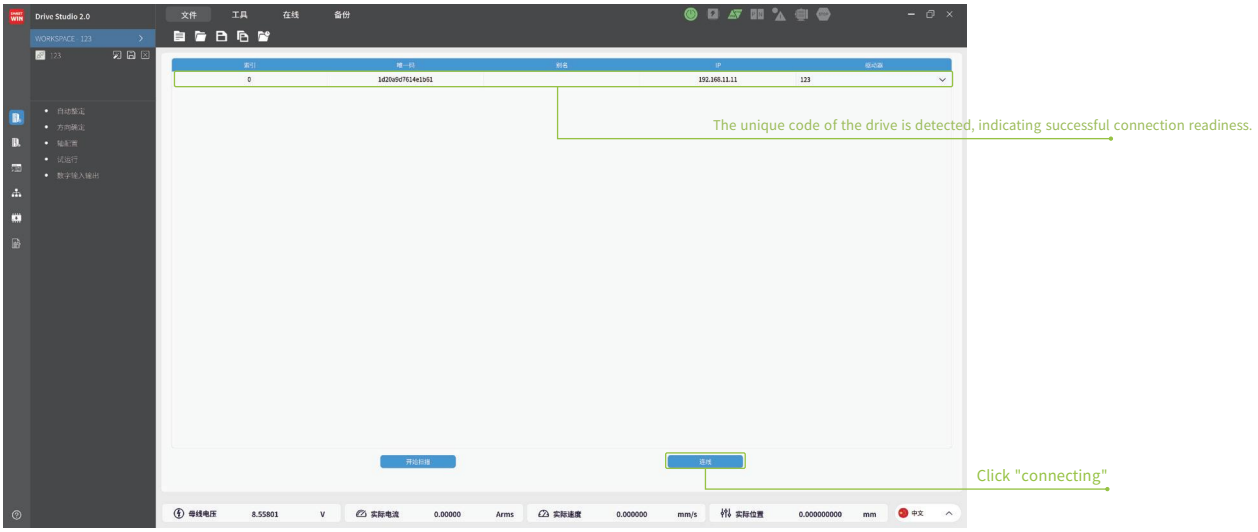


Figure 8-6 Scanning Drive

▶Step 6: Establish communication with the drive. Click Read to upload all drive parameters and finish reading. Up to 8 drives are supported. Connect drives via network switch for networking. Click Start Scan to detect all connected devices. Select target servo drive from dropdown and click Connect to link drive with PC debugging tool.

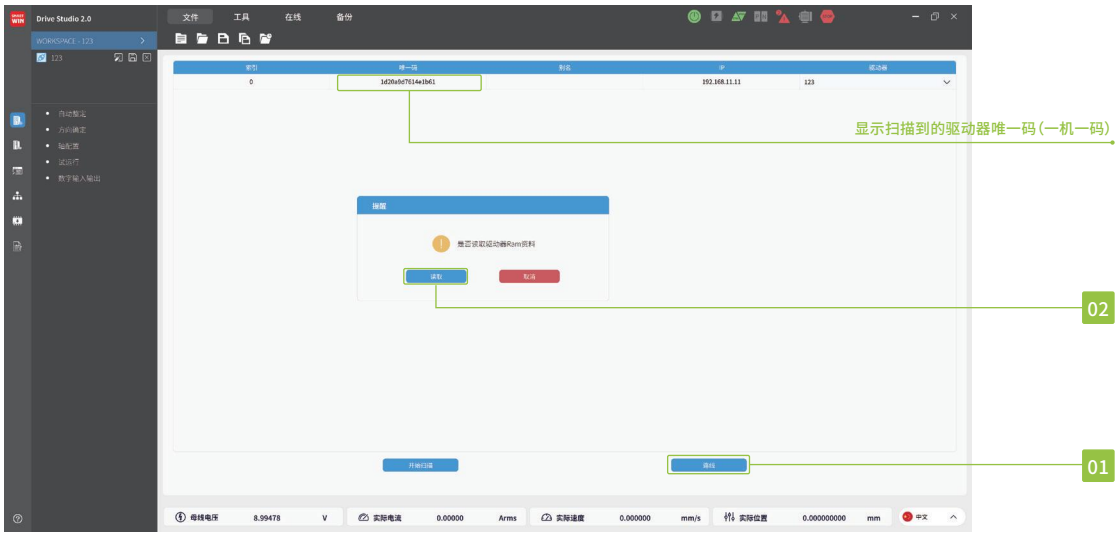


Figure 8-7 Creating communication connecting with drive

9. Drive Parameter Setup

9.1 Drive/Motor Identification

9.1.1 Drive Identification

▶Step 1 ：Single click-Drive Setup；

▶Step 2: Enter the drive information page to view firmware version, specifications, bus voltage, real-time temperature and other data.

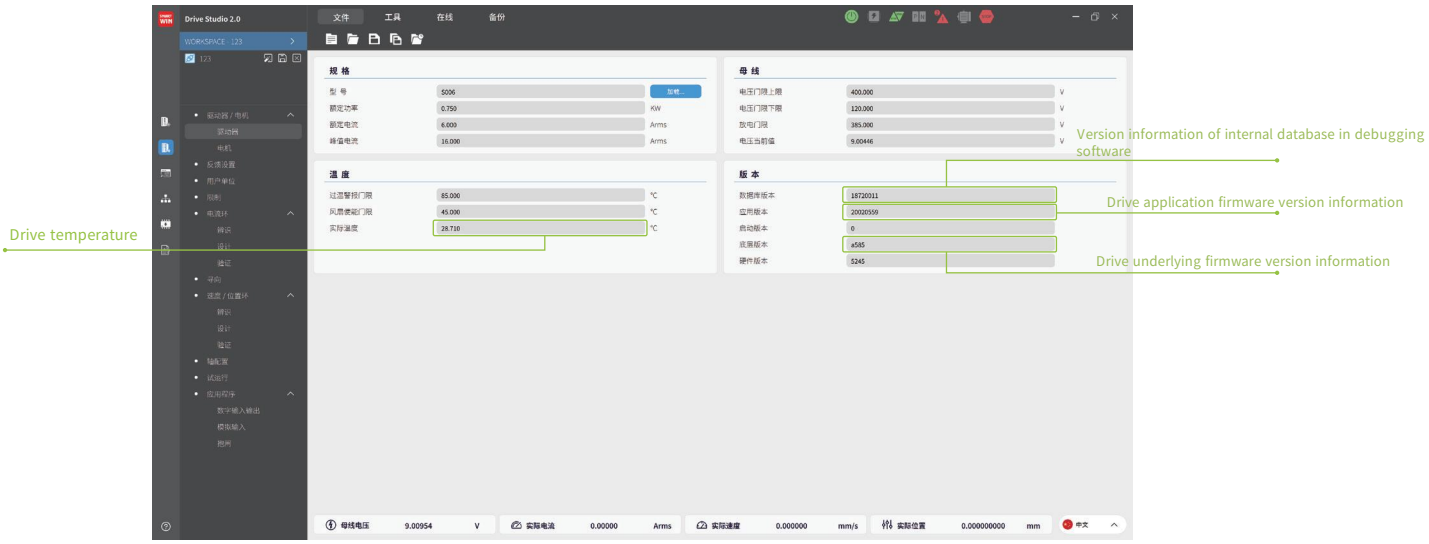


Figure 9-1 Drive Identfication

9.1.2 Motor Parameter Loading

1.Motor Parameter Setup：

- ▶ Step 1: Directly import the matched motor library as shown below；
- ▶ Step 2 ：Click "motor" page;
- ▶ Step 3: Click Load option, select motor type (DDR/Linear Motor), manufacturer and model, then click Load.

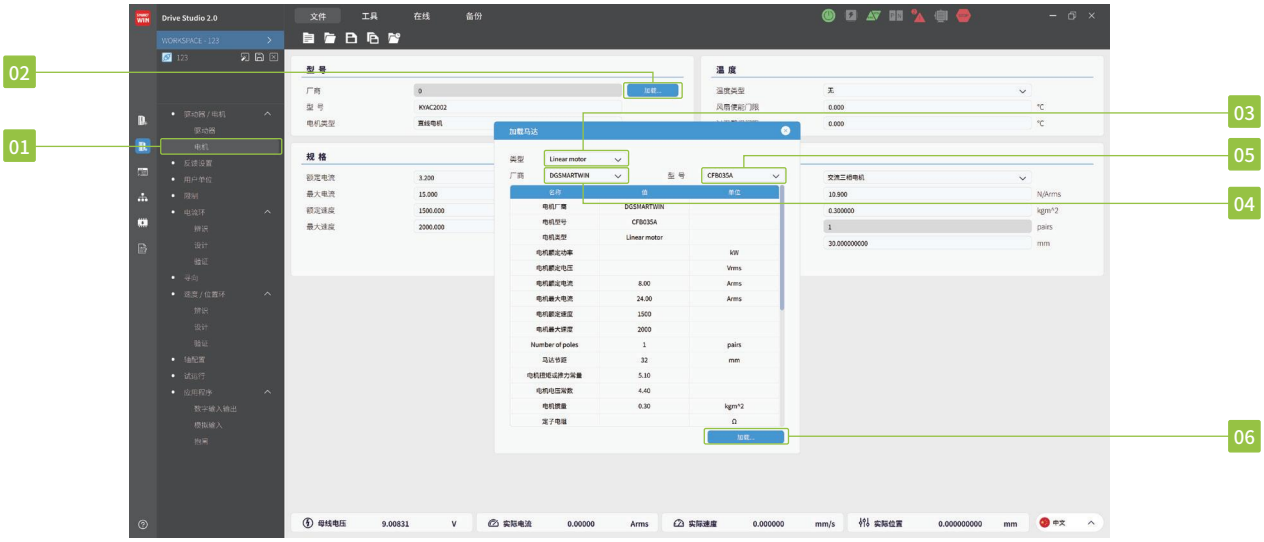


Figure 9-2 Motor Parameter Loading

2. The Third-Party Motor Setup

▶Step 1 ：Setting motor type： rotary motor, linear motor, induction motor；

▶Step 2 ：Setting motor temperature sensor type：KTY，PTC，alarm threshold；

▶Step 3 ：Filling in motor parameter（according to motor data）：motor rated current/max. current /rated speed/max. speed/motor KT constant/linear motor pole pitch/rotary motor pole pairs.

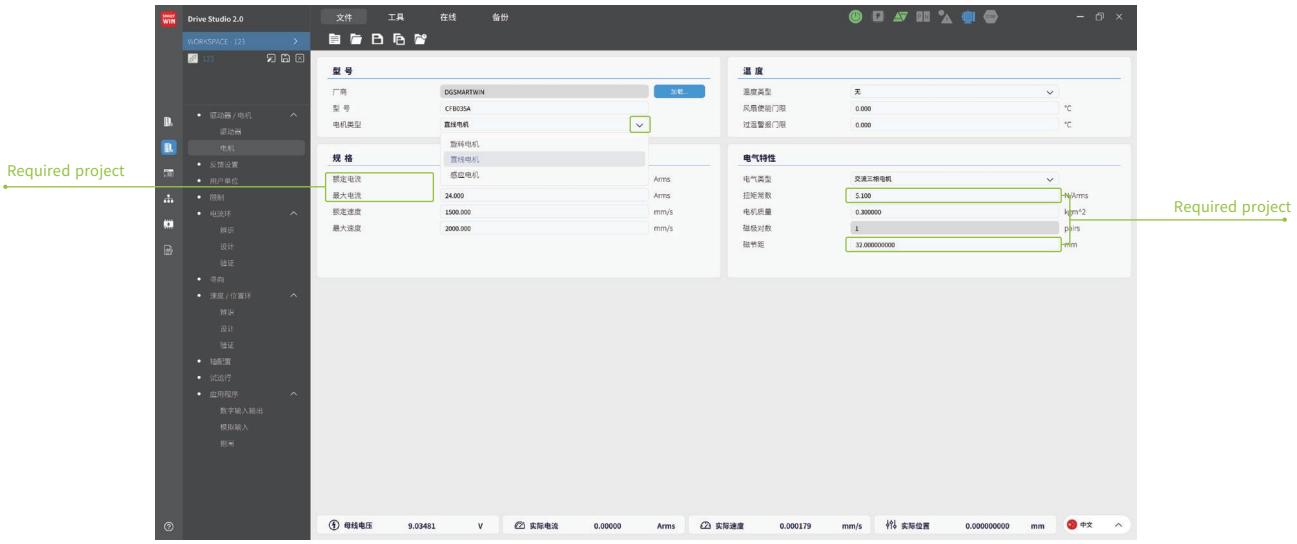


Figure 9-3 The Third-Party Setup

9.1.3 Current & Speed Limitation

▶ Step 1 ：Limit maximum motor operating current；

▶ Step 2 ：Maximum motor operating speed.

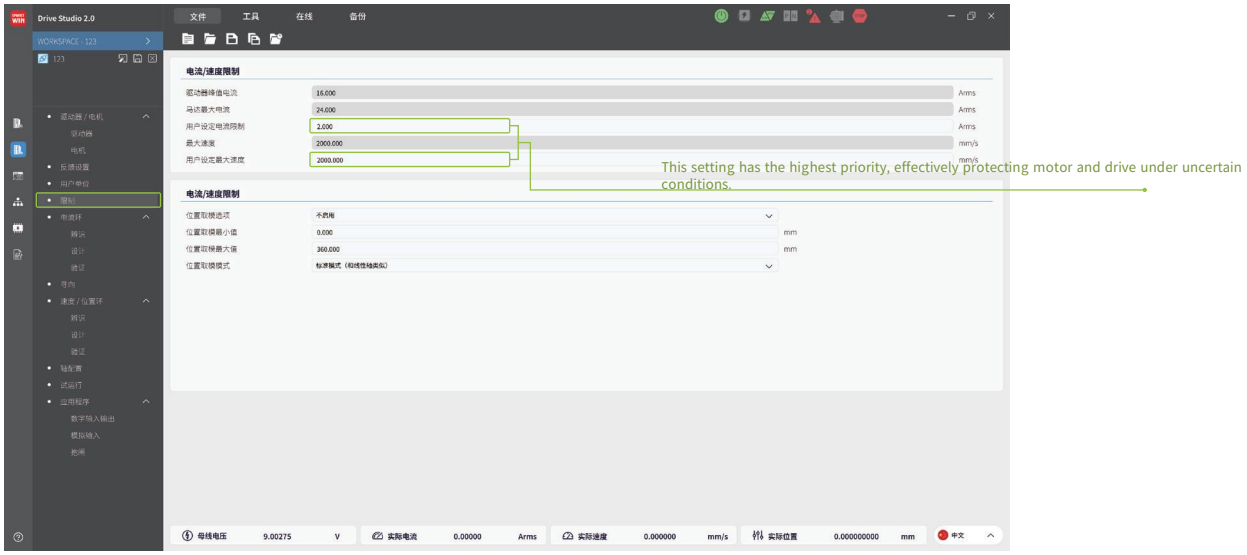


Figure 9-4 Motor & Speed Limitation

9.2 Feedback Setting

9.2.1 Encoder Setting

▶Step 1 ：Encoder feedback type: communication encoder/rotary encoder/pulse encoder/analog encoder;

▶Step 2 ：Encoder filter parameters: Use filter if actual encoder resolution is below 16bit. No filter required for high precision. Default: 800Hz. Input 0Hz to disable signal filtering.

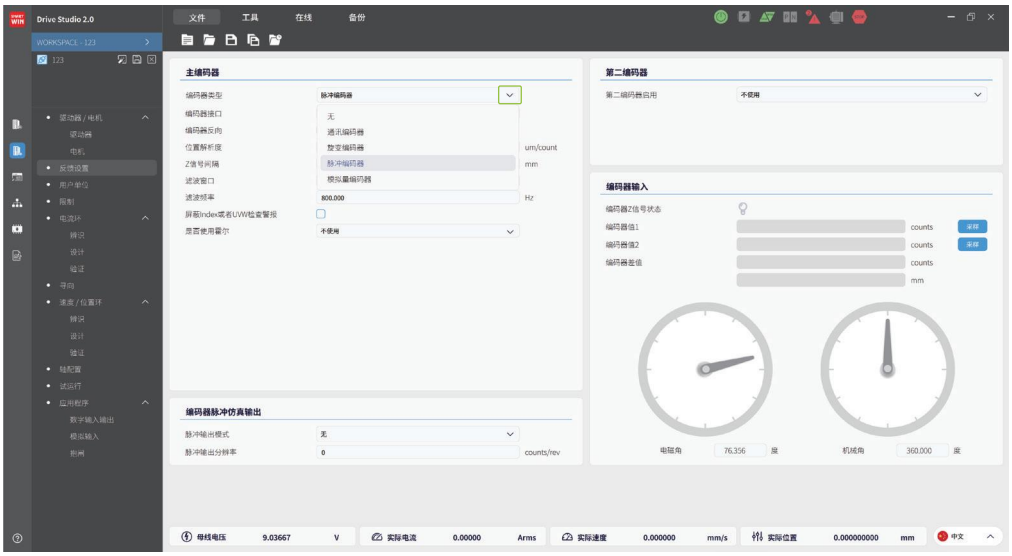


Figure 9-5 Encoder Setting

▶Step 3 ：communication encoder setting: including Hiperface procotol encoder (sick), Endat procotol encoder (HEIDENHAIN), SSI procotol encoder, BISS-C procotol encoder,Tamagawa procotol encoder and Nikon procotol encoder.

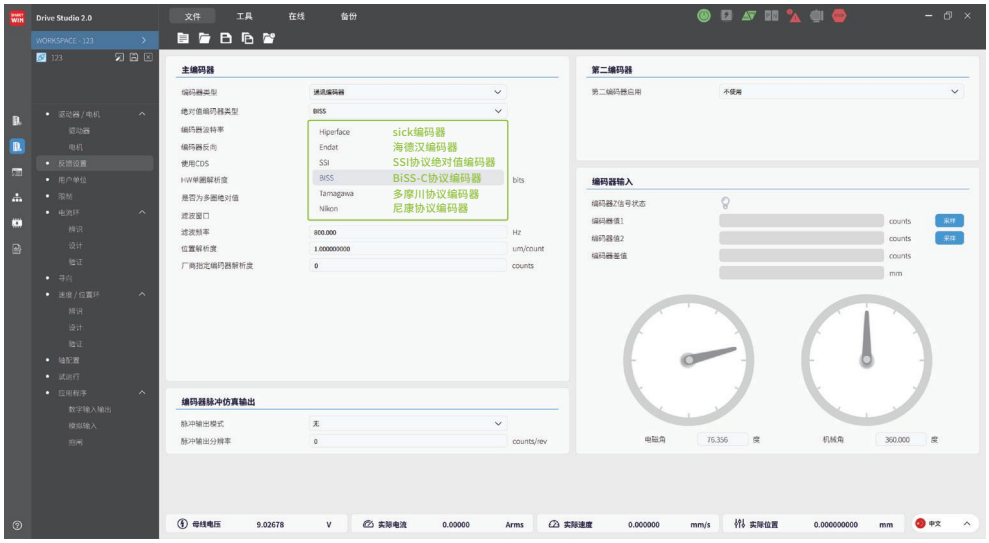


Figure 9-6 Encoder Setting

9.2.2 Common Encoder Setting

1. HEIDENHAIN EnDat linear motor applied setting:

- Step 1: Set encoder communication baud rate to 4M. Use 2M for long cables or interference -prone environments.；
- Step 2: Set position resolution based on encoder parameters, common values include 0.1um, 0.5um.

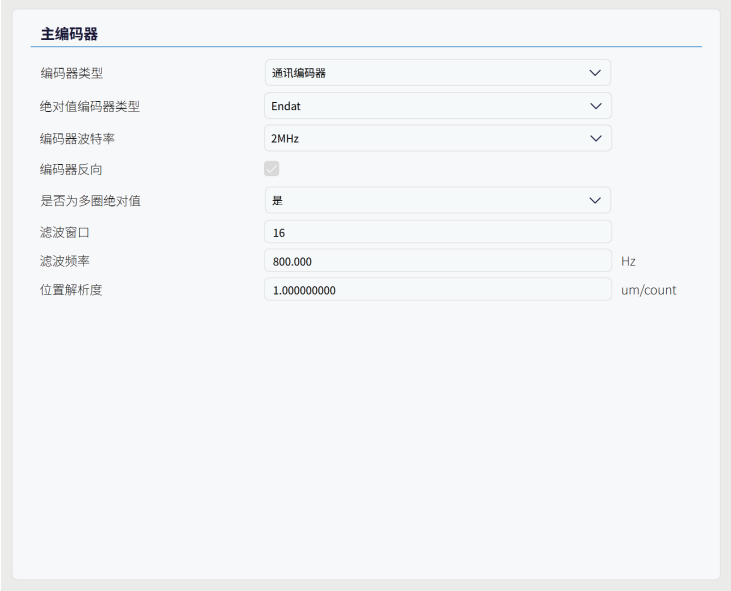


Figure 9-7 HEIDENHAIN EnDat Linear Motor Appied Setting

2. HEIDENHAIN EnDat rotary motor applied setting：

- Step 1：Set encoder baud rate to 4M. Switch to 2M for long cables or heavy interference;
- Step 2 ：EnDat 2.1 requires sincos analog signals;
- Step 3 ：Set single-turn and multi-turn resolution per encoder parameters;
- Step 4 ：Set single-turn resolution based on application requirements.

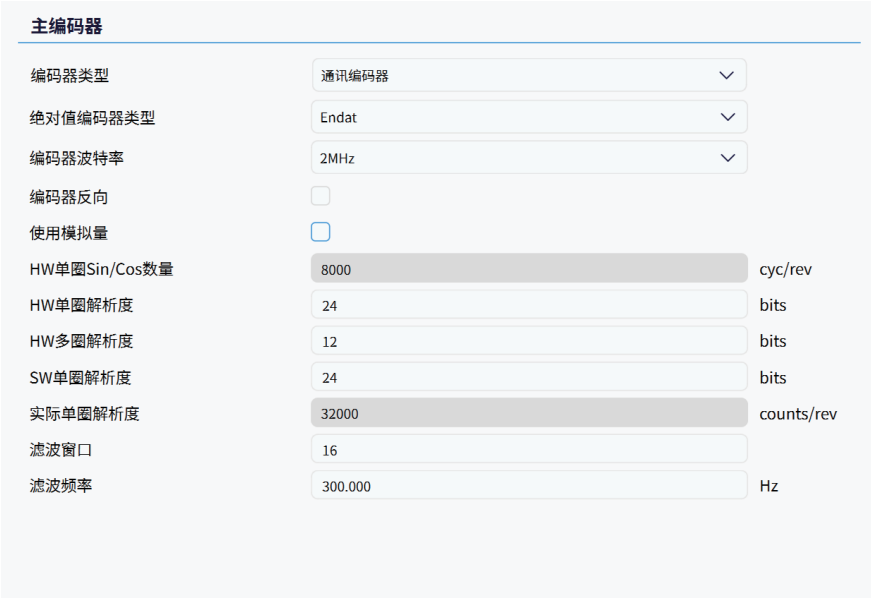


Figure 9-8 HEIDENHAIN EnDat Rotary Motor Applied Setting

3. BISS-C linear motor applied setting:

- Step 1 ：Set encoder communication baud rate to 4M. Set to 2M for long cables or interference -prone surroundings;
- Step 2 ：Set encoder data length and position resolution according to encoder parameters. Common values: 0.1um, 0.5um, 1um.



Figure 9-9 BISS-C Linear Motor Applied Setting

4. BISS-C Rotary Motor Applied Setting：

- Step 1 ：Set encoder baud rate to 4M. Use 2M for long wiring or interference environments;
- Step 2 ：Set hardware single-turn and multi-turn resolution per encoder specifications ;
- Step 3 ：Set software single-turn resolution as required by application.

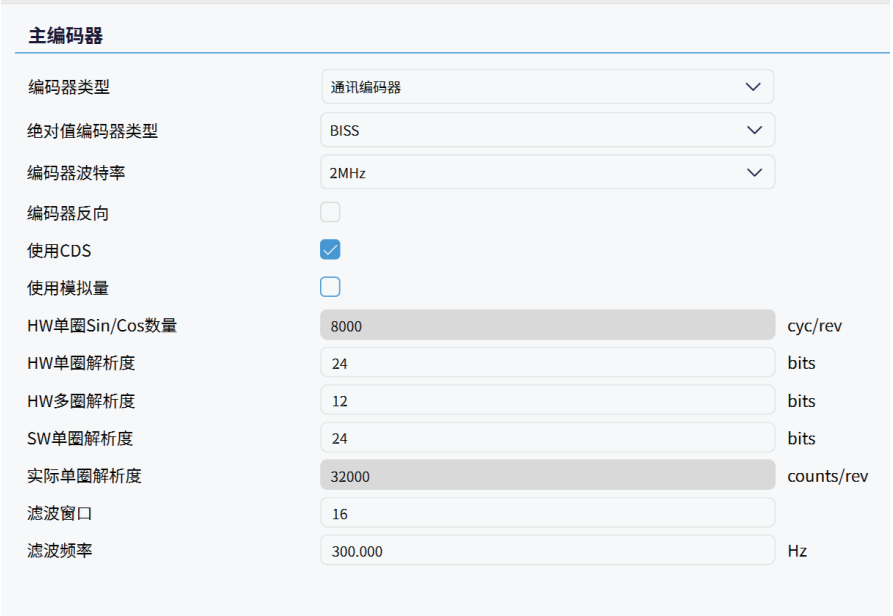


Figure 9-10 BISS-C Rotary Motor Applied Setting

5. Pulse Linear Motor Applied Setting:

- Step 1 ：Set encoder position resolution according to encoder parameters, common values: 0.1 um, 0.5um, 1um.
- Step 2 ：If the motor supports Hall function, you can enable or disable it for incremental encoders.



Figure 9-11 Pulse Linear Motor Applied Setting

6. Pulse Rotary Motor Applied Setting:

- Step 1 ：Set encoder single-turn resolution per specifications, typical values: 2500count, 8000count, 10000count;
- Step 2 ：Enable or disable Hall function for incremental encoder if motor supports it.

Note: AKS monitors all encoder signal lines. Any wire damage triggers A/B line break fault (code: E0150).



Figure 9-12 Pulse Rotary Motor Applied Setting

7. Analog Linear Motor Applied Setting:

- ▶Step 1 ：Set encoder position resolution according to parameters, typical: 5um/count;
- ▶Step 2 ：Enable or disable Hall function for incremental encoder if supported by the motor;
- ▶Step 3 ：Check to mask encoder sin/cos processing alarm and encoder absolute value alarm.



Figure 9-13 Analog Linear Motor Applied Setting

8. Analog Rotary Motor Applied Setting:

- ▶Step 1 ：Set encoder single-turn resolution based on parameters, typical value: 47200cyc/rev
- ▶Step 2 ：Enable or disable Hall function for incremental encoder if motor supports it;
- ▶Step 3 ：Check the options to mask encoder sin/cos processing alarm and absolute value alarm.



Figure 9-14 Analog Rotary Motor Applied Setting

9.2.3 The Second Feedback (Encoder)The

Secondary Encoder Enable Mode：

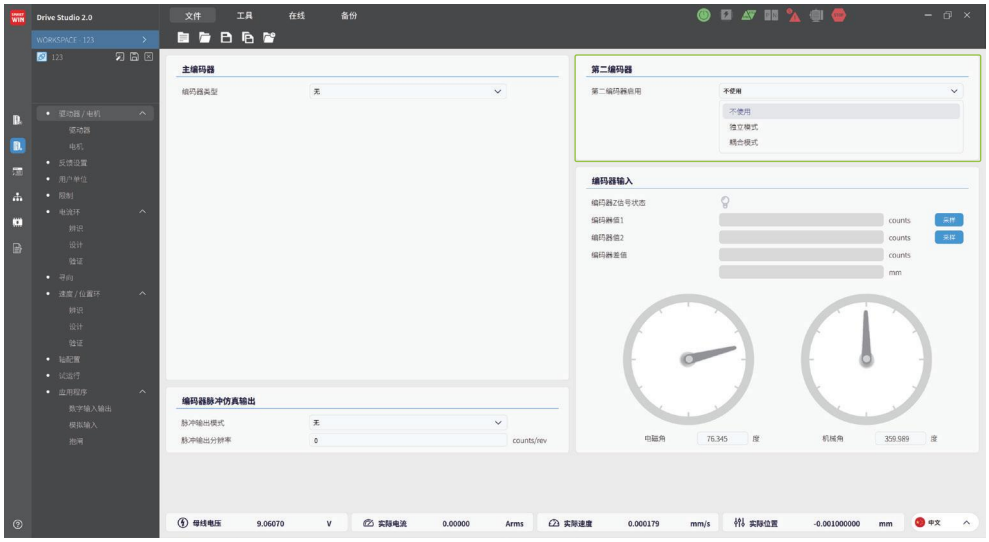


Figure 9-15 The Secondary Encoder Enable Mode



Figue 9-16 The Secondary Encoder Enable Mode

9.2.4 Encoder Pulse Simulation Output

C5 Port Encoder Simulation Output:

▶Step 1 ：Selecting “ by resolution ” ；

▶Step 2 ：Then enter correct pulse output resolution, unit: count/rev. Frequency multiplication and division output supported.

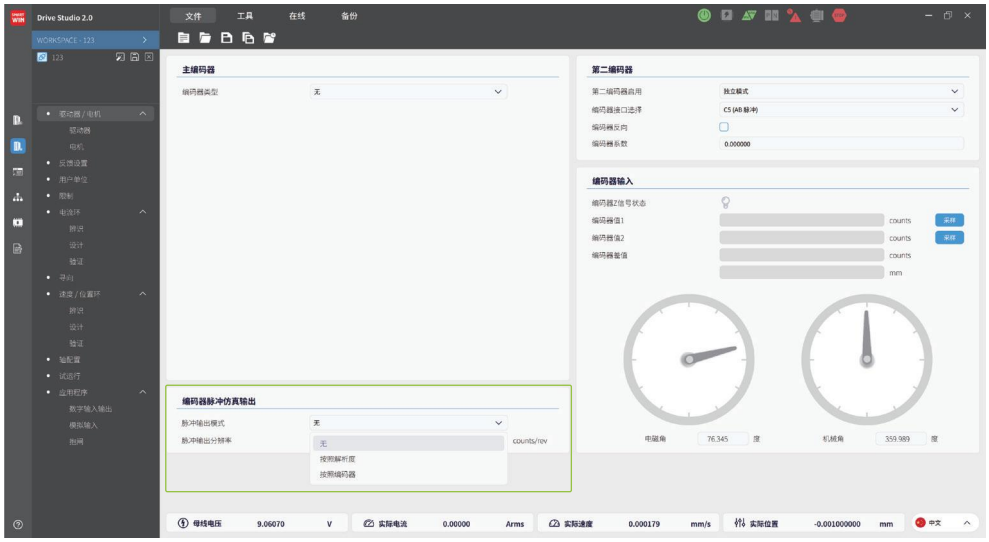


Figure 9-17 Encoder Pulse Simulation Output

9.2.5 Encoder Setting Confirmation

▶Step 1 ：Check if encoder index signal can be detected. Move motor to index position and observe Z-phase status change.

▶Step 2 ：Verify encoder parameter settings. Push or rotate the motor for fixed distance or angle, check position consistency. Deviations indicate incorrect parameters.



Figure 9-18 Encoder Setting Confirmation

9.3 User Unit Setting

9.3.1 Linear Motor User Unit Setting

- Step 1 ：Selecting “ position unit（mm.count.deg.rad） ”；
- Step 2 ：Selecting “ speed&acceleration unit（mm/s.count/s.ded/s.rad/s.rpm） ”；
- Step 3 ：Single-turn incremental setting, match external pulse count of upper PLC. As shown: 1000count=1mm.



Figure 9-19 Linear Motor User Unit Setting

9.3.2 Rotary Motor User Unit Setting

- Step 1 ：Selecting “ position unit（mm.count.deg.rad） ”；
- Step 2 ：Selecting “ speed&acceleration unit（mm/s.count/s.ded/s.rad/s.rpm） ”；
- Step 3 ：Setting deceleration rate；
- Step 4 ：Single-turn incremental setting, match external pulses per revolution of PLC. As shown: 1 rev = 32000count.

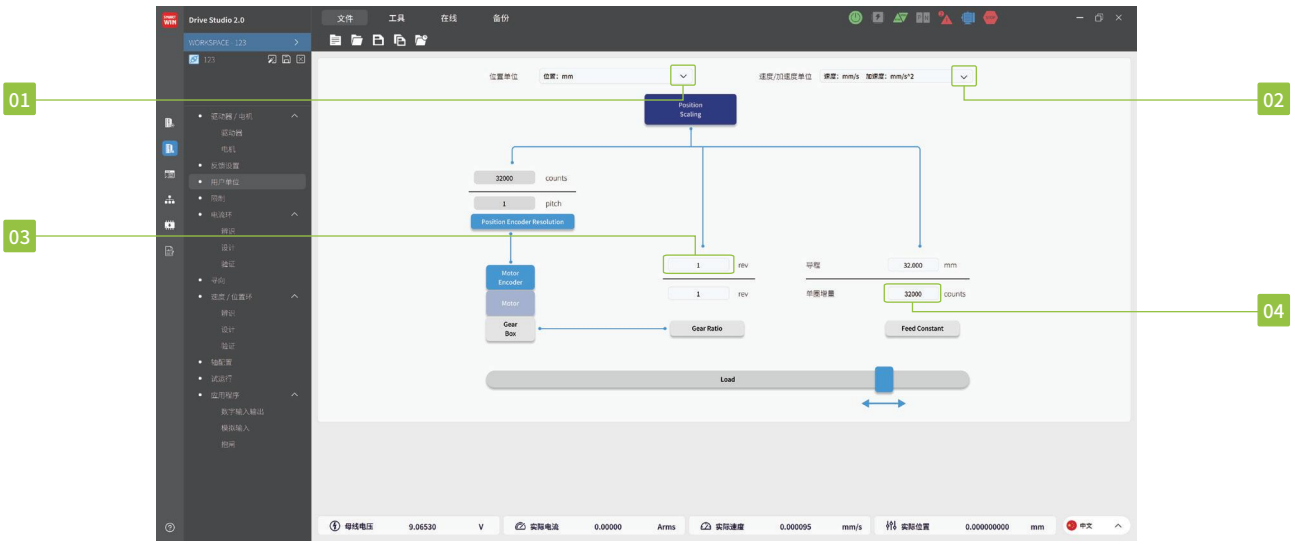


Figure 9-20 Rotary Motor User Unit Setting

9.4 Current Loop

9.4.1 Current Loop Self-learning

▶Step 1 ：Motor identification (enable soft enable in green box in advance)

▶Step 2 ：Click "Apply" to import self-learned resistance and inductance into PID model calculation.

▶Step 3 ：Click to create PID model.

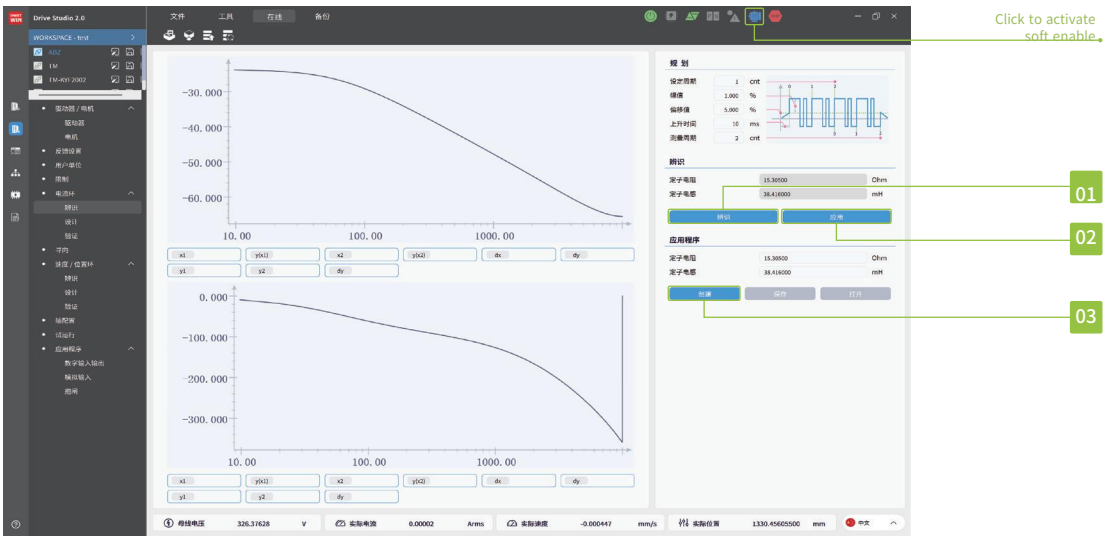


Figure 9-21 Current Loop Self-learning

9.4.2 Current Loop Module Design

▶Step 1 ：Click Design to automatically calculate current loop PID parameters;

▶Step 2 ：Enter the current loop model design page, bandwidth setting: Recommended 1200Hz, phase margin: 45Deg(Phase margin defines closed-loop stability. Excessive open-loop phase lag leads to critical stability. Optimal control systems adopt around 45 ° phase margin. It correlates with overshoot: higher margin reduces overshoot and enhances stability, while slowing down response speed.)

▶Step 3 ：Click Design to calculate proportional and integral constants of current loop, then click Next to enter current loop step response verification page.

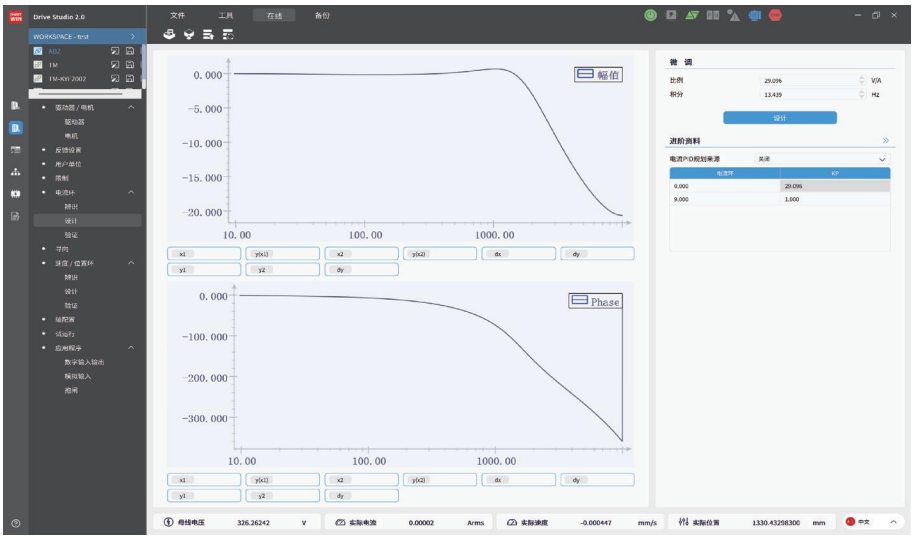


Figure 9-22 Current Loop Module Design

9.4.3 Current Loop Verification

▶Step 1 ：Current Loop Step Verification Page;

▶Step 2 ：Check automatic verification, select Phase A step.Set waiting time: 5ms, amplitude: 30% of rated motor current, offset: 10%, rise time: 0ms, measurement time: 5ms.

▶Step 3 ：Click Start to generate current loop response curve. Adjust KP and KI to obtain ideal waveform. Conduct test on phase A, B and C separately, then click Next to enter speed loop step page.

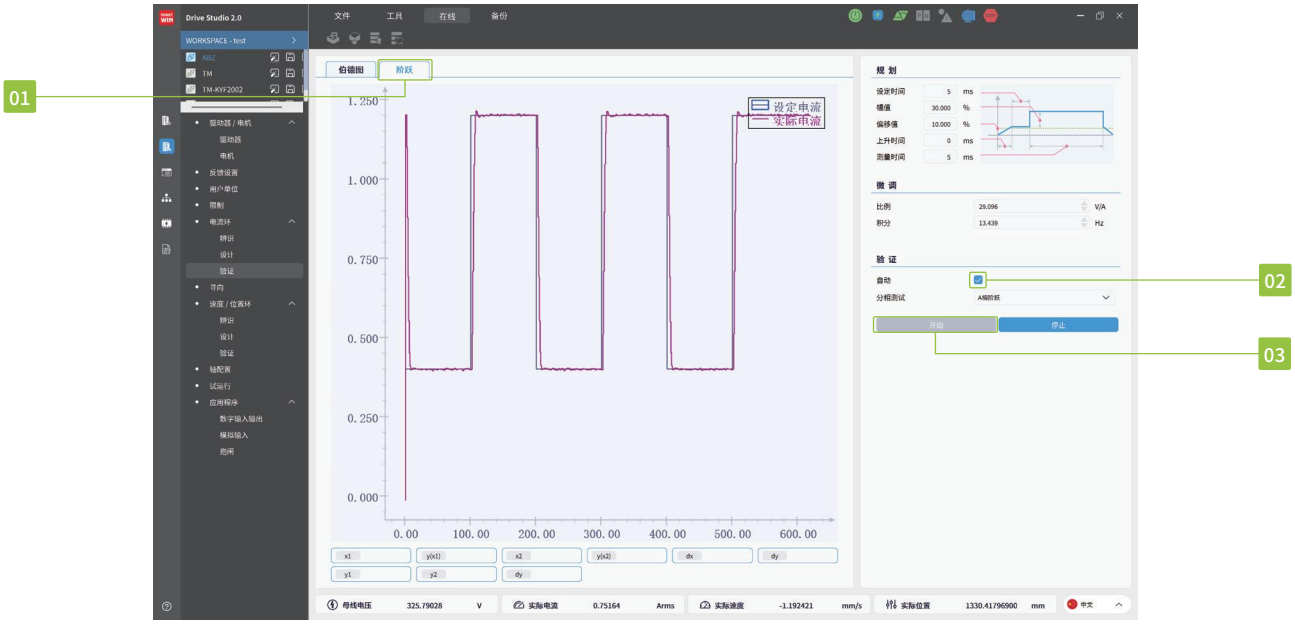


Figure 9-23 Current Loop Verification

9.5 Motor Phase Finding and Motion Direction Judgment

▶Step 1 ：Select encoder phase alignment mode: Absolute: Single-turn and multi-turn absolute encoders; Hall: ABZ/sincos incremental encoders with Hall function enabled; Incremental: ABZ/sincos incremental encoders without Hall function; Direction recognition (short stroke): Match current and encoder signals in limited travel, switch to incremental mode after recognition; Angle recognition (short stroke): Output initial current to avoid vertical axis falling during homing

▶Step 2 ：Set phase search current ratio, default 80% of rated motor current. Adjust value according to load, higher load requires larger current.

▶Step 3 ：Set travel distance and speed for phase search, verify pole pitch, pole pairs and encoder parame -ters. Reduce speed for large inertia load.

▶Step 4 ：Start phase search. Ensure no mechanical interference and sufficient moving range. Activate soft enable first, system will prompt completion or failure status.

▶Step 5 ：Set speed and time for motor direction judgment. Running range: 60mm = 20mm/s × 3s

▶Step 6 ：Initiate direction detection. Confirm actual running direction and select forward or reverse accordingly.

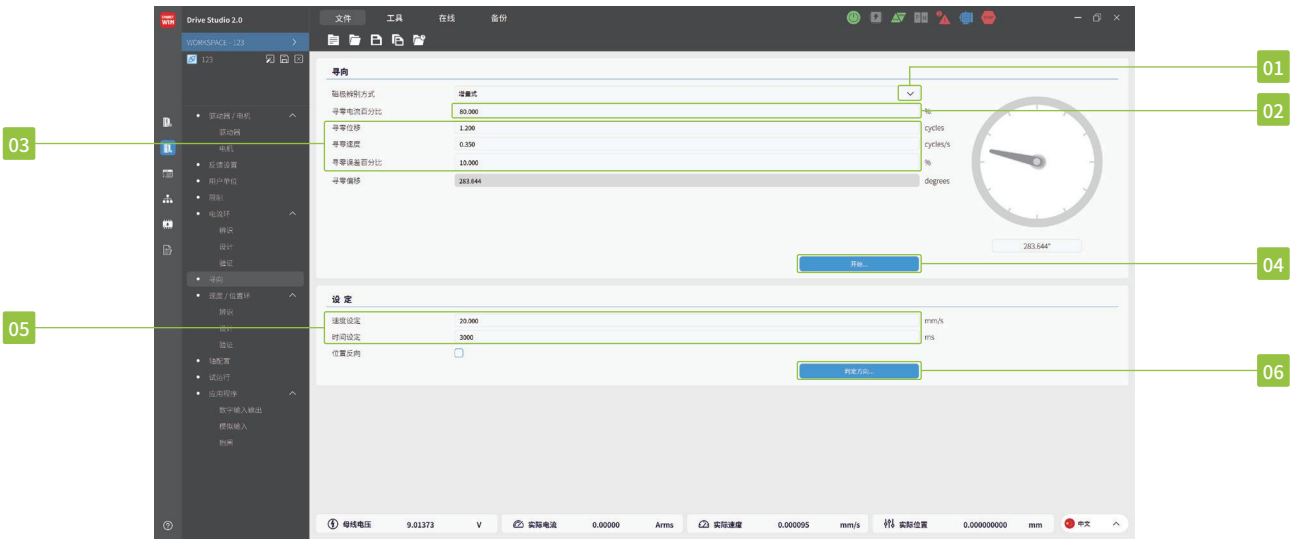
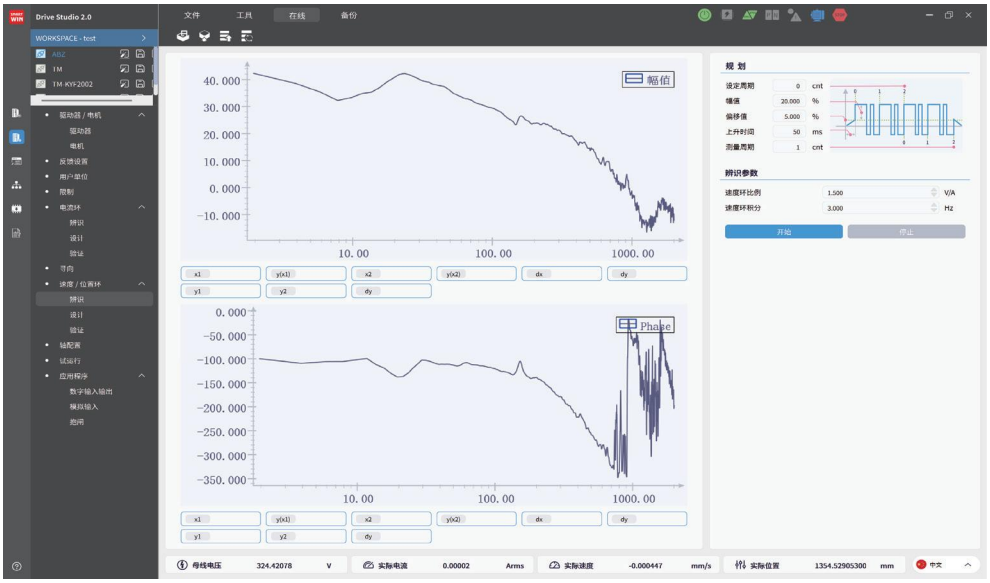


Figure 9-24 Motor Phase Finding and Motion Direction Judgment

9.6 Speed Loop & Position Loop

9.6.1 Speed Loop Frequency Sweep



Figue 9-25 Speed Loop Frequency Sweep

9.6.2 Speed Loop Design

►Step 1: Speed loop PID. Set PID parameters for speed loop and position loop. Refer to the PID tuning instructions below.

►Step 2: Filters: 4 groups with multiple types available. Please refer to the filter application instructions for settings.

►Step 3: Multi-segment PID configuration, designed for special application requirements. Refer to the PID tuning instructions below for setup.

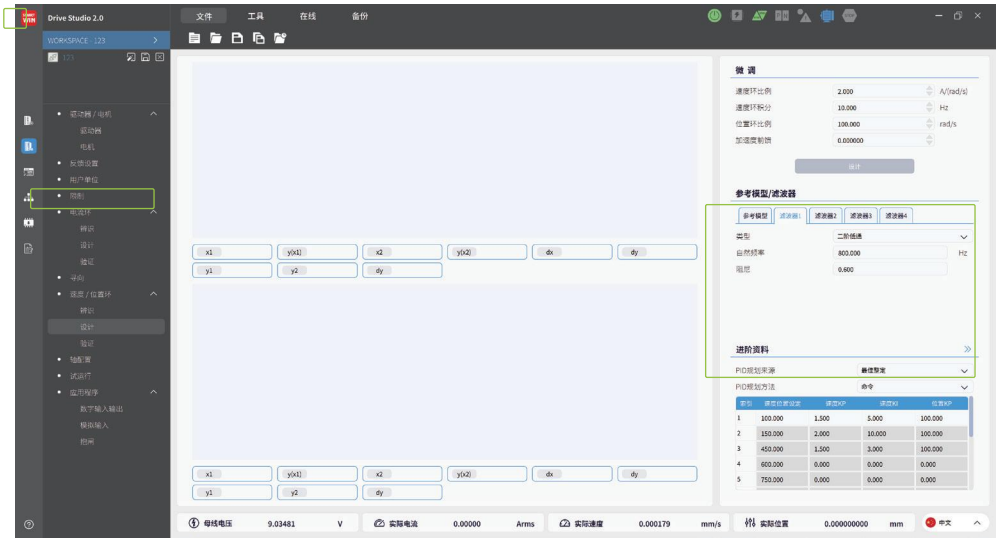


Figure 9-26 Speed Loop Design

9.6.3 Speed Loop Verification

3 Step 1 ：Single-clicking "Motion Step", enter the speed loop step verification page;

3 Step 2 ：Set settling time: 80 ms, amplitude 20% of motor rated current (target current), offset 0%, rise time 50 ms, measurement time 100 ms.

3 Step 3 ：Click Start to generate speed loop step response. (Note: Ensure sufficient moving space for motor rotor/mover.) Adjust speed loop KP and KI to obtain optimal curve, then click Next to enter axis configuration page.

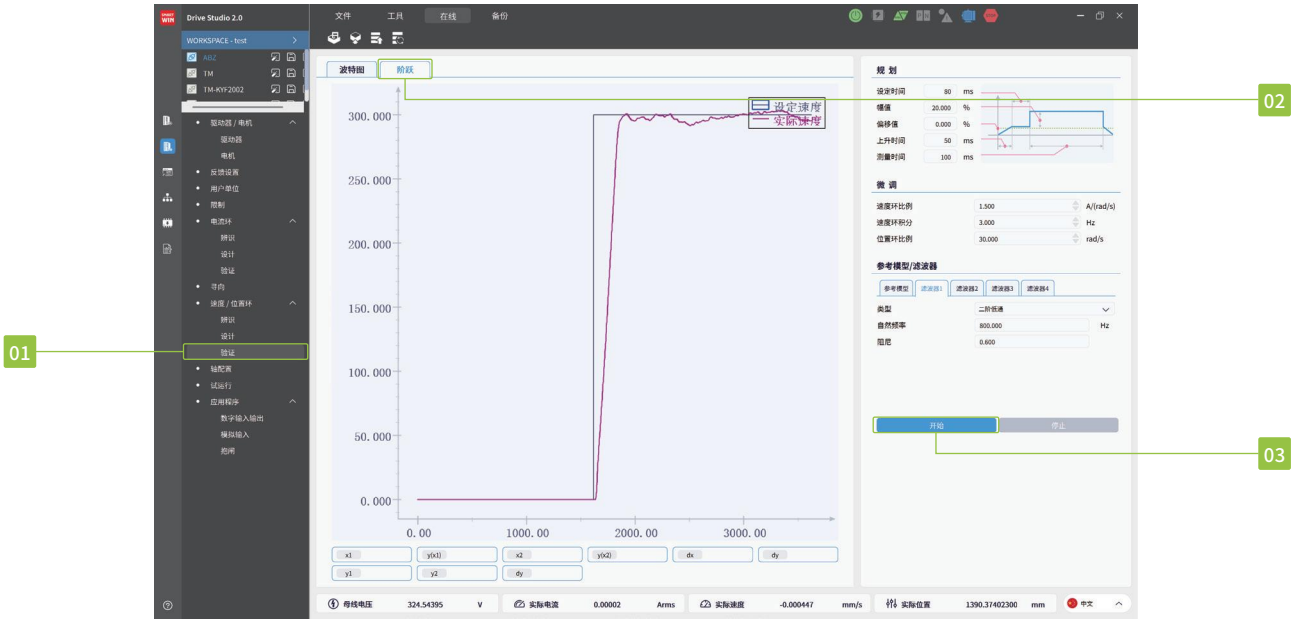


Figure 9-27 Speed Loop Verification

9.7 Axis Configuration Settings

Axis Configuration Settings：

3 Step 1 ：Servo drive control mode, supporting analog/Ethernet/EtherCAT/pulse;

3 Step 2 ：Check Soft Enable on Power-up to automatically activate soft enable after drive power-on and restart;

3 Step 3 ：Set command reception cycle, EtherCAT bus scan cycle and master-slave synchronization parameters;

3 Step 4 ：Set axis type and control mode. (Refer to gantry configuration instructions for gantry applications.)

3 Step 5 ：Set speed control related parameters: acceleration feedforward, positive/negative acceleration feedforward coefficients, speed acceleration/deceleration, speed deviation threshold and duration;

3 Step 6 ：Position control mode, position loop velocity feedforward coefficient, speed deviation threshold and time, in-position range and time, position control interpolation smoothing time.

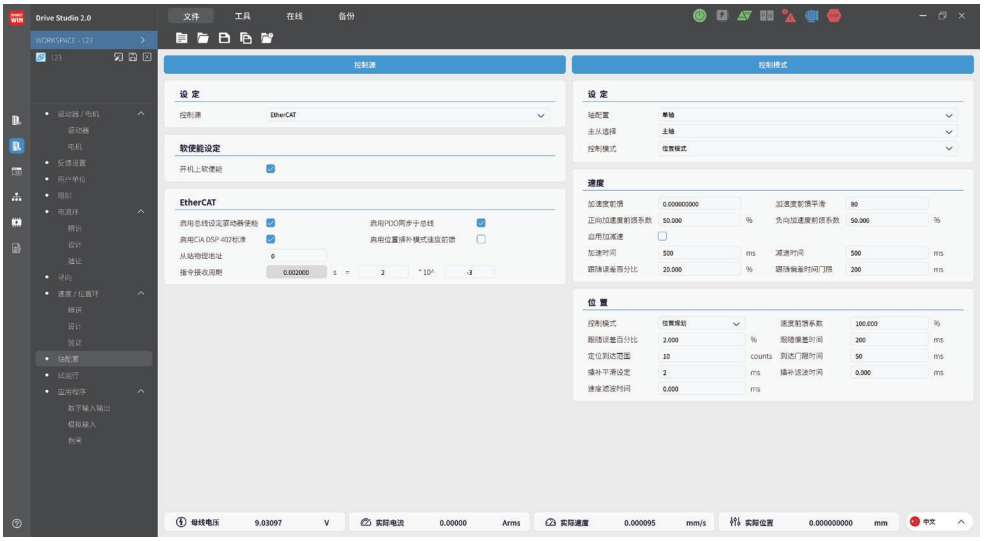


Figure 9-28 Axis Configuration Settings

9.8 Trial Operation Commissioning

3 Step 1 ：Enable trial run to obtain control authority for PC operation;

3 Step 2 ：Select control mode: Position, Speed or Torque;

3 Step 3 ：Set motion parameters including positioning mode, travel distance, speed, acceleration/ deceleration and dwell time;

3 Step 4 ：Click Enable. (Check the alarm icon above and clear all alarms first if any exist.)

3 Step 5 ：Motion mode, JOG mode, single-operation, multi-operation;

3 Step 6 ：Select JOG mode, then click Forward or Reverse to move the motor mover appropriately. If the travel range is sufficient for the preset distance, switch to Single-cycle or Reciprocating mode and operate accordingly. Ensure enough travel stroke in the corresponding direction before operation.

3 Step 7 ：Tune motion parameters. Optimize performance by adjusting parameters of current loop, speed loop and position loop in advanced settings.；

3 Step 8 ：Enable curve tracking function. There are six standard curves for trial operation. Adjust rigidity parameters in advanced settings based on actual operation curves.

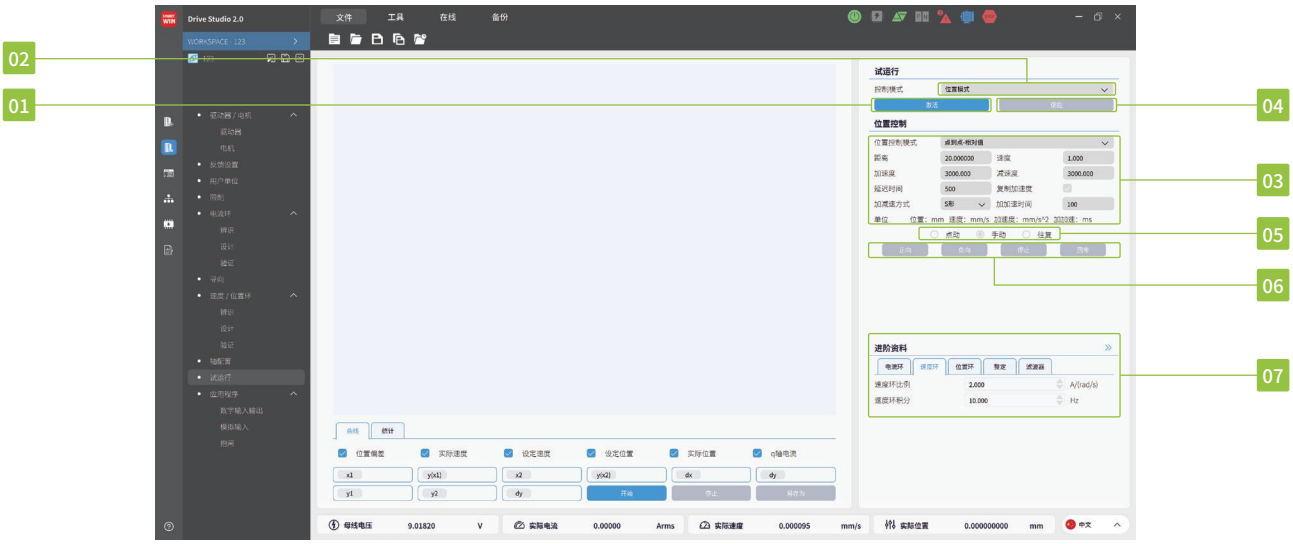


Figure 9-29 Trail Operation Commissioning

9.9 Application Program

9.9.1 Digital IO Configuration Digital

INPUT configuration:

3 Step 1 ：Set External Signal Filter: normally set to 0. For long cables prone to interference, set the filter time to 50 – 100 ms accordingly.

3 Step 2 ：Set External Signal Polarity for logic inversion. Select low or high level inversion per actual application requirements.

3 Step 3 ：Configure Function Definition, map corresponding function commands according to actual application requirements.

3 Step 4 ：Check the rear status indicators to verify whether the input wiring is normal.

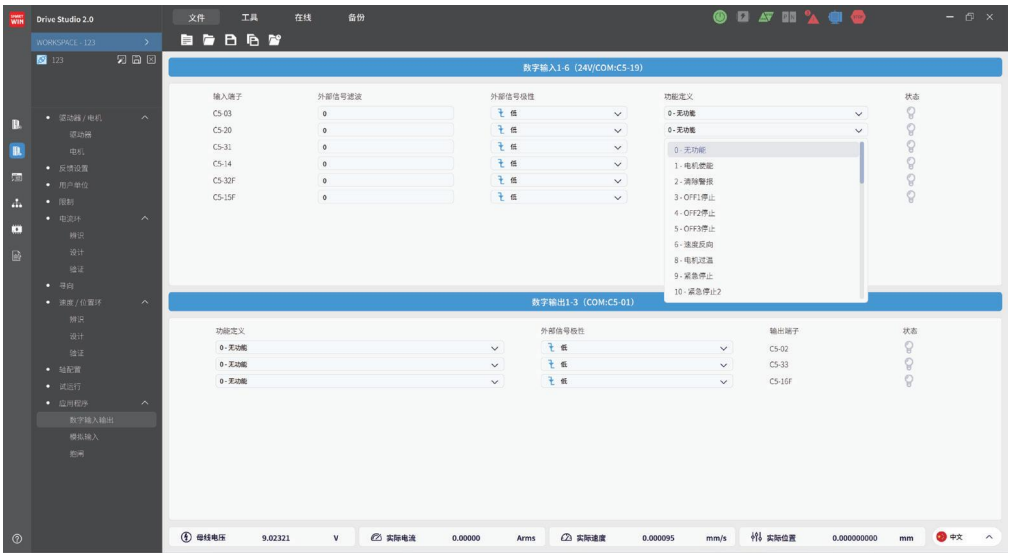


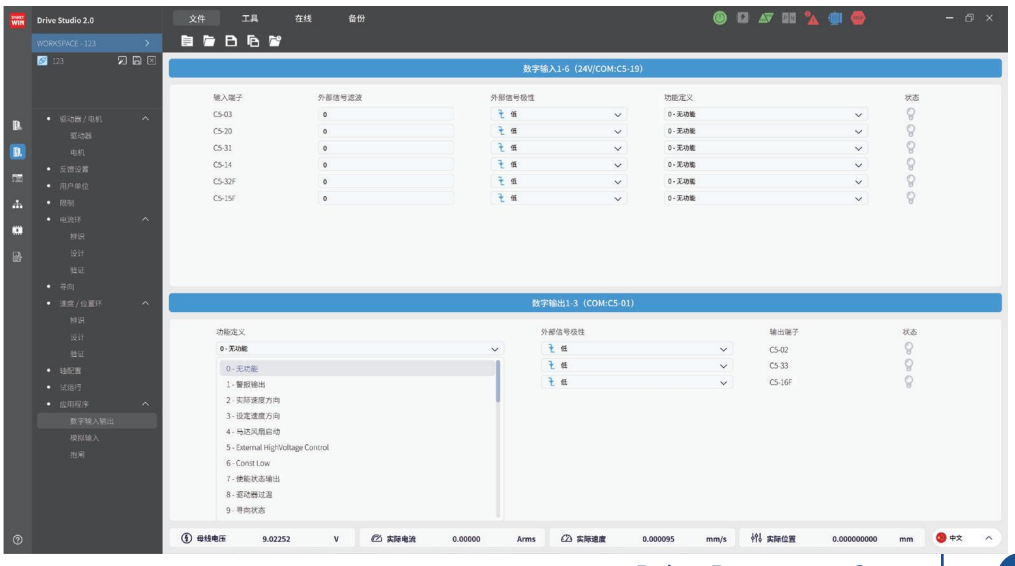
Figure 9-30 Digital INPUT Configuration

Digital OUTPUT configuration:

3 Step 1 ：Configure Function Definition to map relevant function commands as required by practical applications.

3 Step 2 ：Set External Signal Polarity to invert logic level. Select low or high level inversion based on on-site requirements.

3 Step 3 ：Check the rear status indicators to verify output status.



9.9.2 Analog IO Configuration

3 Step 1 ：Set the offset value of analog channel as required;

3 Step 2 ：Set analog channel scaling according to application requirements.；

3 Step 3 ：Set the filter frequency for analog channel input voltage based on the operating environment；

3 Step 4 ：Configure analog channel function commands according to actual application requirements.

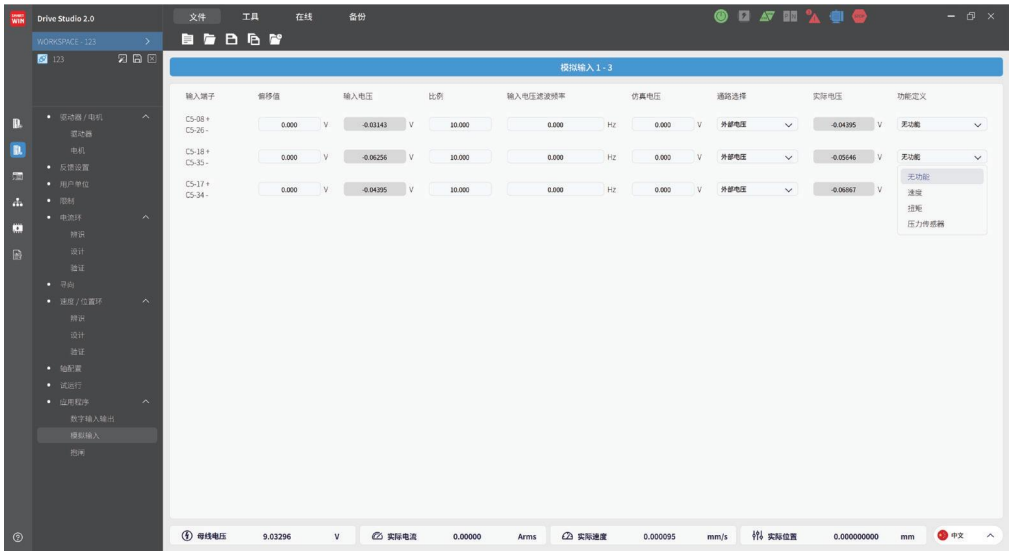


Figure 9-32 Analog IO Configuration

9.9.3 Motor Brake Output

3 Step 1 ：Enable dynamic braking if needed. It can be disabled for medium-voltage drives with native hardware support, while high-voltage drives may enable it per application conditions.

3 Step 2 ：Enable motor brake function as required. This function is used for motors equipped with mechanical brake pads (e.g., vertical axes) to realize brake control via the drive.,

3 Step 3 ：Configure brake output parameters, including brake output terminal, delay time for brake opening/closing, and pre-torque current for vertical axes.

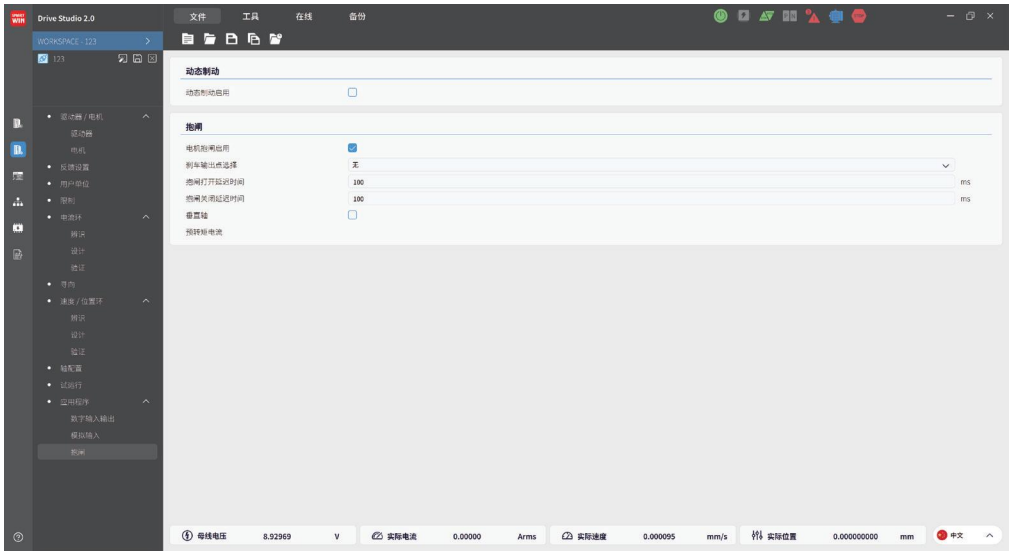
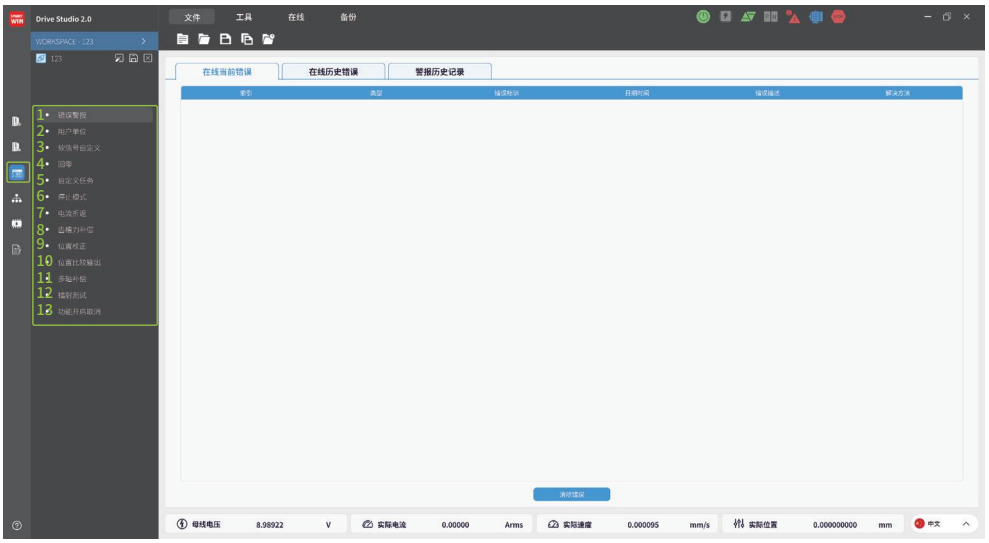


Figure 9-33 Motor Brake Output

10. Application Setting

10.1 Axis Application Setup Wizard



Figue 10-1 Axis Application Setup Wizard

1	错误警报	Error message	Alarm codes and statuses under connection
2	用户单位	User units	Unit Setting
3	软信号自定义	Soft signal definition	Absolute Encoder Soft Signal Custom Configuration
4	回零	Homing	Homing Setting
5	自定义任务	Motion task	Custom Task Settings
6	停止模式	Stop mode	Stop mode setting
7	电流折返	Current foldback	I2T Setting
8	齿槽力补偿	Cogging Compensation	Cogging compensation setting
9	位置校正	Position calibration	Position compensation setting
10	位置比较输出	Position compare output	Fly shooting setting
11	多轴补偿	Multiple axes compensation	Multi-axis compensation setting
12	镭射测试	Laser test	Laser testing setting
13	功能开启取消	Function choice	Function enable/disable setting

10.2 Application Instrucment

10.2.1 Error Warning

Current Alarms: Connect the PC to the drive to view all active alarms, including alarm type, occurrence time, description and troubleshooting measures.Historical Alarms: The drive records all past alarms before alarm reset or device restart, enabling effective analysis of historical faults.

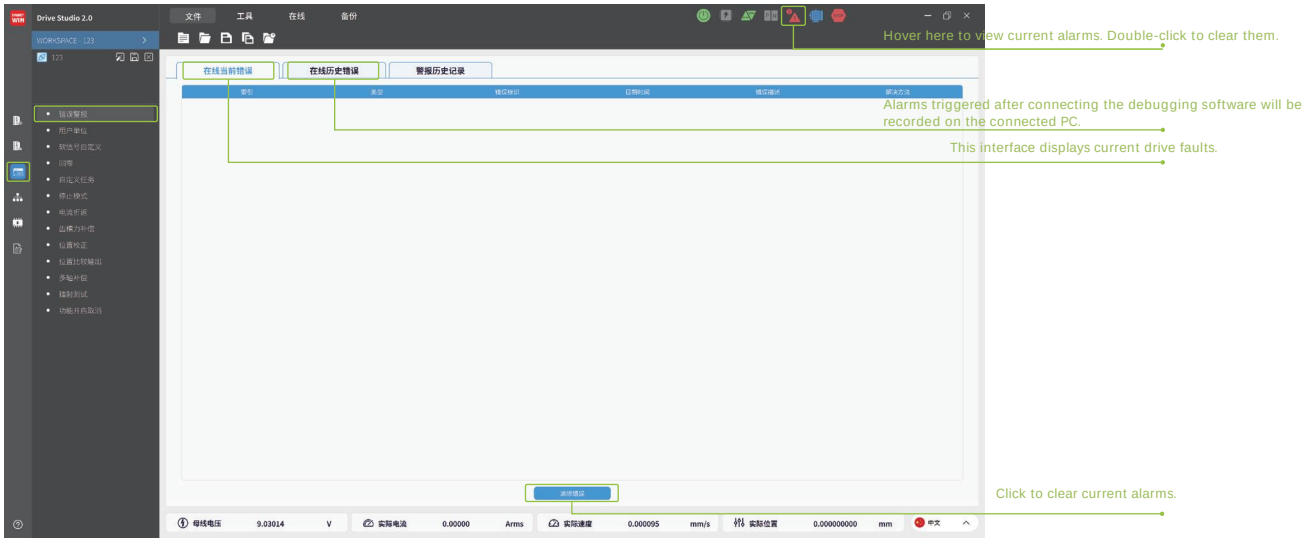


Figure 10-2 Error Warning

10.2.2 User Unit

This section uses the same user units as the drive setting interface.

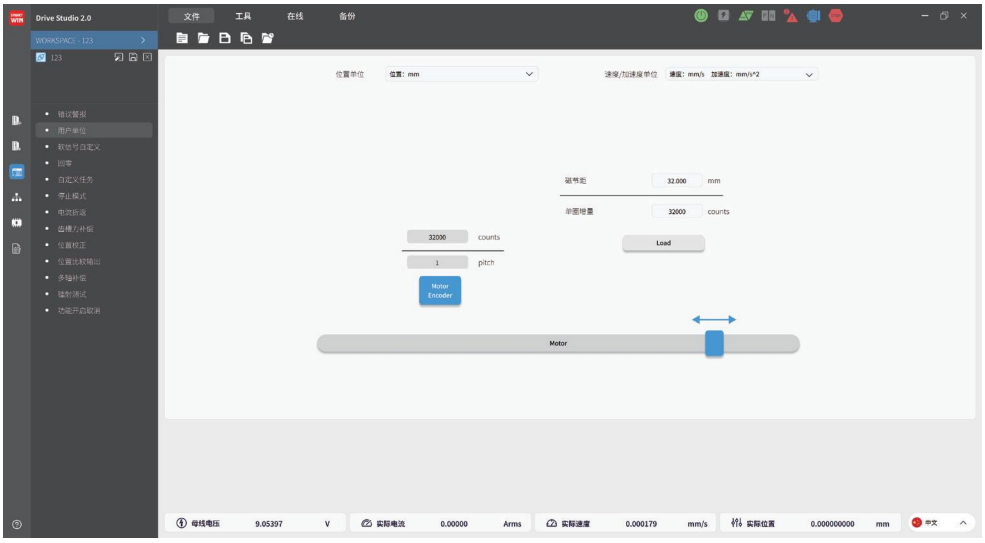


Figure 10-3 User Unit

10.2.3 Soft Signal Customization

Applicable to absolute encoders. The absolute position of the encoder can be set as soft positive/ negative limits and home switch to realize physical limit functions and reduce wiring. When enabled, the upper computer can normally read the corresponding limit status via PDO mapping.An absolute encoder generally requires no homing. Move the slider to the desired positive/negative limit position, then click to set the current position as the positive or negative limit.

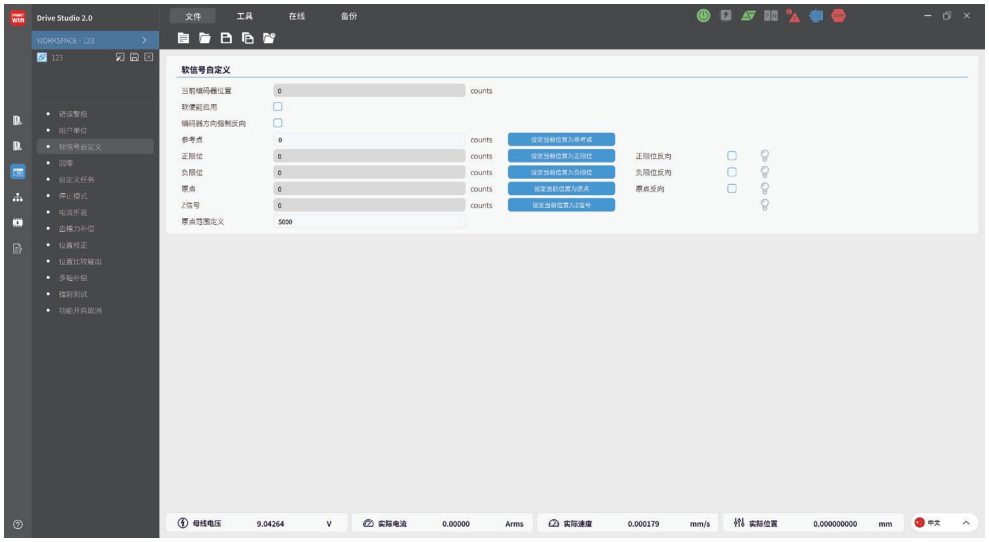


Figure 10-4 Soft Signal Customization

10.2.4 Homing

1. Homing Setup

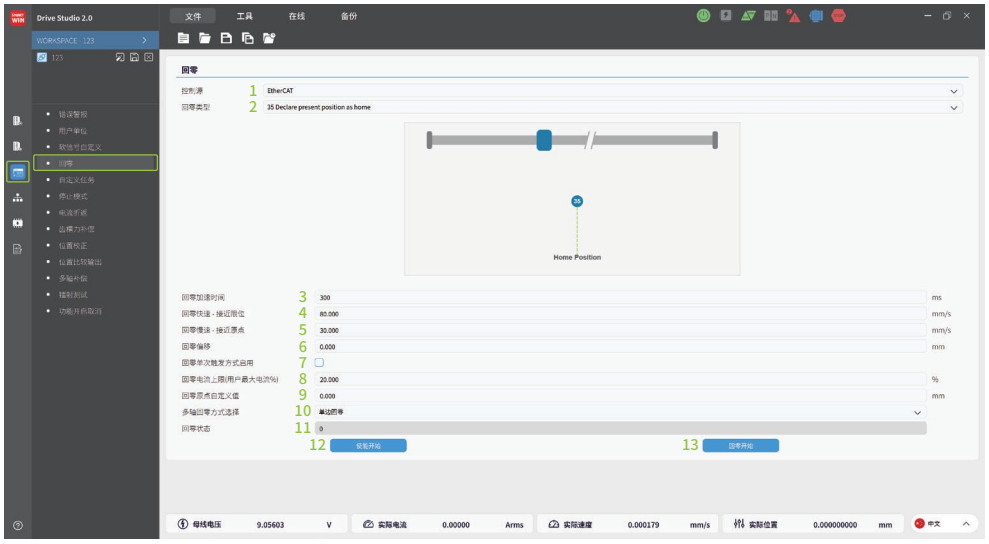


Figure 10-5 Homing Setup

No	Project	Instrucment
1	Control source	Select the required control source via dropdown.
2	Homing type	Select the required homing mode from the dropdown (meet EtherCAT CiA402 standard)
3	Homing acceleration time	Jerk time for acceleration and deceleration during homing
4	Homing fast-approach limit	Speeds used during homing with limit switch, home switch and hard limit
5	Homing slowly-approach origin	Speed for homing trigger search (index pulse, limit switch, home switch or other sources defined by HOMETYPE)
6	Homing offset	Set home offset (counts)
7	Enable single-shot home trigger	Supplement
8	Home current limit	Set homing current limit (percentage of user-defined maximum current)
9	Home origin custom value	Modify homing position into target position
10	Multi-axis home method	Divided into single-side homing and dual-side homing; select for gantry applications.
11	Homing state	Display homing status: 0 = Not homed, 1 = Homing completed
12	Enable start	Click to enable the motor
13	Home start	The motor starts homing. Stop the homing action once completed.

2. Common Homing Methods for Drive via Host Computer

2.1 Controller PDO Mode Homing Process

Detailed step are as follows:

- ▶Step 1 ：Setting homing mode and homing parameter;
- ▶Step 2 ：Setting drive object dictionary 0x6060=6 (homing mode);
- ▶Step 3 ：Setting bit 4 of drive object dictionary 0x6040 to 1 to start homing；
- ▶Step 4 ：Check 0x6041 Bit10=1 & Bit12=1;
- ▶Step 5 ：Setting drive object dictionary 0x6040 bit 4=0;
- ▶Step 6 ：Setting drive object dictionary 0x607A=0x6064；
- ▶Step 7 ：Setting drive object dictionary 0x6060=8 (Cyclic Synchronous Position Mode)

2.2 Controller SOD Mode Homing Process

Detailed step are as follows:

- ▶Step 1 ：Setting homing mode and homing parameter;
- ▶Step 2 ：Setting drive object dictionary SDO 0x6060=1;
- ▶Step 3 ：Setting drive object dictionary SDO 0x29AE=1024；
- ▶Step 4 ：Waiting 3-5s;
- ▶Step 5 ：Setting drive object dictionary SDO 0x29AE=0;
- ▶Step 6 : Setting drive object dictionary0x607A=0x6064；
- ▶Step 7 ：Setting drive object dictionary 0x6060=8 (Cyclic Synchronous Position Mode).

10.2.5 Custom Task

1. Detailed Instrucment

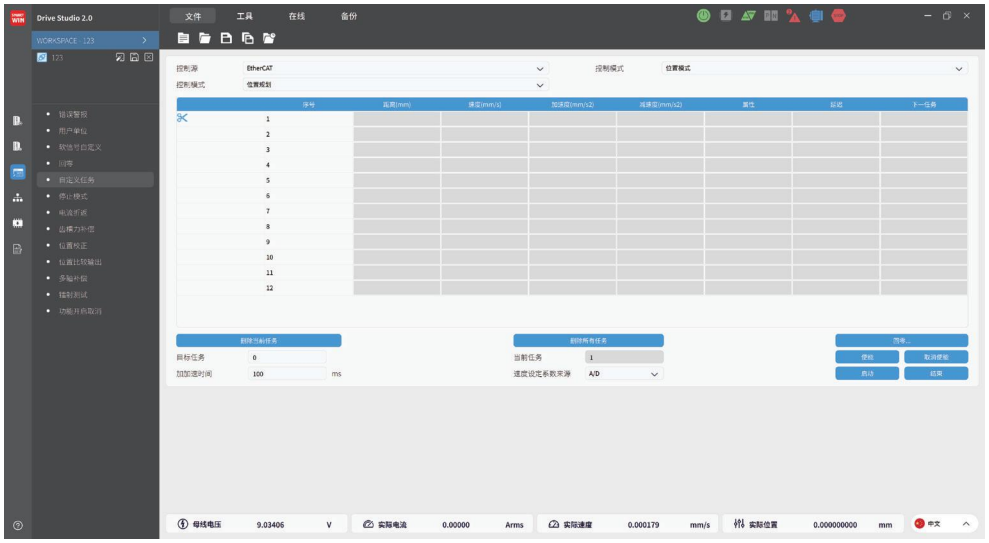


Figure 10-6 Custom Task

No	Project	Instrucment
1	Control Source	Select the desired control source (typically Analog)
2	Control Mode	Control mode: Current mode
3	Control Mode	Position Profile - Drive Internal Positioning Mode
4	No.	Task No.
5	Distance (mm)	Setting current task operating distance, unit: mm
6	Speed (mm/s)	Setting current task operating speed, unit: mm/s
7	Acceleration (mm/s²)	Setting current task operating acceleration, unit: mm/s²
8	Deceleration (mm/s²)	Setting current task operating deceleration, unit: mm/s²
9	Attribute	Operation profile mode of current task
10	Delaying	Execute the next task with delay after current task completes
11	The Next Task	Task number of the next task after current task completion
12	Target Task	PC control assigned task
13	Acceleration Time	Position Profile Jerk-
14	Delete Current Task	Deleting Current Selected Task
15	Delete All Tasks	Deleting all edited task
16	Current Task	Current operating task No.
17	Speed setting coefficient	Set maximum speed percentage for AD channel
18	Homing	Drive operating homing (meeting EtherCAT402 standard)
19	Enable/Disable	Motor Enable/Disable
20	Start/End	Task Start/End

2. Task Setting

Open the setting window for the selected task number (double-click the target task number in the task table).

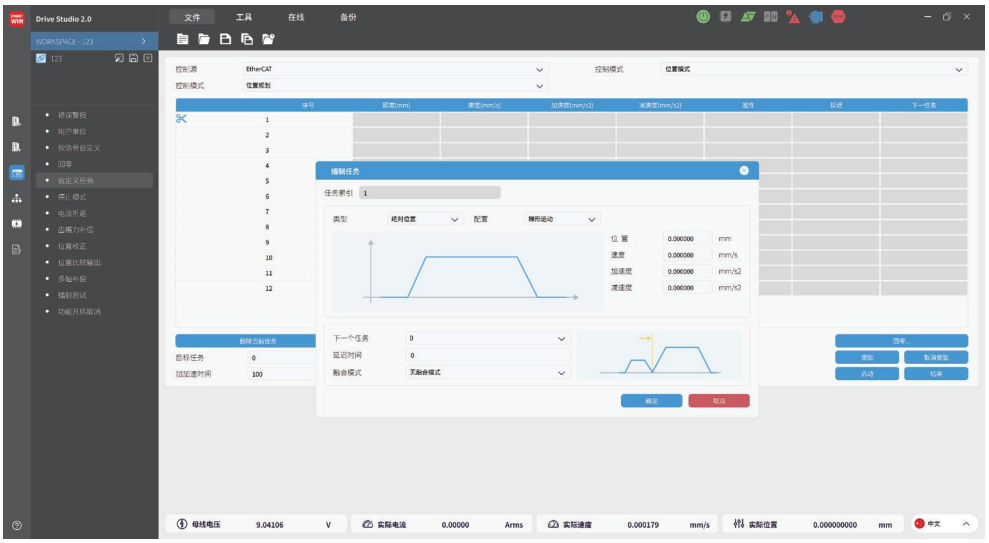


Figure 10-7 Task Setting

3. Task Execution

Configure task bits 0~3. These four bits determine the target task to be executed. Activate the task to trigger motor motion, and set the speed percentage for analog channels.

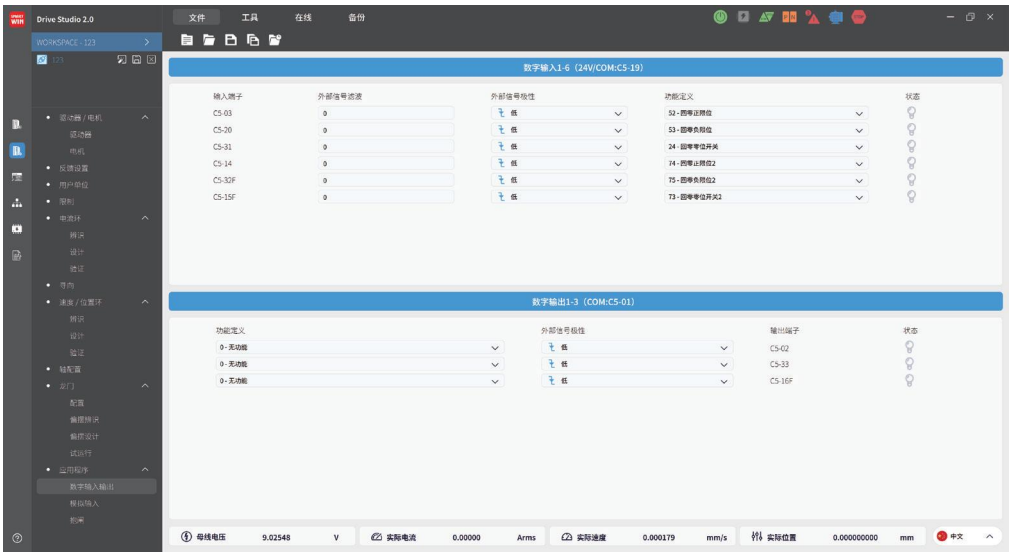


Figure 10-8 Task Execution

10.2.6 Stopping Mode

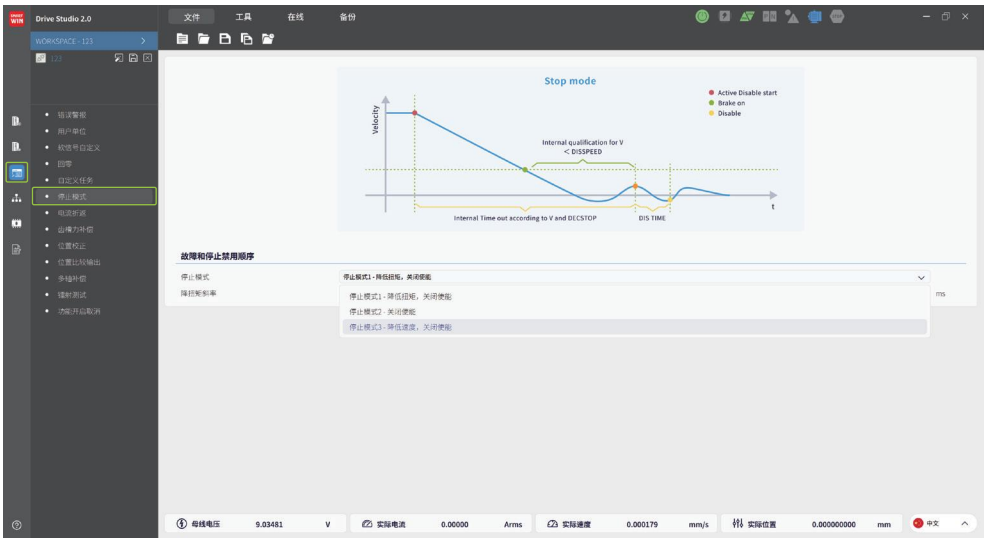


Figure 10-9 Stopping Mode

No	Project	Instructment
1	Stop Mode 1: Reduce torque, disable drive	If the drive generates an alarm or fault, reduce torque to zero as configured before disabling the drive.
2	Stop Mode 2: disable drive	When the drive triggers an alarm or fault, disable the drive. (Default mode)
3	Stop Mode 3: Reduce speed, disable drive	When a drive alarm or fault occurs, ramp speed down to zero as configured, then disable the drive.

10.2.7 Current Foldback

Current foldback is a safety mechanism of AKS, which protects the drive and motor from overheating caused by continuous excessive current. You can set the foldback current values for the drive and motor respectively.

When a motor is selected from the motor library in Drive Studio, the preset foldback current parameters will be downloaded to the drive. These parameters are defined by the manufacturer and cannot be modified by users. Please contact technical support if parameter adjustment is required.

Use the Current Foldback task screen in Drive Studio to view and configure the foldback current parameters for motor and drive.

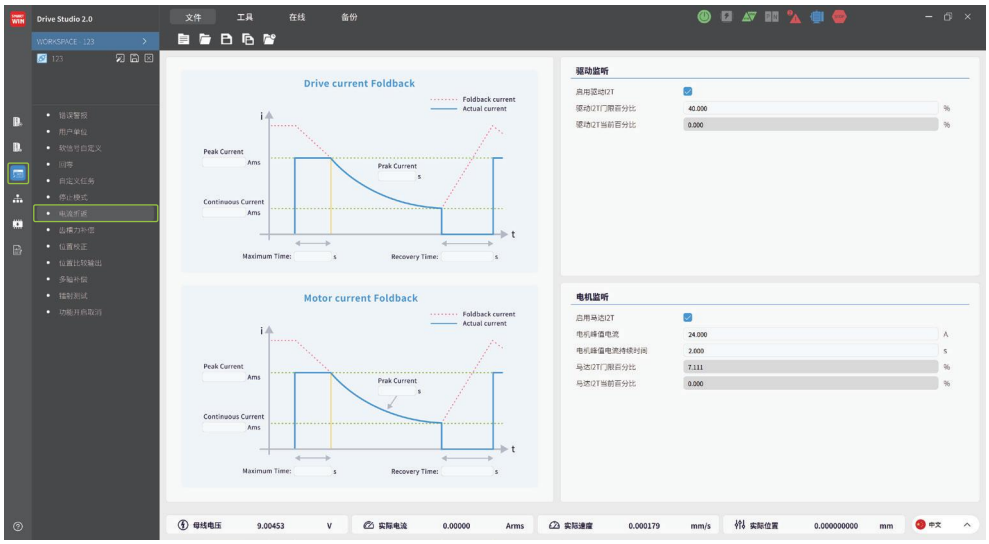


Figure 10-10 Current Foldback

10.2.8 Cogging Force Compensation

1. Cogging Force Compensation Introduction

Enable cogging force compensation for applications requiring low speed fluctuation and motors with prominent cogging force to meet operational demands. This function is unnecessary for general applications.

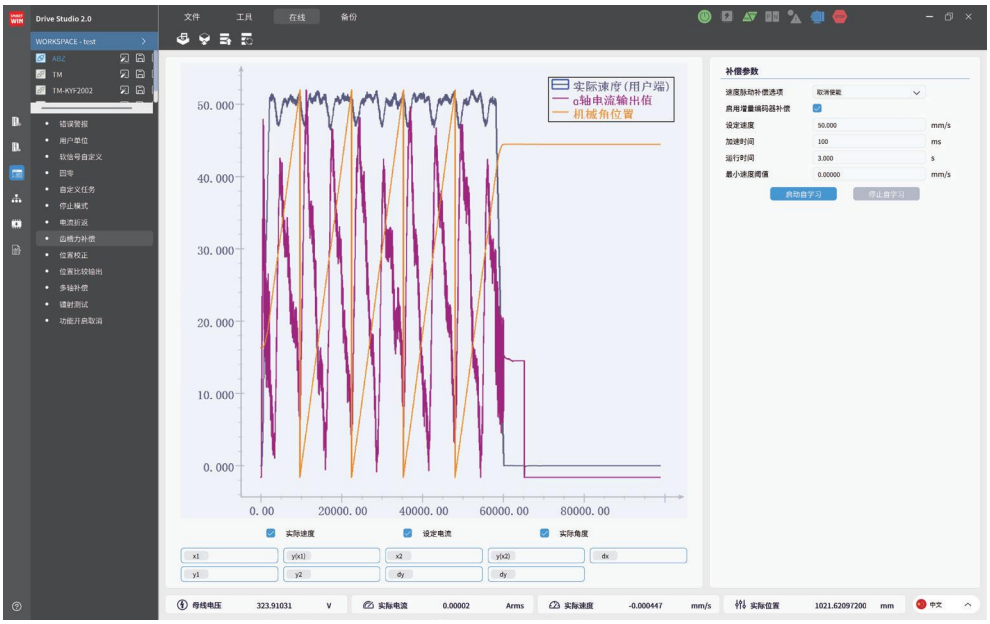


Figure 10-11 Cogging Force Compensation Screen

2. Auto-Tuning Procedures：

►Step 1: First disable the Speed Ripple Compensation option and select Disable Enable;

►Step 2: For incremental encoders such as Incremental and Sin/Cos types, check Enable Incremental Encoder Compensation. Leave this option unchecked for single-turn or multi-turn absolute encoders;

►Step 3: Speed, acceleration and running time can use default values:

1、Set speed: Low speed is recommended. For linear motors, 50 mm/s is suggested; for rotary motors, 50 rpm is recommended.

2、Set running time: Verify that the mechanical travel is sufficient for the set speed within the configured time. This is critical. If travel is insufficient, reduce the speed or shorten the running time.Successful auto-tuning requires the rotary motor to complete at least one full rotation, and the linear motor to travel at least one magnetic pole pitch during the process.

►Step 4: After completing the above settings, click Start Auto-Tuning. Click Stop Auto-Tuning anytime to abort the process. Auto-tuning will stop automatically once fully completed.

3.Enable Cogging Compensation：

After completing the auto-tuning steps 2.1 to 2.4, enable the cogging force compensation function.



Figure 10-12 Enable Cogging Force Compensation

■You may select Set Speed or Actual Speed. Set Speed is recommended for better repeatability.

■ If Actual Speed is selected, it is advised to enable Minimum Speed Threshold as well. When the motor is stationary, actual speed may fluctuate (e.g. ±0.3 rpm). Set the threshold value to 0.3 to suspend compensation during speed fluctuation and prevent compensation errors.

4. After completing the above operations, you can compare the performance before and after cogging force compensation. Examples are shown below.：

The curve before compensation is shown below. With a set speed of 50 mm/s, the actual speed fluctuation reaches ±2.4 mm/s.

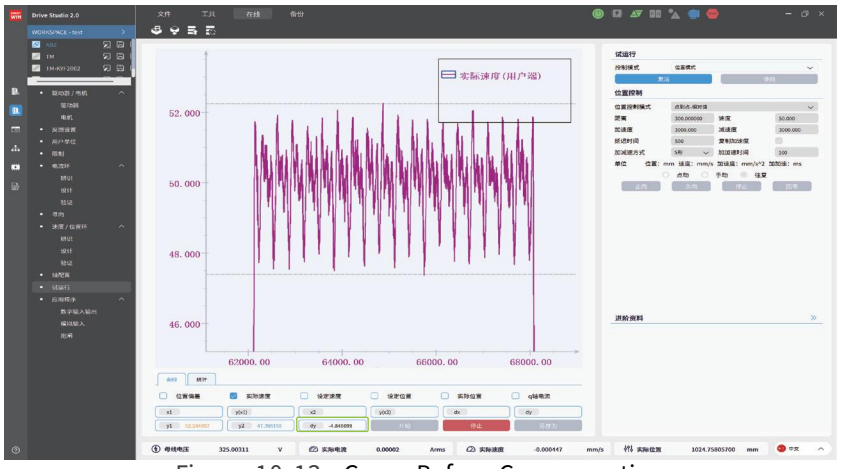


Figure 10-13 Curve Before Compensation

The curve after compensation is shown below. At a set speed of 50 mm/s, the actual speed fluctuation is reduced to less than ±1.2 mm/s.

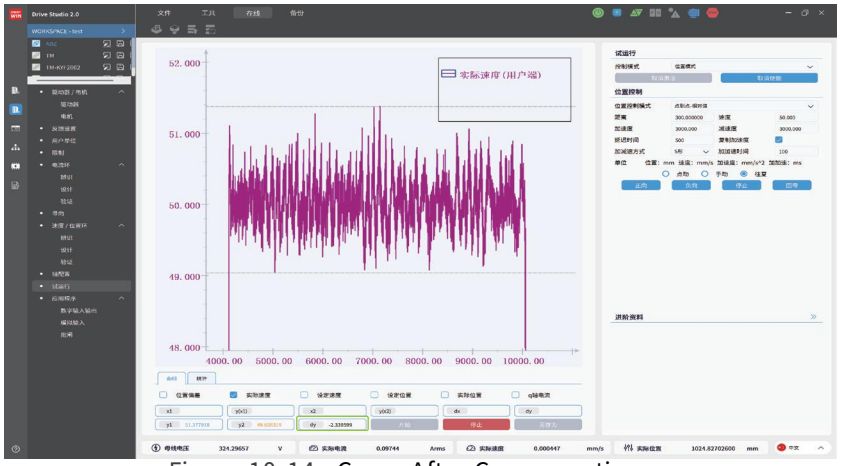


Figure 10-14 Curve After Compensation

10.2.9 Position Correction

1. Position Compensation Introduction

The AKS drive supports position compensation function to effectively improve motor positioning accuracy.Generate an error map using external measuring devices such as laser interferometers, and store the map in the drive ROM. The drive retrieves correction values in real time according to the actual position and performs on-the-fly dynamic correction.

2. Position Compensation Sample Table

In this example, the load position on the linear axis is measured by a laser interferometer. The total travel of the axis is 1100 mm. The drive software issues motion commands every 110 mm, moving the motor across 10 position intervals. As the motor drives the load, the laser system rec -ords the actual travel distance. The absolute position at each measuring point is compared with the position from the motor encoder, and the difference between the two values is the position compensation value.

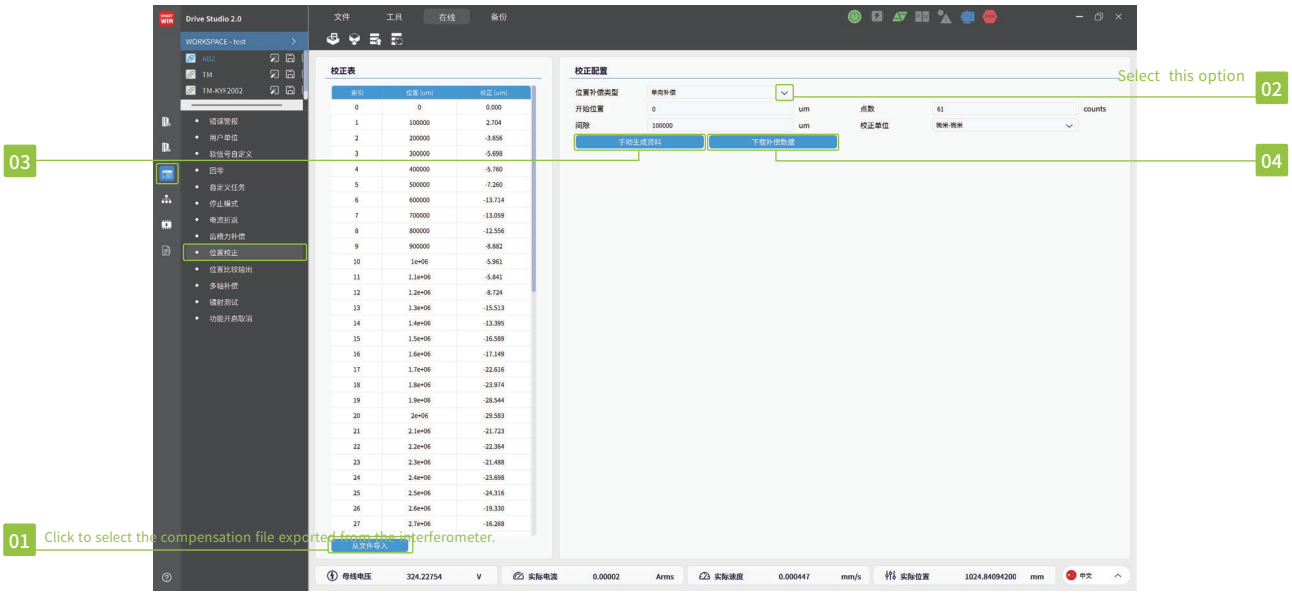


Figure 10-15 Position Compensation

3. Position Compensation Function

The position compensation function of AKS is applicable to various motor system types:

- Direct drive linear platform;
- Direct drive rotary platform;
- Linear stage driven by servo drive (rotary motor, linear error) ;
- Rotary stage driven by servo motor (motor and load rotate). The position compensation function works with all absolute and incremental encoders supported by AKS.

Notes: Home operation is required when using position compensation in incremental systems.

Notes: For dual control loop systems, the AKS position compensation corrects readings from the secondary (load) encoder. For single control loop systems, it corrects readings from the primary (motor) encoder.

10.2.10 Flying Shot

1. Introduction to PCO (Position Compare Output):

PCO (Position Compare Output) rapidly outputs digital I/O signals when the motor passes predefined position points during movement. It triggers external devices such as cameras, loading/unloading equipment, robots and measuring instruments to start subsequent processes.

2. Function Enable Instructions：

2.1 Host computer homing, compare and output based on relative position.

■After the host computer completes homing, enable the position compare output function of the drive in real time at the current home position. (No drive homing is required; the drive will record the position offset internally.)

■Set drive object dictionary 0x6060 = 8 (Profile Position Synchronous Mode).

■Perform homing via the host computer.

■After host computer homing, enable the drive position compare output in real time at the current position (home position) by setting bit12 = 1 in drive object dictionary 0x6040. Note: Bit12 must remain set to 1 at all times while the position compare output function is in use.

2.2 Drive Homing (Refer to the above homing procedure for drive homing.) Compare and output based on absolute position.

■The host computer performs drive homing. (In some applications, even after host computer homing, the drive does not complete homing internally, so the position compare output function cannot be activated. Therefore, the drive must complete homing.)

■Configure the position compare output parameters via CoE SDO and write them into the relevant object dictionary of the drive. Configuration parameters for the position table:

►Step 1: Set drive object dictionary 0x2F6F = 0 to disable the position compare output function.

►Step 2: Write position points 1 – 16 to drive object dictionary 0x2F73 ~ 0x2F82.

►Step 3: Set the direction for each compare point by writing to drive object dictionary 0x2F71. Every 2 bits represent the direction of one position point:

Bit 0 – 1: Position point 1 direction (0 = Negative, 1 = Positive, 2 = Both directions)

Bit 2 – 3: Position point 2 direction

Bit 4 – 5: Position point 3 direction

►Step 4: Set position table units via 0x2F6E (0 = count, 1 = mm, 2 = deg, 3 = rad).

►Step 5: Set pulse output hold time via 0x2F70 (unit: 0.1 μs).

►Step 6: Write 1 to 0x2F6F to enable the position compare output function.

Prerequisite: Drive homing must be completed.

Position Compare Setting（DINT）			Object Dictionary 0x2F71: Output Direction Setting（UDINT）				
Index	Object Dictionary	Compare Position (count)	Bit	Negative	Positive	Bi-directional	No Output
Position Compare Output 1	0x2F73	***	bit0~bit1	00	10	01	11
Position Compare Output 2	0x2F74	***	bit2~bit3	00	10	01	11
Position Compare Output 3	0x2F75	***	bit4~bit5	00	10	01	11
Position Compare Output 4	0x2F76	***	bit6~bit7	00	10	01	11
Position Compare Output 5	0x2F77	***	bit8~bit9	00	10	01	11
Position Compare Output 6	0x2F78	***	bit10~bit11	00	10	01	11
Position Compare Output 7	0x2F79	***	bit12~bit13	00	10	01	11
Position Compare Output 8	0x2F7A	***	bit14~bit15	00	10	01	11
Position Compare Output 9	0x2F7B	***	bit16~bit17	00	10	01	11
Position Compare Output 10	0x2F7C	***	bit18~bit19	00	10	01	11
Position Compare Output 11	0x2F7D	***	bit20~bit21	00	10	01	11
Position Compare Output 12	0x2F7E	***	bit22~bit23	00	10	01	11
Position Compare Output 13	0x2F7F	***	bit24~bit25	00	10	01	11
Position Compare Output 14	0x2F80	***	bit26~bit27	00	10	01	11
Position Compare Output 15	0x2F81	***	bit28~bit29	00	10	01	11
Position Compare Output 16	0x2F82	***	bit30~bit31	00	10	01	11

- Configure PCO parameters on the PC tool interface, then save drive parameters.

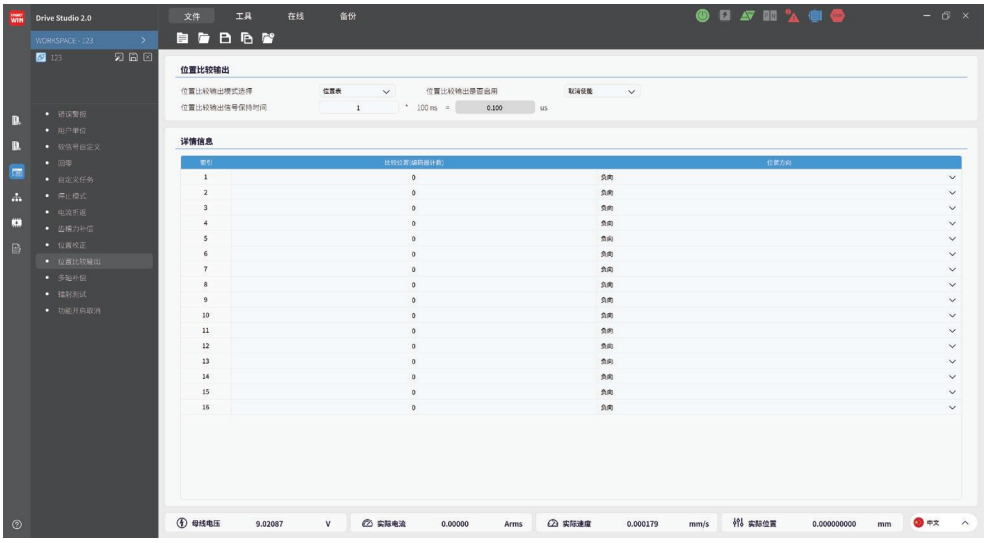


Figure 10-16 Position Compared Output

2.3 Configure Output Signals

Configure IO function mapping on the following screen, as shown below:

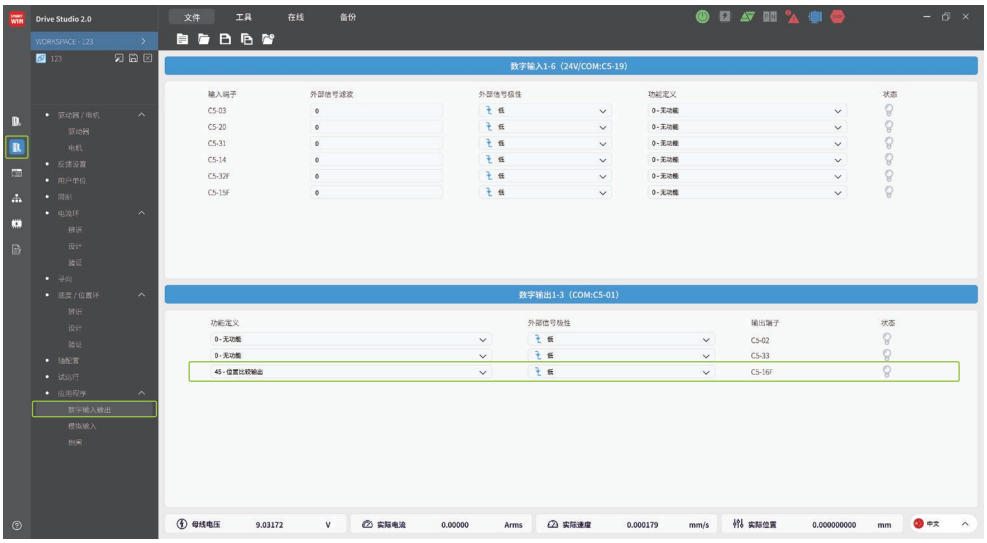


Figure 10-17 Configure Output Signals

10.2.11 Laser Testing

1. Function Introduction

1.1 Motor Offline Burn-in: Run the motor continuously for an extended period in accordance with preset parameters.

1.2 Position Compensation Acquisition: Laser Interferometer

2. Parameter Setup

2.1 Positon mode

- ▶Step 1 ：Control source-selecting “ Ethernet offline ” ；
- ▶Step 2 ：Testing mode： Positon mode ；
- ▶Step 3: Position mode: point-point-relative PTP-Relative、point-point-absolute PTP-Absolute ；
- ▶Step 4 ：Control mode: Position block ；
- ▶Step 5 ：Acceleration & deceleation mode: Sin-curve、 T-curve、 S-curve ；
- ▶Step 6 ：Smoothing time: Interpolation smoothing for position control, Unit: ms ；
- ▶Step 7：Interval points: Set pause points for full travel ；
- ▶Step8：End position: Displays the configured travel end position = Displacement interval x interval points.
- ▶Step9：Displacement interval: Travel distance for each pause segment ；
- ▶Step10：Interval time: Pause time between individual running points;
- ▶Step11：Speed：operating speed ；
- ▶Step12：Acceleration：operating acceleration&deceleration ；
- ▶Step13：Overtravel: Overtravel distance between start and end positions ；
- ▶Step14：Recycle time: Pause time after reaching the end position before return;
- ▶Step15：Recycle cycles: Number of cyclic runs ；
- ▶Step16：Homing type: Homing mode for laser testing ；
- ▶Step17：Measurement times: Displays the current running cycles.

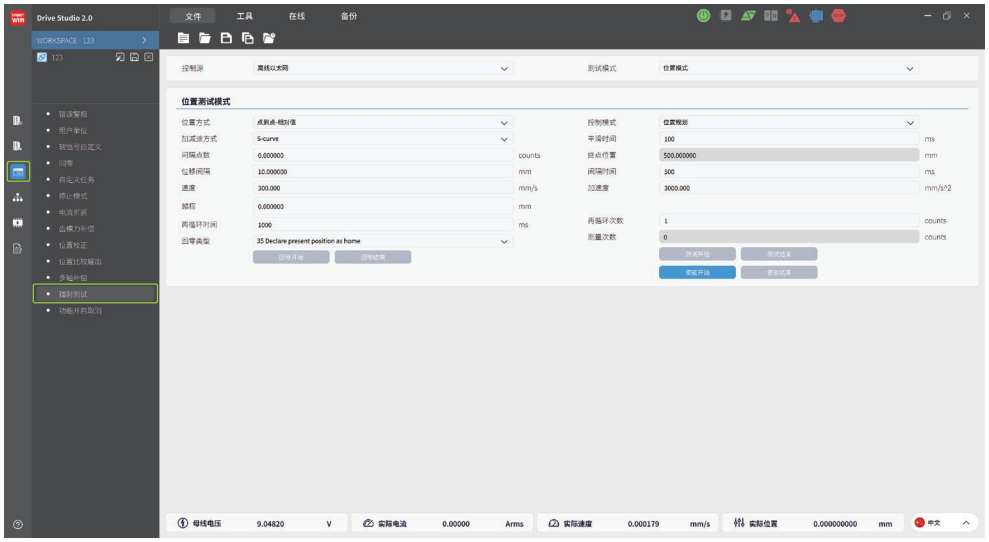


Figure 10-18 Parameter Setup-Position Mode

2.2 Current Mode: I/F Mode, Drive Burn-in Testing

►Step1：Control source-selecting "Ethernet offline";
►Step2：Testing mode: I/F mode;
►Step3：Current Setting: Set the burn-in current, and ensure the long-term operating；
►Step4：Speed Setting: Configure the burn-in speed. Three-phase output switching must be enabled during burn-in operation.

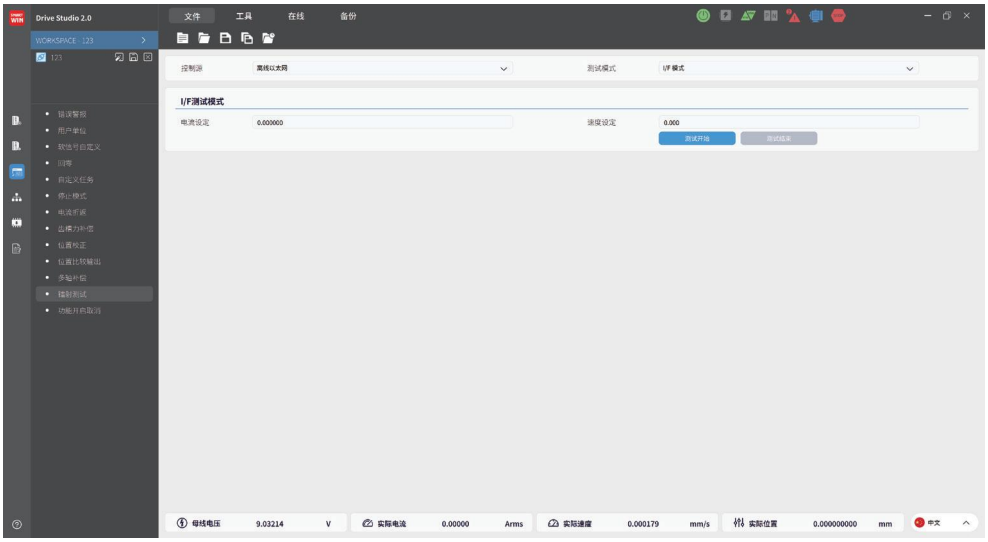


Figure 10-19 Parameter Setup-Current Mode

10.3 Gantary System

10.3.1 Gantary System Overview

A gantry system adopts two parallel axes (Y1 and Y2) to drive a single linear axis, which is perpendicular to the system X-axis.

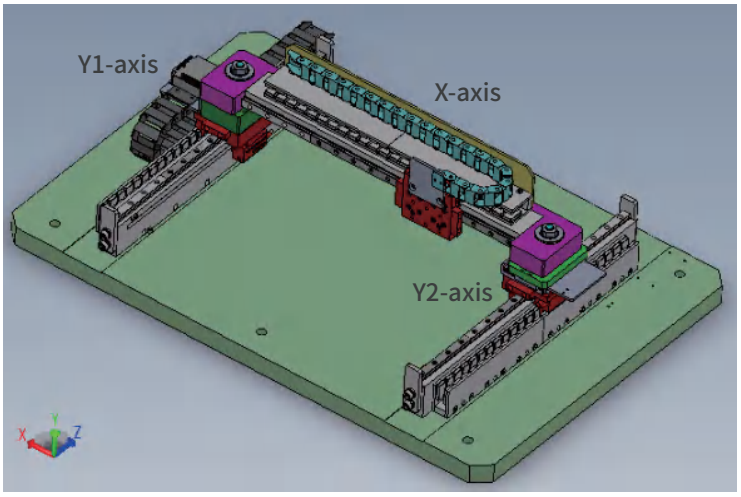


Figure 10-20 Gantary System

In most gantry systems, the two parallel linear axes (Y1 and Y2) are controlled independently, which may cause mutual interference. Typical symptoms include vibration, high current consumption, low acceleration and velocity, as well as degraded positioning accuracy.

The AKS gantry system synchronizes the two Y axes via paired AKS drives that operate in coordination. Leveraging high-speed communication, these drives generate and regulate Y-axis motion. Each Y axis can be driven by either a linear or rotary motor. Both Y axes must be equipped with feedback devices of the same type and resolution.

10.3.2 Gantary Type

The AKS gantry system supports two types of gantry structures: rigid and flexible. The two types differ in the orthogonality of gantry axes and the elasticity of axis coupling. The gantry type parameter (GANTRYTYPE) must be configured to ensure proper homing and effective offset compensation.

1. Rigid Gantry System

In a rigid gantry structure, the X-axis and Y-axis are strictly orthogonal with no elasticity in mechanical coupling. When one Y-axis moves, the other follows synchronously without any axis offset.

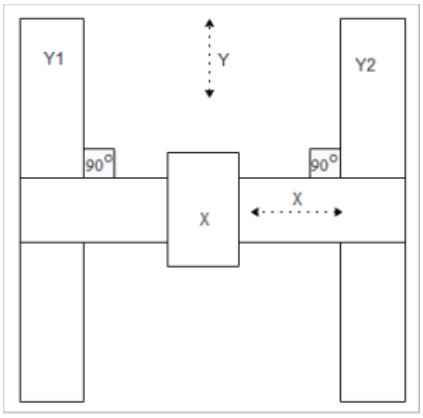


Figure 10-21 Rigid Gantry System

2. Flexible Gantry System:

For a flexible gantry structure, the mechanical coupling between one or both Y-axes and the X-axis features a certain degree of elasticity. Differential motion between the two Y-axes will generate rotational movement around the gantry center, namely yaw. The AKS system treats this rotation as a differential axis. Accordingly, the flexible gantry system requires position difference control between Y1 and Y2 axes.

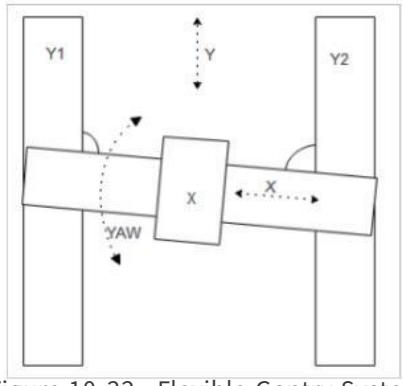


Figure 10-22 Flexible Gantry System

3. Flexible Gantry Yaw Alignment

In a rigid gantry system, the yaw value is zero, and the X-axis is perfectly aligned orthogonally with the Y-axis, so yaw alignment is not required. However, offset will occur in flexible gantry systems. The parameter GANTRYOFFSET is used for yaw alignment. This parameter represents the distance difference between the reference points of Y1 and Y2 axes. Typically, the reference points refer to motor index signals or home switches.

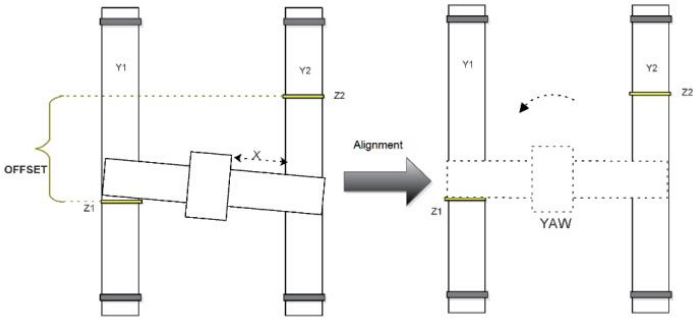


Figure 10-23 Flexible Gantary Yaw Alignment

If the motor manufacturer specifies the index positions of Y1 and Y2, the GANTRYOFFSET can be simply calculated as the difference between the Y1 index position and Y2 index position.If the value of GANTRYOFFSET is unknown, determine it via one of the following methods:

Manual measurement with offset measuring devices (e.g., laser interferometer). This method delivers the highest accuracy and is recommended.

Automatic measurement via the GANTRYOFFSET parameter. Use the command GANTRYFINDOFFST to monitor the measurement status. The value of GANTRYFINDOFFST will change to 1 once the measurement is completed, and the measured result will be stored in GANTRYOFFSET. Run the measurement repeatedly to guarantee accuracy. Set GANTRYOFFSETST = 1 to activate and retain the offset value.

Save the final result to the drive memory upon completion.

The system GANTRYOFFSET only needs to be configured once. Subsequent homing commands will apply the valid offset value. In addition, the flexible gantry system will perform alignment based on GANTRYOFFSET every time the system is powered on or switched from disabled to enabled state.

10.3.3 Gantry Control Mode

The AKS gantry system is controlled by two drives interconnected via communication cables, known as paired drives.

The drive for axis Y1 acts as the master drive. It receives position commands from the motion controller and runs the position loop based on the virtual Y-axis, which is calculated as the average value of Y1 and Y2.

The drive for axis Y2 serves as the slave drive, and it follows the output status of the master drive. For rigid gantry systems, it maintains consistent current output between the master and slave axes during operation of Y1 and Y2.

10.3.4 Gantry Setting

1. Current Loop Parameter：

Configure master-slave current verification for AKS drives. Normally, select motors and drives with identical specifications for master and slave axes to ensure consistent current loop parameters on both axes.

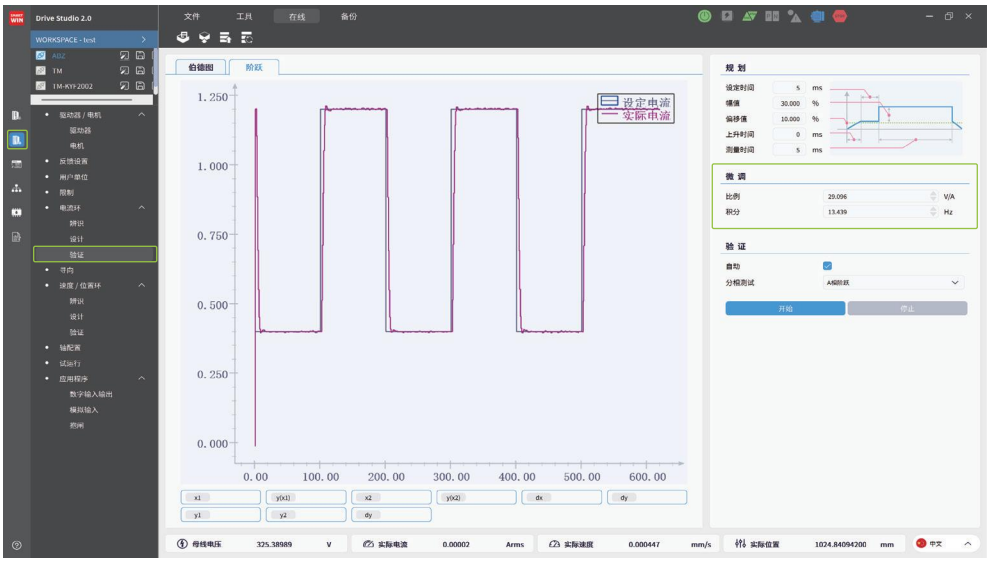


Figure 10-24 Current Loop Parameter

2. Set Master-Slave Axis Direction

Verify the current direction of master and slave axes. Perform manual phase searching to ensure consistent movement direction. If directions differ, invert the slave axis current in the gantry configuration interface. This step is critical.

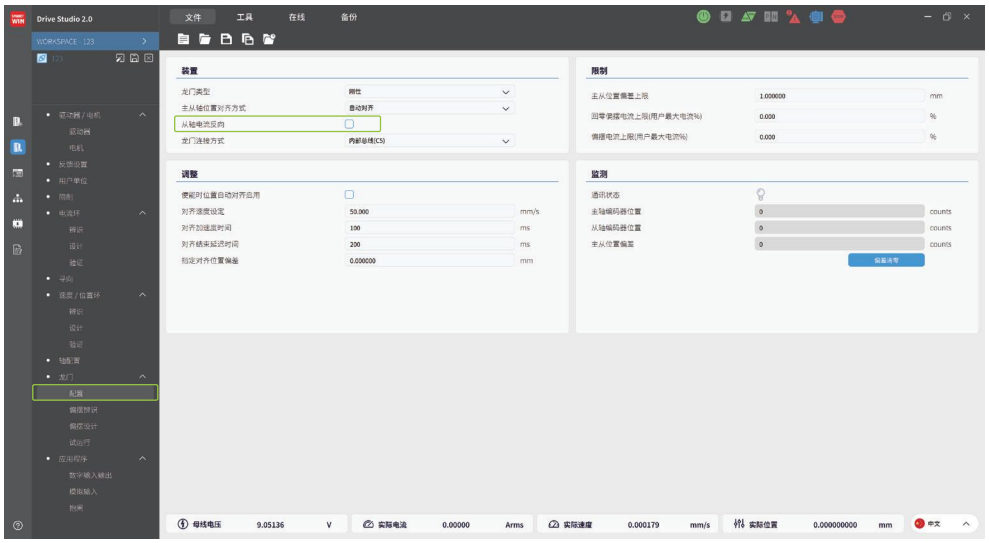


Figure 10-25 Set Master-Slave Axis Direction

3. Master Axis Mode Setup

Control source for master axis-EtherCAT, axis configuration-gantry, master/slave selection-master axis, control mode-position mode

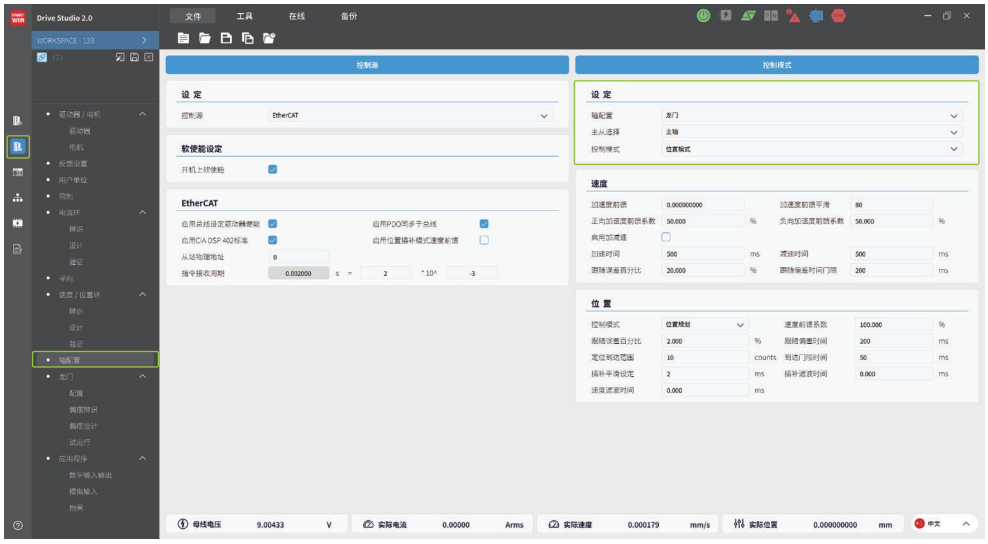


Figure 10-26 Master Axis Mode Setup

4. Slave Axis Mode:

Control source for slave axis-analog, axis configuration-gantry, master/slave selection-slave axis, control mode-current mode.No further adjustments are required after completing slave axis configuration.

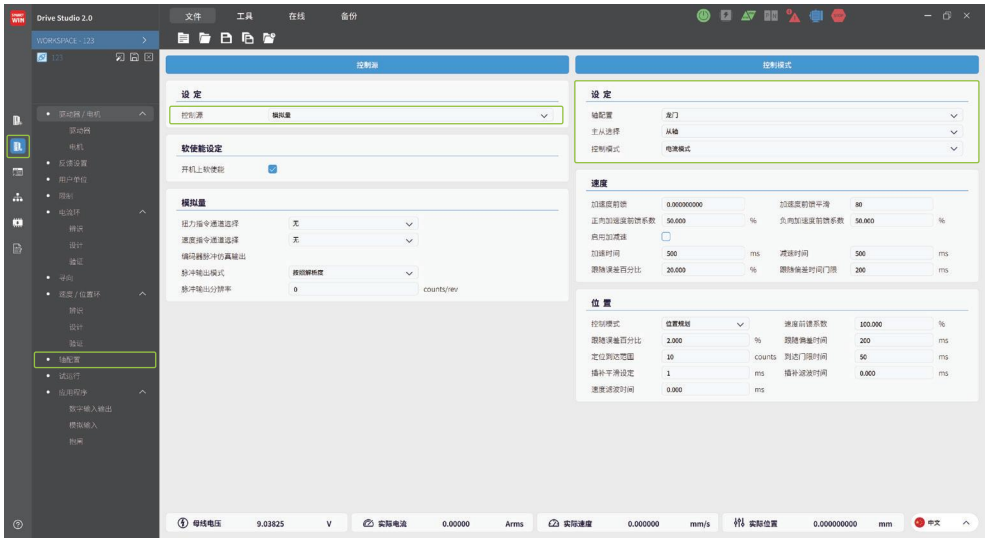


Figure 10-27 Slave Axis Mode

5. Gantry Configuration:

Device: Select rigid gantry for rigidly connected master and slave axes; select flexible gantry for flexibly connected or disconnected axes. Set master-slave alignment mode to automatic alignment.

Limits: An alarm E0228 (Master-Slave Position Deviation) will be triggered if the master-slave position deviation exceeds the threshold.

Homing Yaw Current Limit: The maximum current for yaw compensation during homing of flexible gantry systems.

Yaw Current Limit: The maximum current for yaw compensation against position deviation between master and slave axes. This function is generally disabled for rigid gantries.

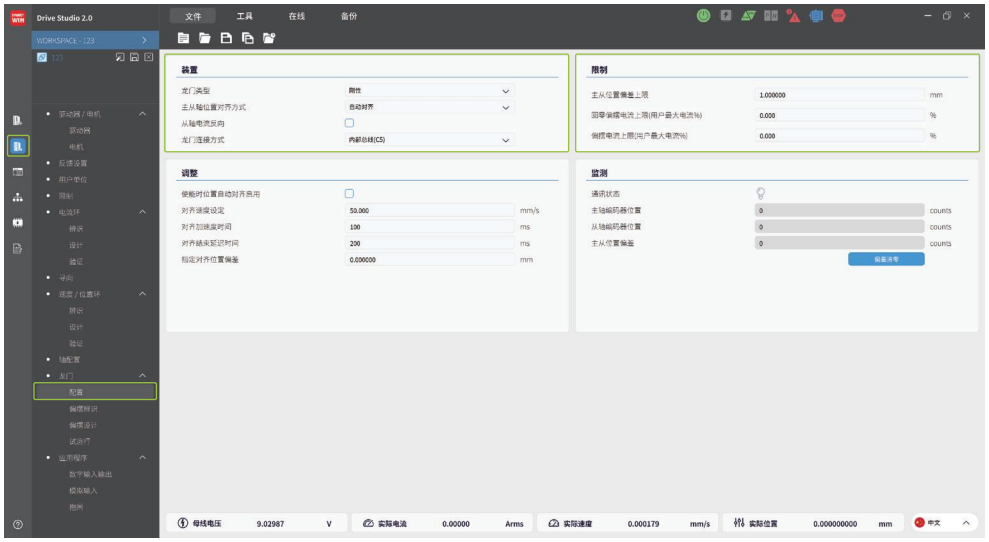


Figure 10-28 Gantry Configuration

Adjustment: Enable automatic position alignment when the drive is activated. This function applies to flexible gantries: it aligns the master and slave axes upon re-enable if position deviation occurs after deactivation post homing. Homing must be performed first on initial power-up after power loss.

Monitoring：Communication status，displaying indicates normal communication between the master and slave axes.Regarding master and slave encoder positions: A large position deviation is acceptable before homing, and the deviation must be cleared after homing. The values can be read for comparison. During normal gantry operation, the position deviation between the two axes stays near zero.

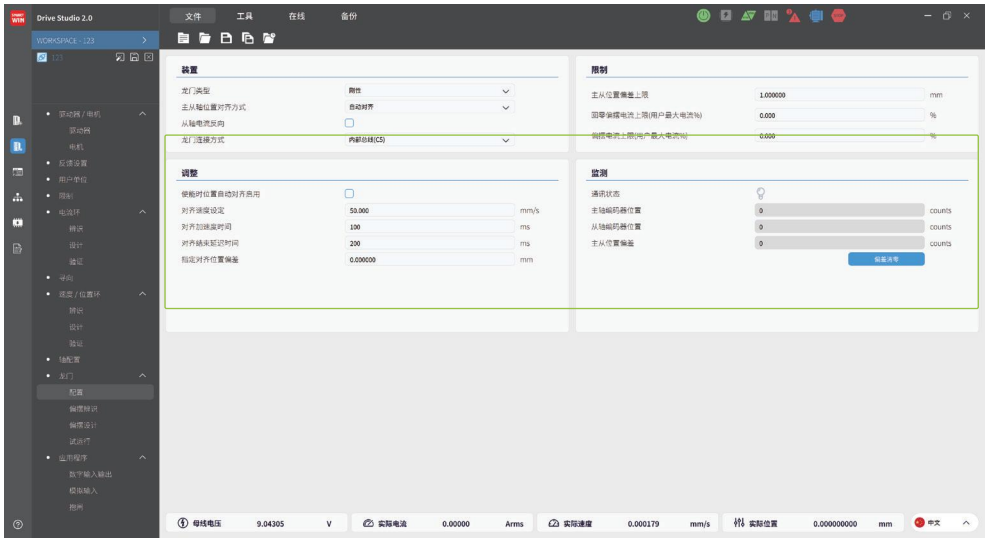


Figure 10-29 Gantry Configuration

6. Master-Slave Yaw Tuning:

Tune the parameters of yaw speed loop and position loop to reduce position deviation and improve synchronization between master and slave axes. For flexible gantries with yaw current, an additional filter shall be configured.

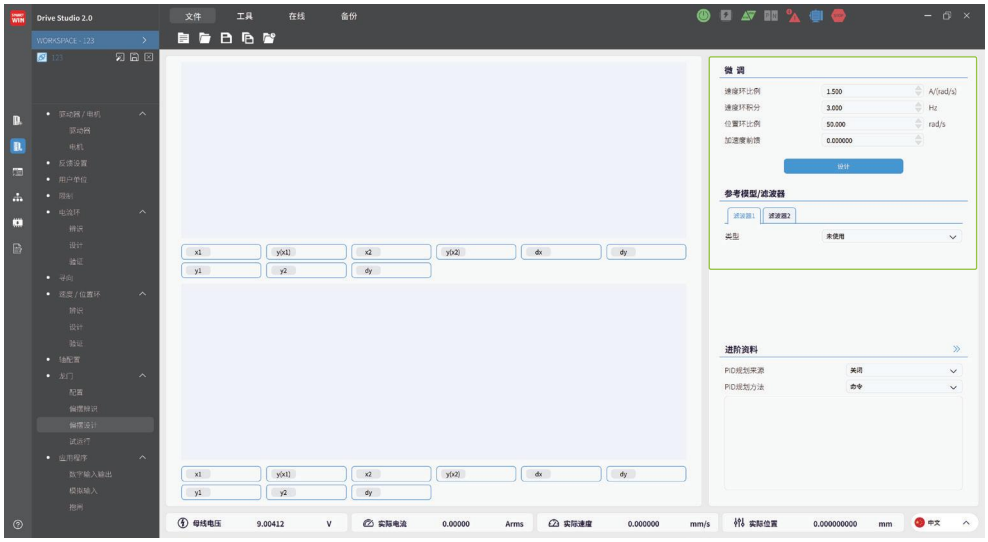


Figure 10-30 Master-Slave Yaw Tuning

7. Trial Operation:

This operation is the same as single-axis operation. Tune the yaw axis and monitor master-slave position deviation and yaw current in real time.

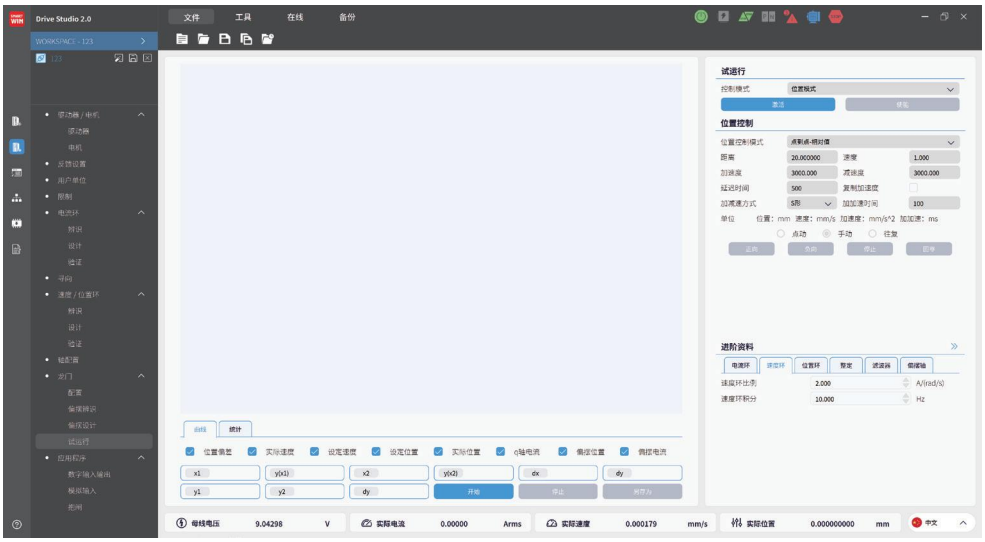


Figure 10-31 Trial Operation

8. IO Application Configuration:

This system is equipped with two sets of forward/reverse limit switches and two sets of home switches for the master and slave axes. Home Switch corresponds to the master axis, and Home Switch2 to the slave axis. The second set of limit switches are assigned to the slave axis.

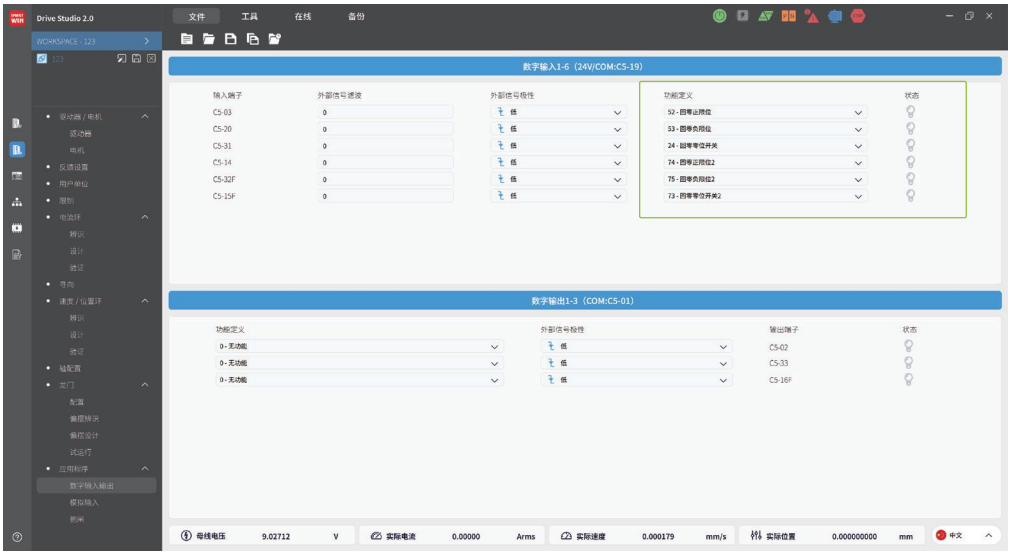


Figure 10-32 IO Application Configuration

9. Homing Application:

Homing mode: Select according to actual application requirements.

Homing speed: Set a relatively low value for gantry homing to avoid force conflict between master and slave axes.

Homing torque limit: Set to 50%.

Homing axis: Select single-side homing for rigid gantries and dual-side homing for flexible gantries.

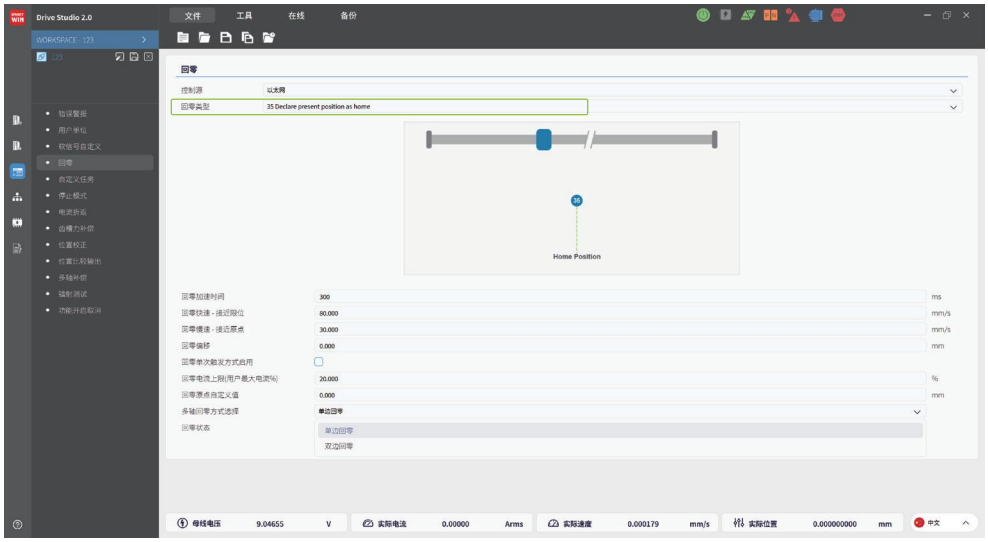


Figure 10-33 Homing Application

10. Gantry Application Warning

Warning Code	Warning Description	Solutions
E0227	Multi-axis wiring error	1.Check the communication cables between master and slave axes. 2.Verify that the slave axis is set to gantry mode.
E0228	Master-slave axis position deviation	1.Check for mechanical jamming. 2.Verify the upper limit of master-slave position deviation.
E0229	Slave axis error occurs	1.Check for alarms on the slave axis and clear them accordingly
E0230	Slave axis enable timeout	1.Slave axis enable timeout 2.Check if there are other abnormalities on the slave axis

11. Drive Control Mode

11.1 Overview of Drive Control Modes

The drive supports multiple control modes, mainly current control, speed control and position control.

11.2 Current Control

11.2.1 PC Current Control

Trial operation, control mode-torque mode, current setup-output current

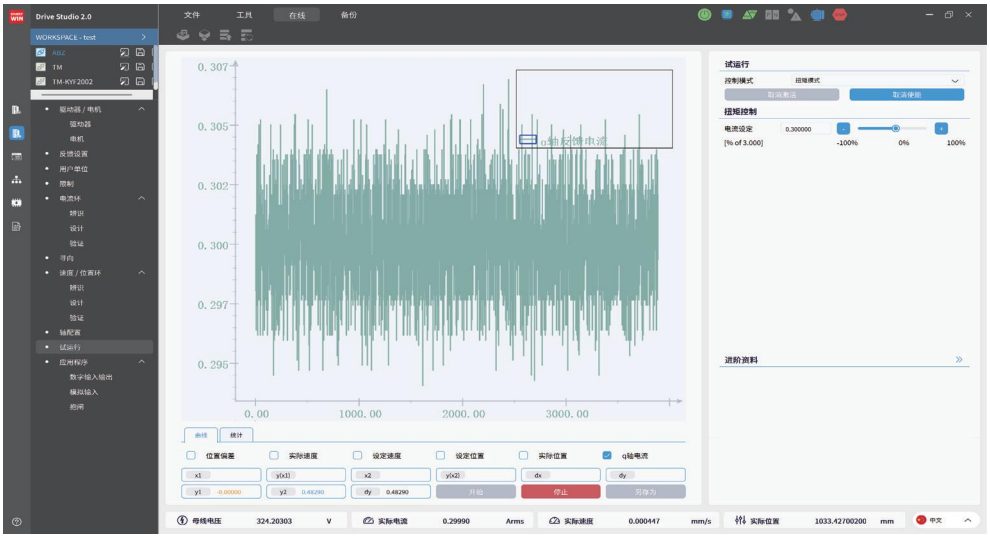


Figure 11-1 PC Current Control

11.2.2 AD Current Control

Setting control source-analog, torque demand access selection-analog input 1, control mode-current mode.

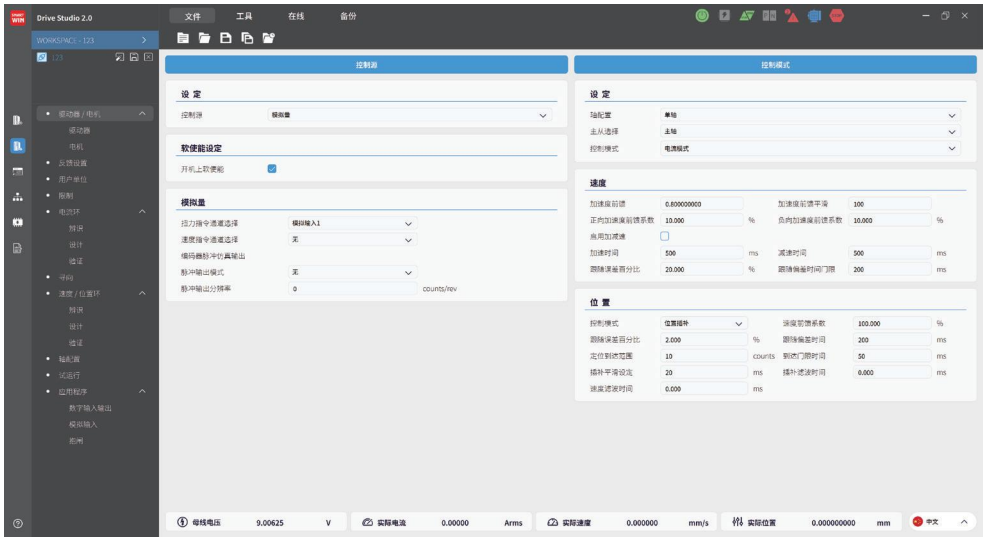


Figure 11-2 AD Current Control

11.2.3 Bus Current Control

Supporting CiA DSP 402.

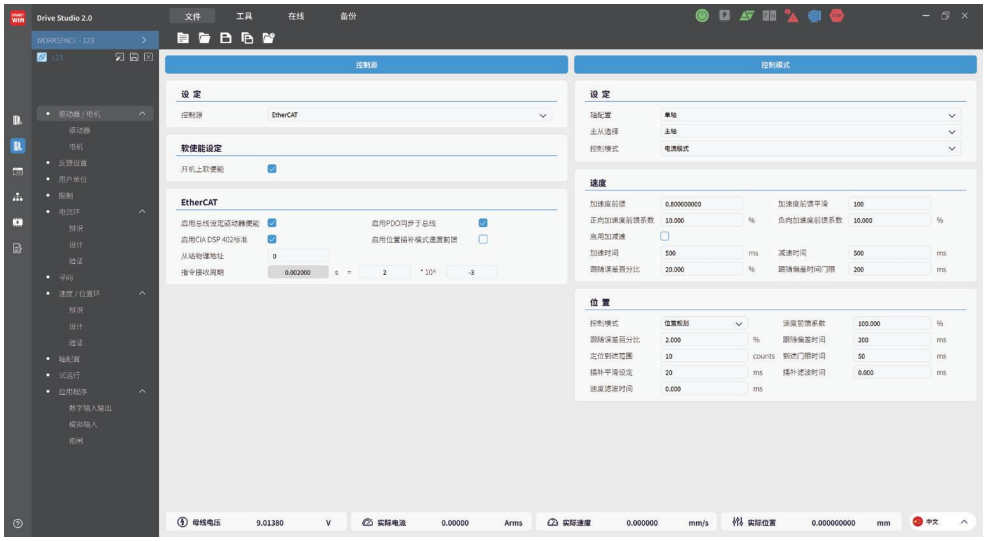


Figure 11-3 Bus Current Control

11.3 Speed Control

11.3.1 PC Speed Control

Trial operation, control mode-speed mode, speed steup-operation speed.

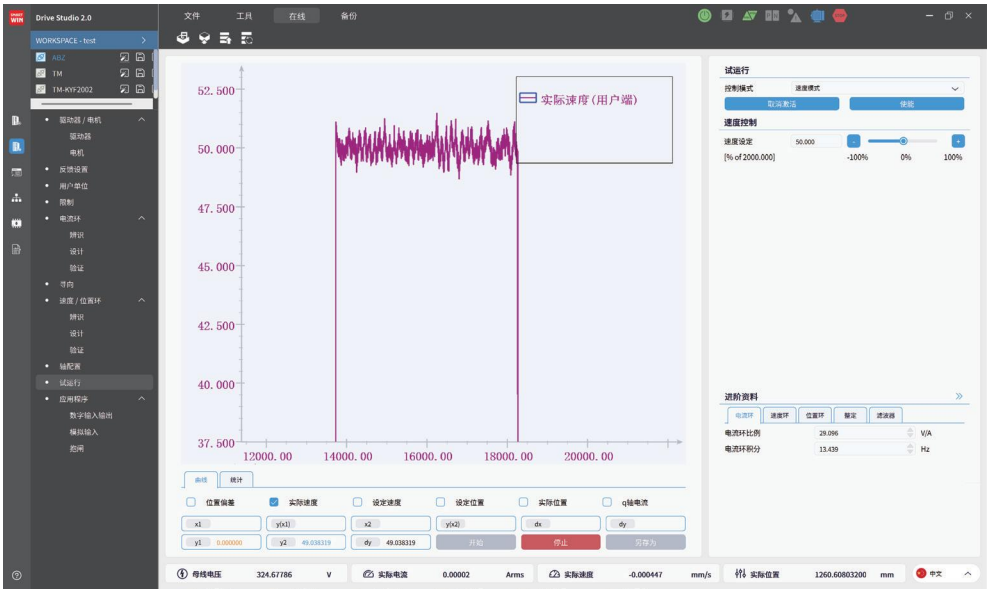


Figure 11-4 PC Speed Control

11.3.2 AD Speed Control

Setting control source-analog, torque demand access selection-analog input 1, control mode-speed mode

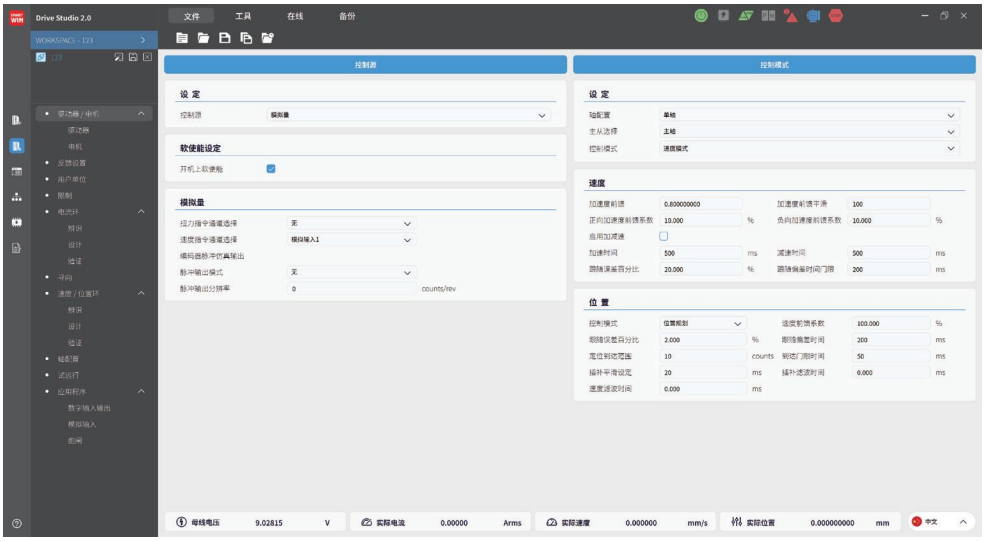


Figure 11-5 AD Speed Control

11.3.3 Bus Speed Control

Supporting CiA DSP 402.

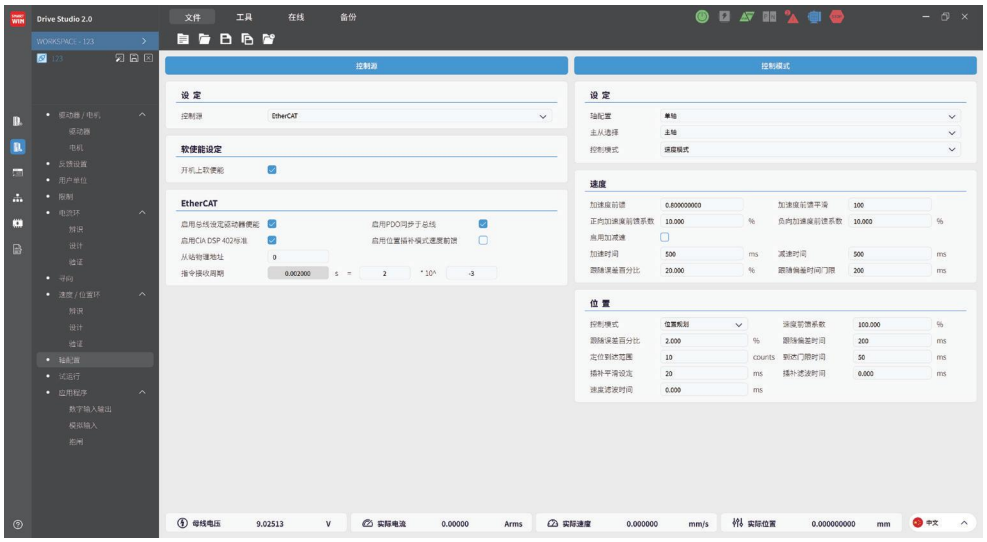


Figure 11-6 Bus Speed Control

11.4 Position Control

11.4.1 PC Position Control

Trial operation, control mode: Position mode. Refer to trial run settings for relevant configurations.

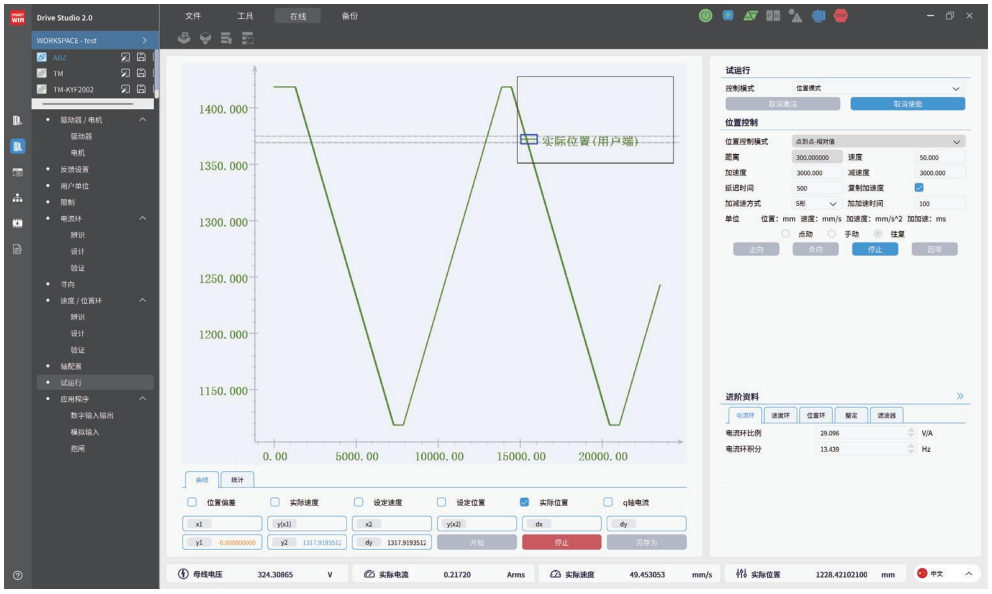


Figure 11-7 PC Position Control

11.4.2 Bus Position Control

Supporting CiA DSP 402.

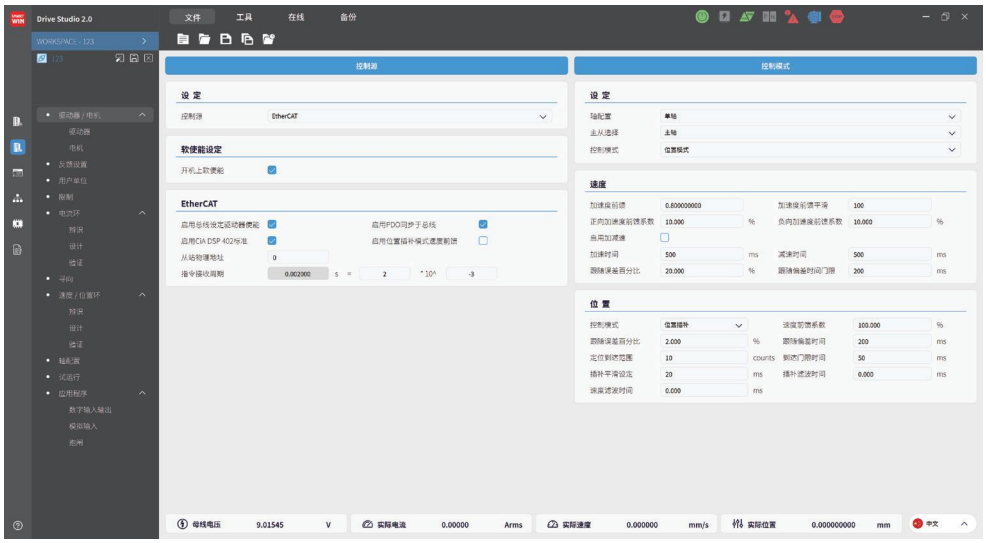


Figure 11-8 Bus Position Control

11.5 Pulse Direction Control

11.5.1 Pulse & Direction

In Pulse plus Direction (P+D) position control, the drive synchronizes with external pulse and direction command signals. The drive increments or decrements the external position counter by one count on each rising edge of the pulse, subject to the direction signal. The counter value is processed by the gear ratio module and used as the position command for the motor.

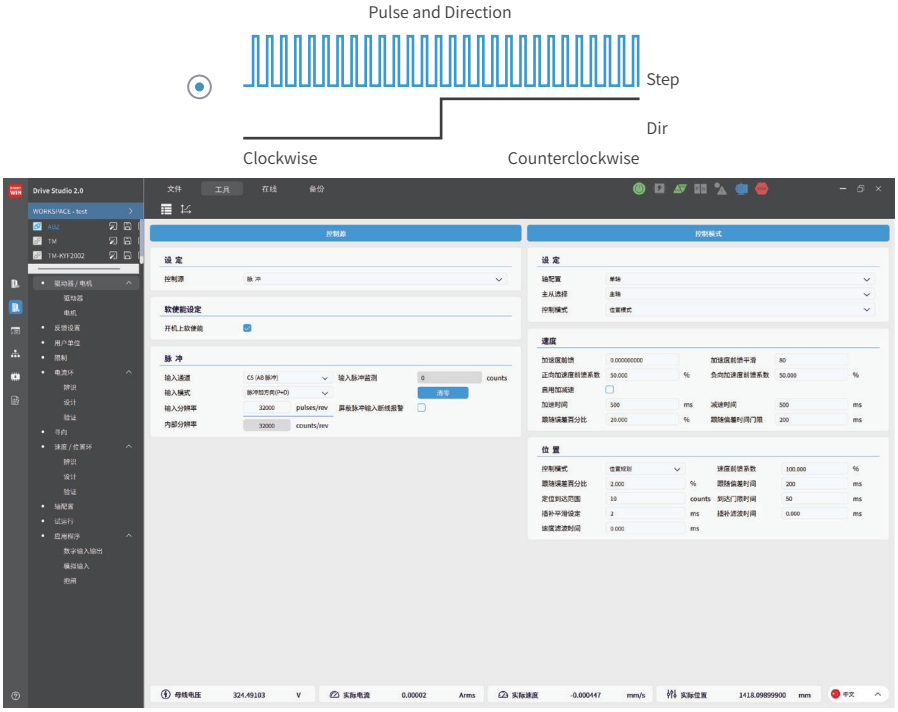


Figure 11-9 Pulse & Direction

11.5.2 CW/CCW Counting

In the dual-pulse (CW/CCW pulse) system, pulses on one signal increase the motor position, while pulses on the other decrease it. Pulses sent to Channel A increment the external position counter and rotate the motor forward. Pulses sent to Channel B decrement the counter and rotate the motor in reverse. The pulse frequency and gear ratio determine the axis moving speed and travel distance.

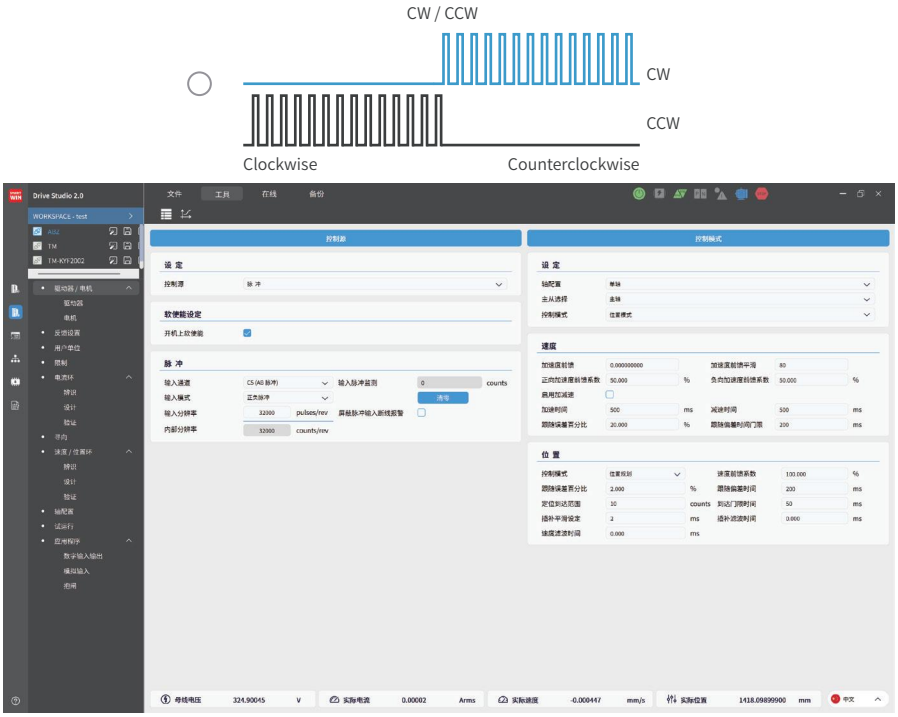
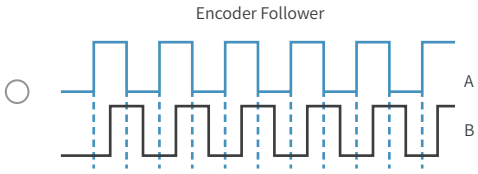


Figure 11-10 CW/CCW Counting

11.5.3 AB Pulse Follower

In incremental (AB) follower mode, the drive tracks the quadrature encoder signals from the master device. The motion direction is determined by the phase sequence of the quadrature signals (A leading B or B leading A).



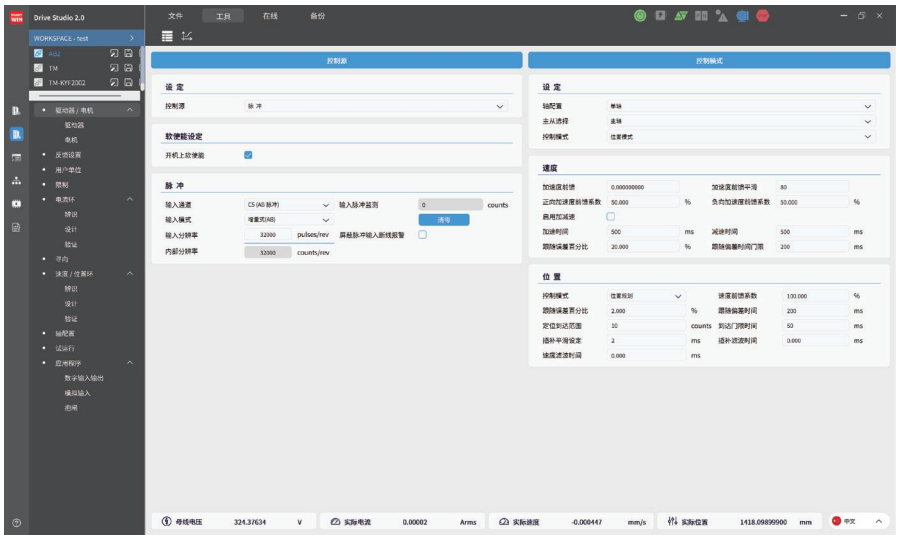


Figure 11-11 AB Pulse Follower

12. Drive Commissioning

12.1 PID Adjustment

12.1.1 Current Loop Commissioning

1. Weak current loop commissioning：

As shown in the figure, when PID parameters are set too low, the actual value lags behind the setpoint. Appropriately increase the proportional gain KP. The system current loop exhibits slow response.

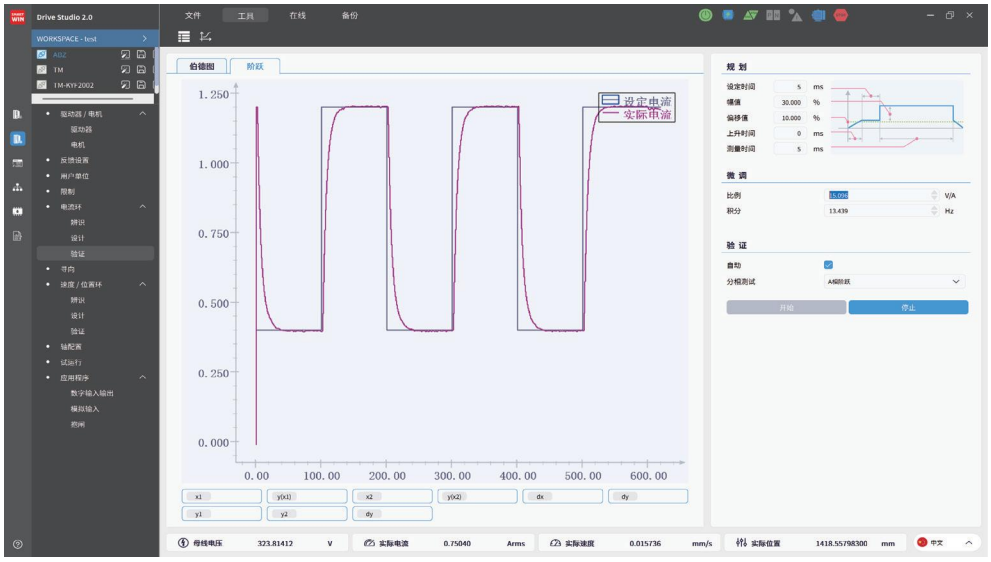


Figure 12-1 Current Loop Commissioning-Weak Current Loop Commissioning

2. Current Loop Commissioning Completed Properly：

As shown in the figure, with properly tuned PID parameters, the actual value tracks the setpoint well without obvious overshoot. The current loop delivers excellent response.

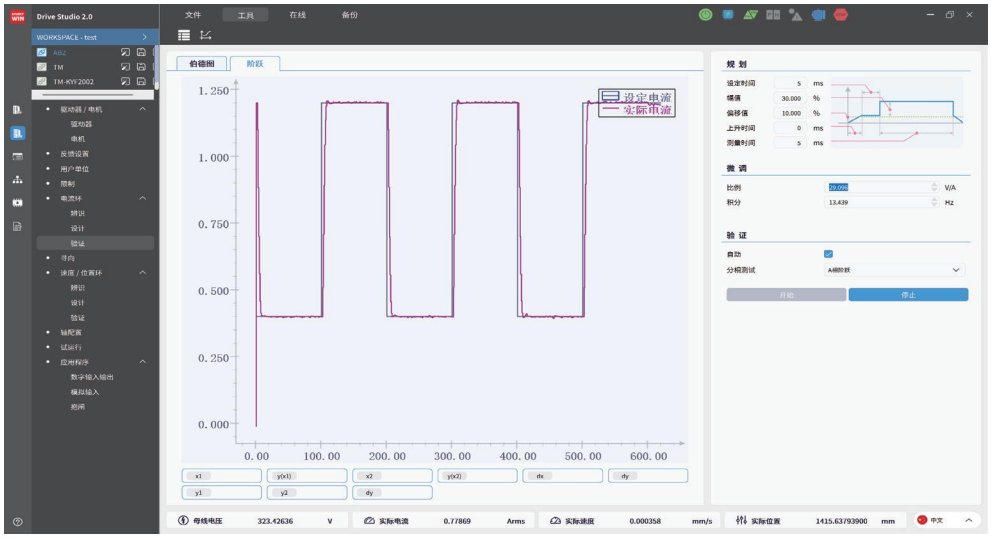


Figure 12-2 Current Loop Commissioning-Current Loop Commissioning Completely Properly

3. Strong current loop commissioning：

As shown in the figure, when PID parameters are set excessively high, the actual value exceeds the setpoint with obvious overshoot. While the current loop responds rapidly, the system becomes unstable and the motor emits high-frequency noise.

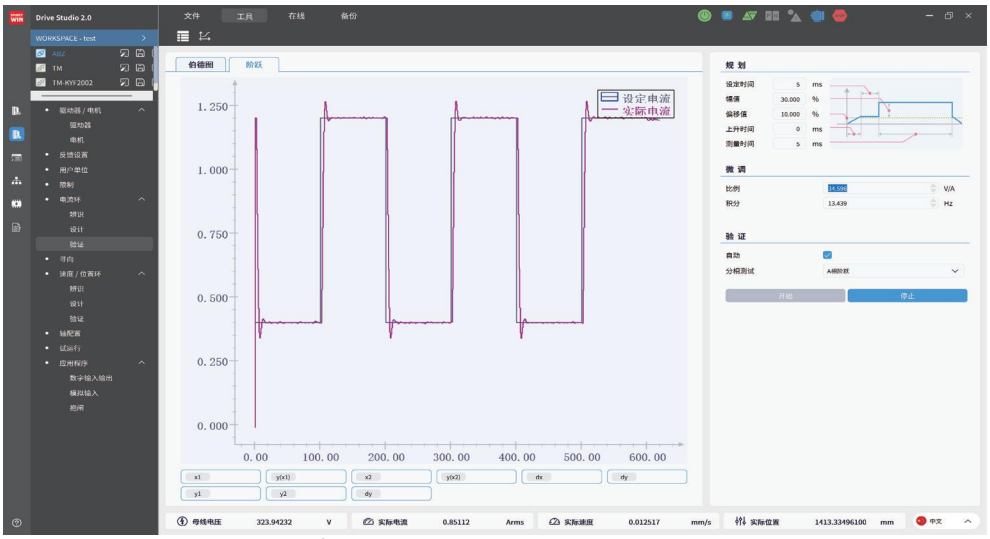


Figure 12-3 Current loop tuning-Strong Current Loop Commissioning

12.1.2 Speed Loop Commissioning

1. Control Model Loop Circuit:

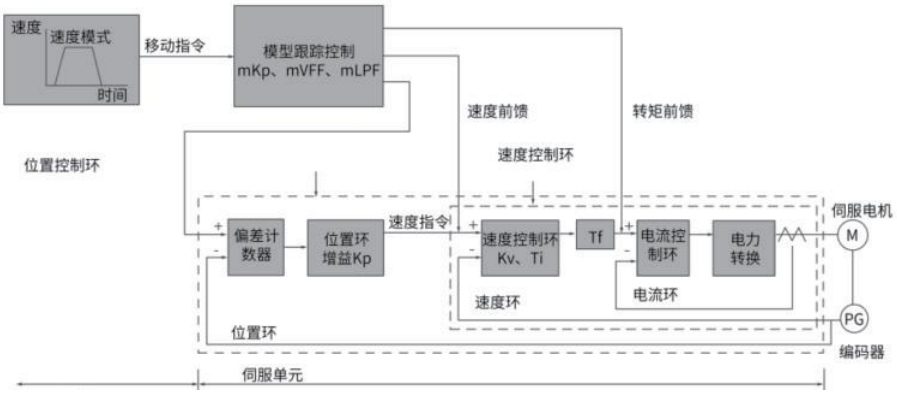


Figure 12-4 Control Model Loop Circuit

2. Speed Loop Rate:

This parameter defines the maximum frequency of dynamic speed commands trackable by the speed loop. When the average load inertia ratio is set correctly, it governs the tracking performance of the speed loop.

Adjustment method: Increase the parameter within the range without noise or vibration to shorten positioning time and improve speed stability and tracking performance. Reduce the setting if noise occurs. In case of mechanical vibration, activate the mechanical resonance suppression function.

►Step 1: Set KP to half of the default value.

►Step 2: Set KI to zero.

►Step 3: Increase the speed loop proportional gain until position error (PE) begins to oscillate. As the gain rises, the position error waveform becomes square, showing steady values during acceleration and deceleration. With higher proportional gain, the position error stabilizes throughout all motion phases: acceleration, constant speed and deceleration.

Optimal tuning: The waveform should be as square as possible. This means the position error stays constant in each motion phase, with no oscillation during transitions (acceleration to constant speed, constant speed to deceleration, deceleration to stop).

3. Speed Loop Integral:

Parameter Function: Eliminates speed loop deviation.

Adjustment Method: Increasing the value enhances integral action and shortens positioning time. Excessively high settings may cause mechanical vibration, while overly low values will leave speed loop deviation uneliminated.The speed loop PI control is a single feedback controller without pre-filtering. The proportional gain (KP) tunes the system, and the integral gain (KI) compensates for steady-state error. Both gains are adjusted via trial and error.



Figure 12-5 Speed Loop Integral

12.1.3 Position Loop Rate

Parameter Function: Determines the maximum frequency of dynamic position commands that the position loop can track.

Adjustment Method: Increasing this parameter reduces position deviation during motion, shortens positioning time and improves the motor's anti-disturbance capability at standstill. An excessively high setting may cause system instability and oscillation.



Figure 12-6 Position Loop Rate

12.2 Filter Application Instructions

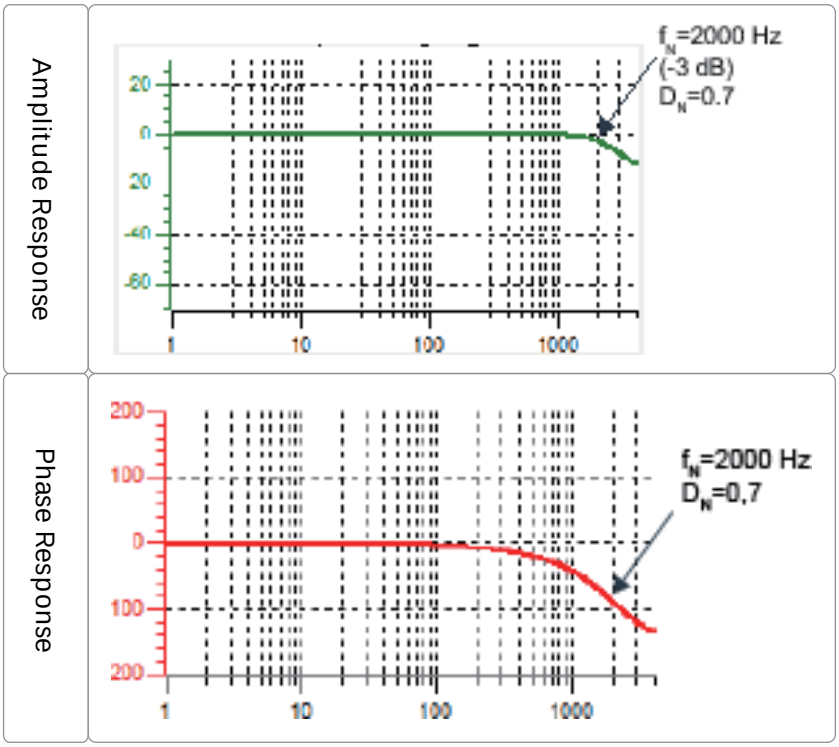
12.2.1 Current Demand Low-Pass Filter

1. Low pass PT2，second-order low-pass filter：



Figure 12-7 Second-Order Low-Pass Filter Parameter Setting

The corresponding frequency response diagram of the above parameters is shown below:



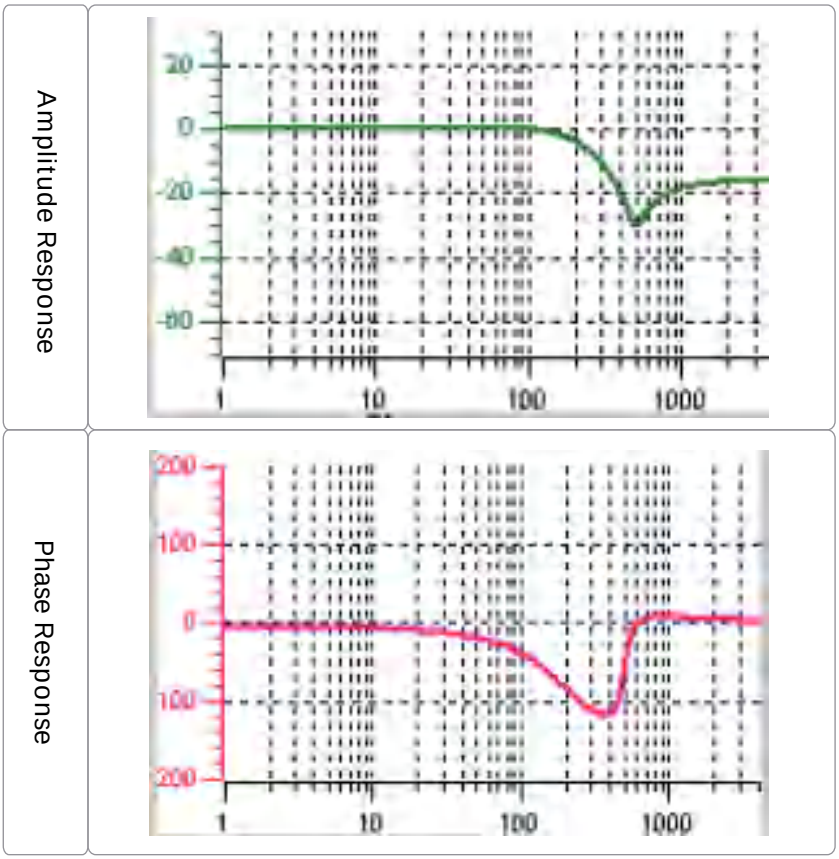
2. General 2nd-order filter，common second-order filter：

This filter is the general form of all second-order filters, and it can be equivalently converted to band-stop filters, notch filters and second-order low-pass filters.



Figure 12-8 Common Second-Order Filter Parameter Setup

The frequency response graphs for the above parameters are shown below:



3. Band stop，filter with resistor：

Band-stop filters are commonly used to attenuate signals within a specific frequency band and block signal transmission in abnormal conditions.



Figure 12-9 Filter with Resistor Parameter Setup

The frequency response plots corresponding to the above parameters are presented below:

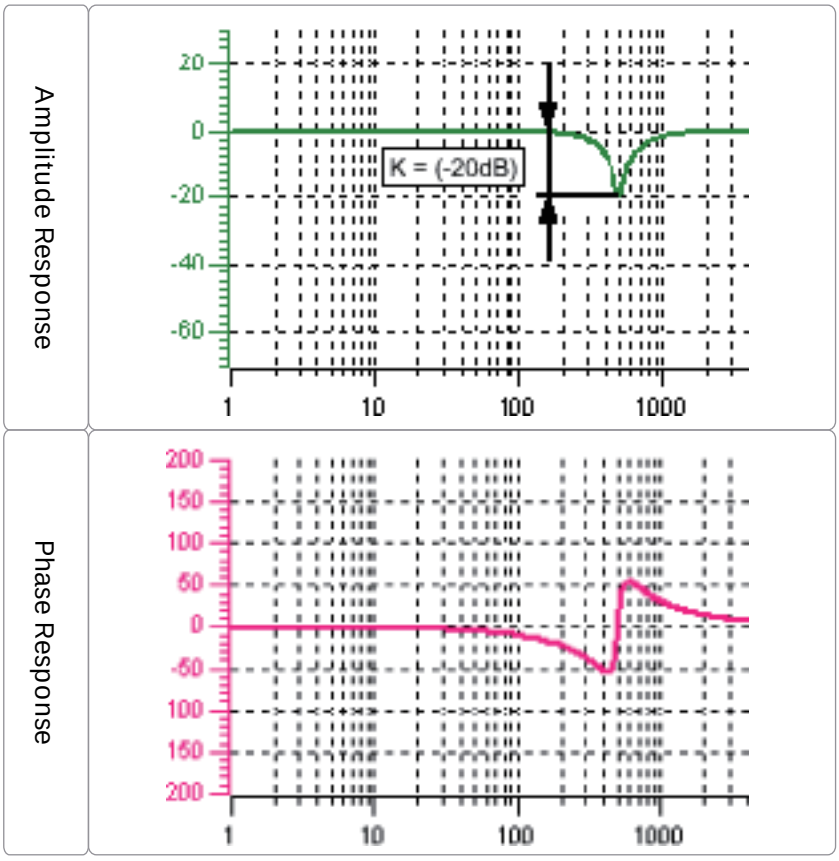
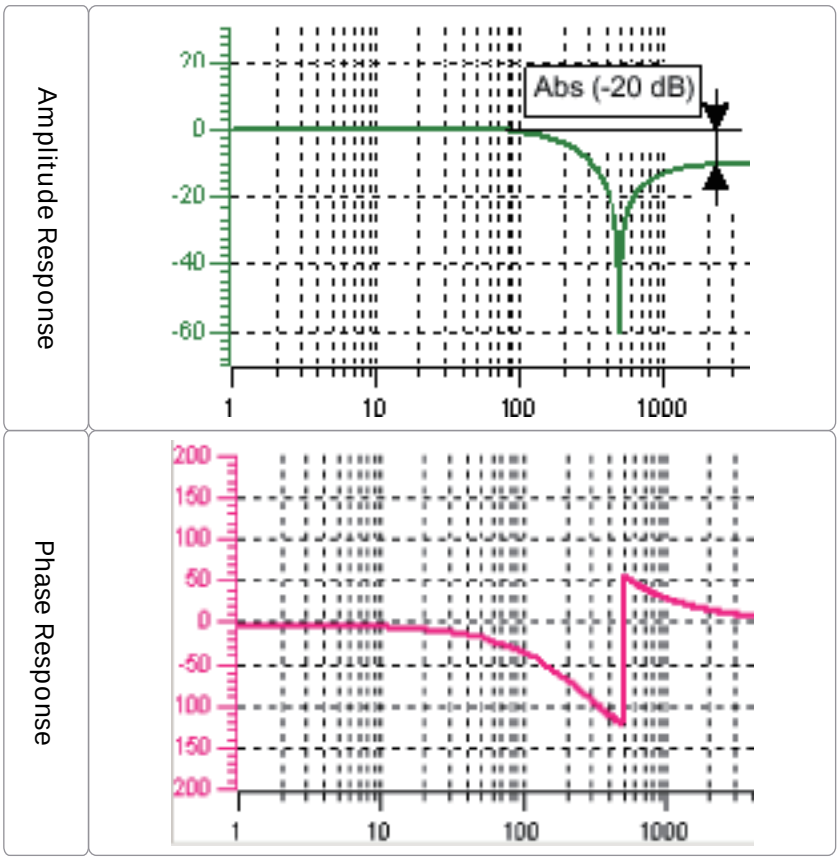




Figure 12-10 Filter with Resistor Parameter Setup

The frequency response curves of the above parameters are shown below:



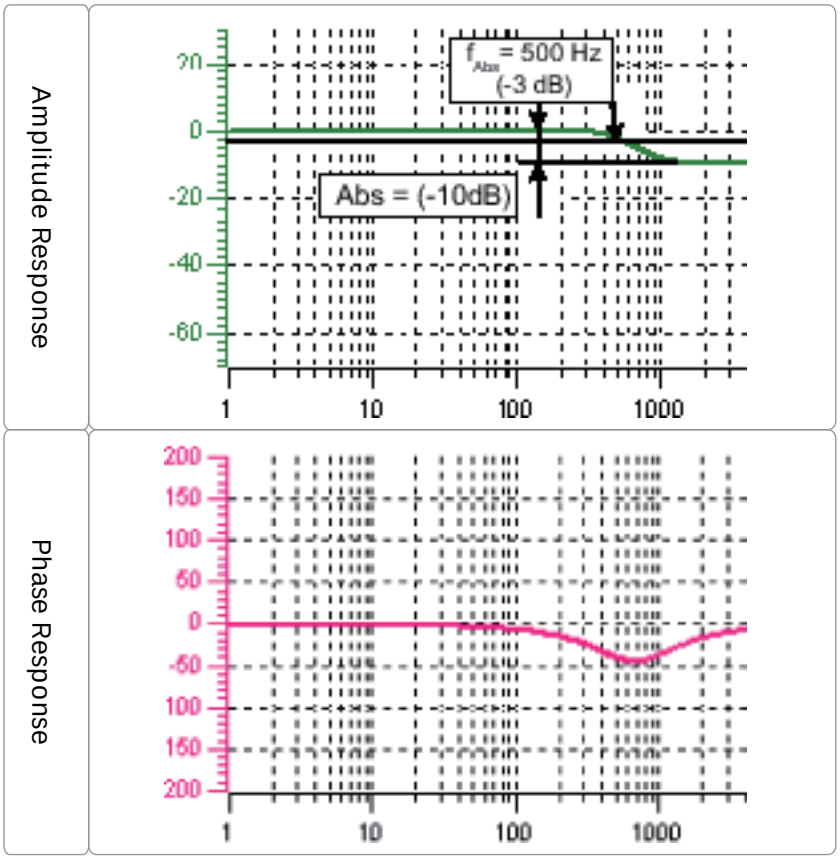
4. Low pass with reduction filter：

A low-pass filter allows signals below the cutoff frequency to pass through and attenuates those above it. When using a low-pass filter with roll-off characteristics, the amplitude response at high frequencies will be reduced accordingly. This can be regarded as a special type of second-order low -pass filter.



Figure 12-11 Low Pass with Reduction Filter Parameter Setting

The frequency response diagrams for the above parameters are as follows:



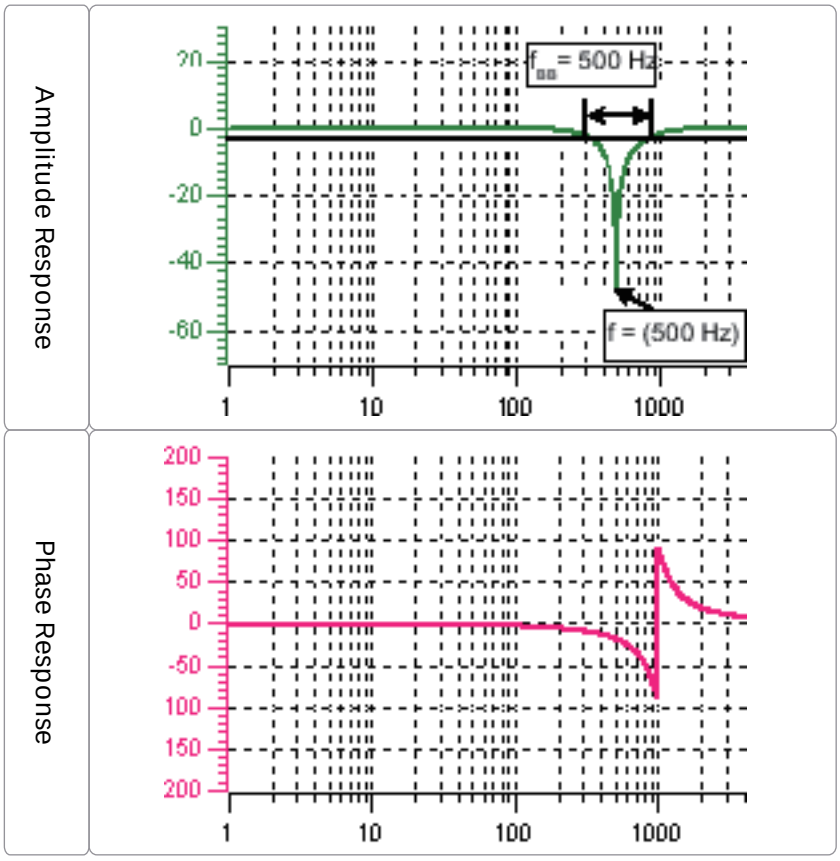
5. Notch Filter

A notch filter is a special case of a band-stop filter, commonly used to suppress vibration at a specific frequency.



Figure 12-12 Notch Filter Parameter Setup

The frequency response diagrams for the above parameters are as follows:



12.2.2 Velocity Feedforward Filter

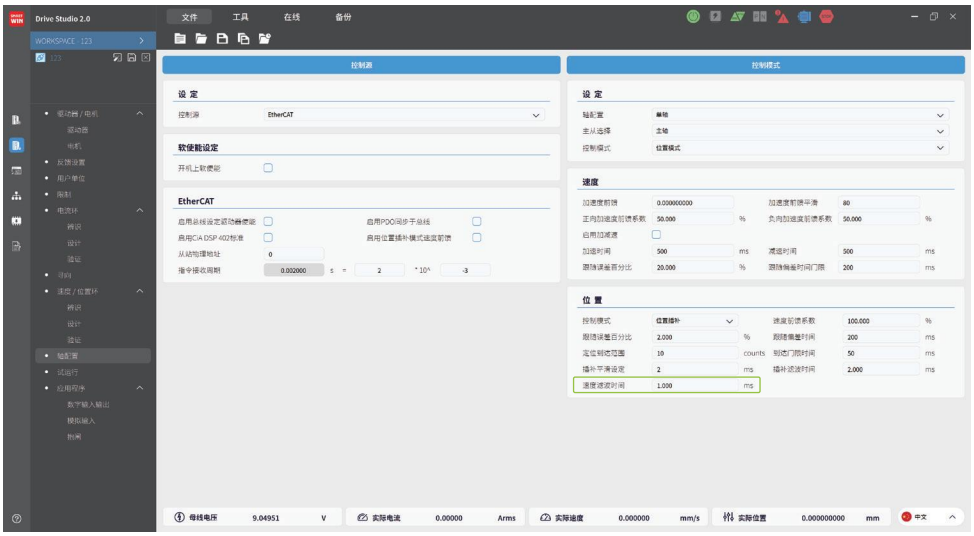


Figure 12-13 Velocity Feedforward Filter Parameter Setup

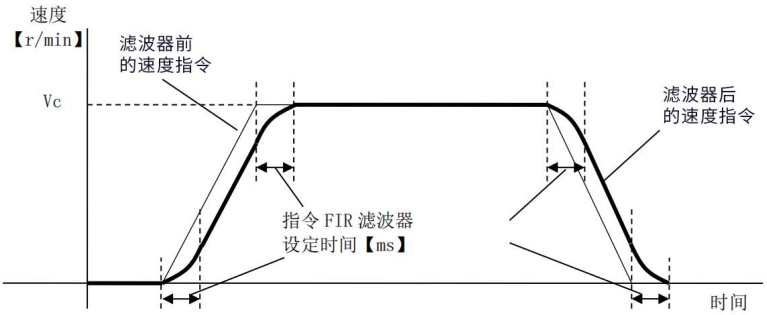


Figure 12-14 Velocity Feedforward Filter Diagram Curve

12.2.3 Position Demand Filter1. Moving Smoothing Filter：

Smoothing Filter Parameter Setting: Interpolation smoothing range: 1 ~ 150 ms

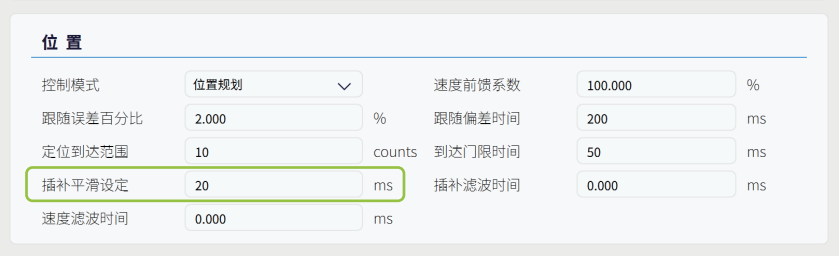


Figure 12-15 Moving Smoothing Filter Parameter Steup

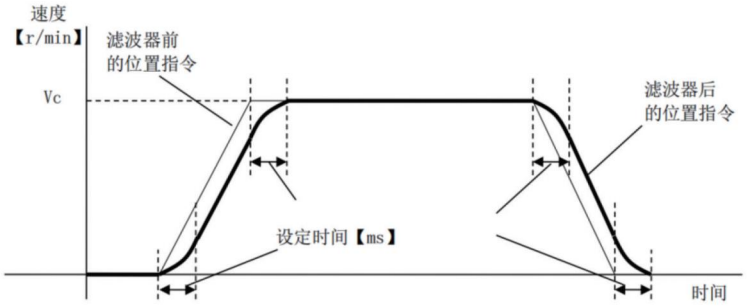


Figure 12-16 Moving Smoothing Filter Diagram Curve

2. Low-Pass Filter：



Figure 12-17 Low-Pass Filter Parameter Setup

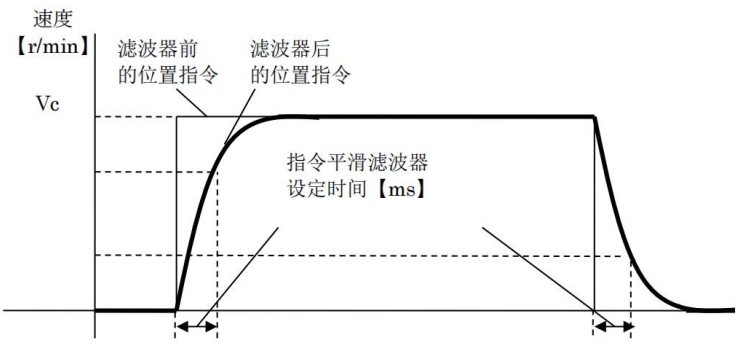


Figure 12-18 Low-Pass Filter Diagram Curve

13. Fireware Update

13.1 Fireware Update Preparation

Important Notice:Before updating the firmware, please complete the following steps:

■Refer to the release notes and all other documents supplied with the new firmware.
■Parameter settings may be lost during the update. Please back up parameters of all drives in the system and keep the backup files properly. You can restore the parameters after the update.
■To back up parameters, navigate to the Backup & Restore page in Drive Studio and click Save Ba ckup.

13.2 Fireware Update Selection

Open the firmware update screen in the Drive Studio drive commissioning software.

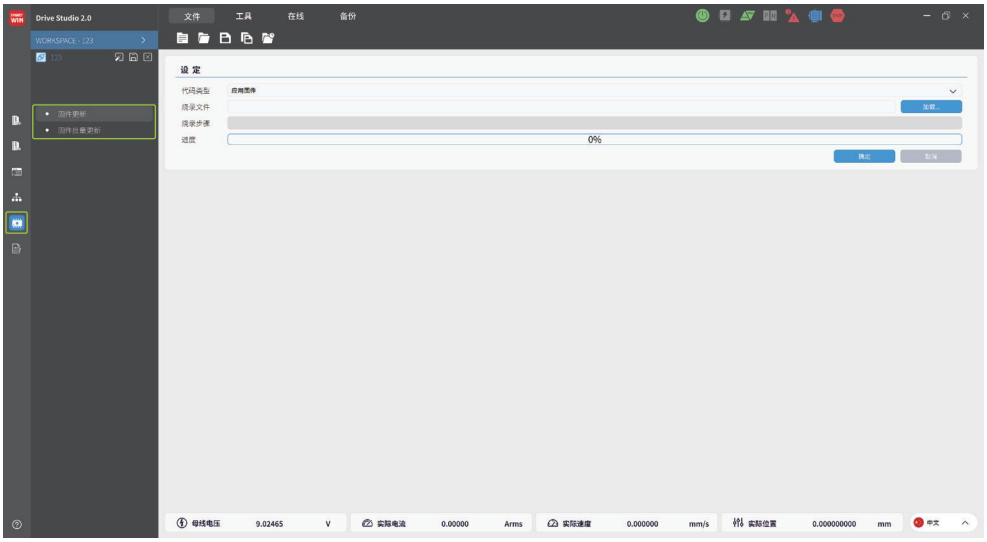


Figure 13-1 Firmware Update Selection

Firmware Update	Firmware update screen via NET debug port
Firmware ECAT Update	Firmware update screen via EtherCAT bus
Code Type	Select the corresponding firmware type: 1.Low-level firmware2.Application firmware
Flash File	It displays the path and name of the firmware update file. The file name indicates the firmware version. Click Browse to select another path.
Flash Step	Displays the real-time flashing position.
Progress	Displays flashing progress
Confirm	Start Firmware Update
Cancel	Cancel Firmware Update

13.3 Update firmware via NET debug port

►Step 1: Select the correct firmware type, then click Load.

►Step 2: Choose the target firmware file in the pop-up window and click Open. You can check the file path and name in the flash file field to verify the selection.

►Step 3: After confirming all settings, click OK to start flashing.

►Step 4: A pop-up window will appear, showing the current internal firmware version of the drive and the version to be updated. Click OK if the information is correct.

►Step 5: A prompt will appear once flashing is completed. Click OK to finish the process.

Note: The progress bar will display the flashing progress after startup. Do not cut off the power during flashing! Please keep this in mind!

Note: After flashing completes, power off and restart the drive before normal operation.

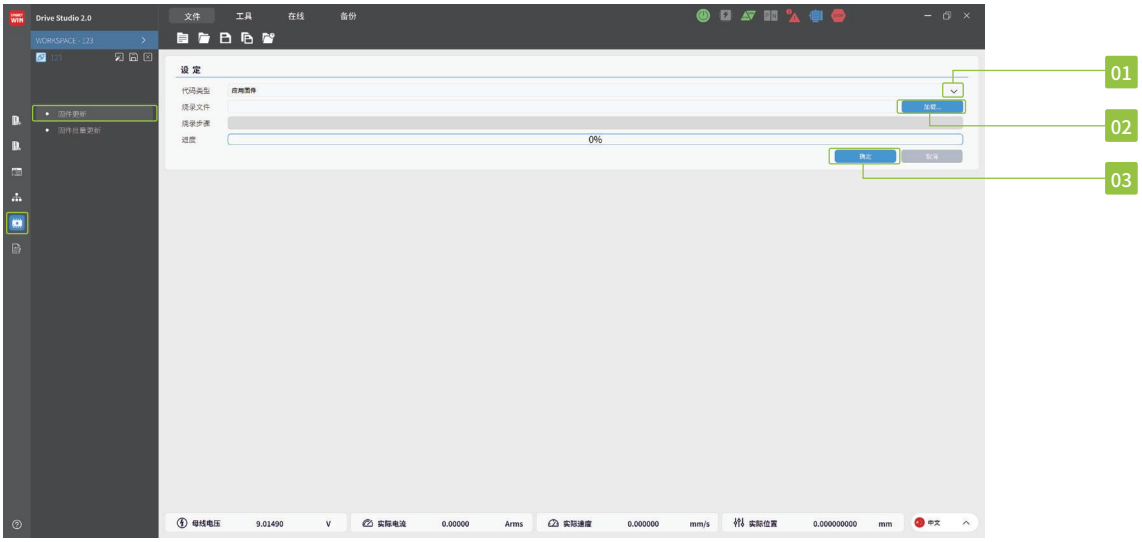


Figure 13-2 Update Firmware via NET Debug Port

13.4 Update Firmware via EtherCAT Bus Interface

1. This method supports simultaneous firmware update for multiple drives.

2. Wiring: Connect the drives in daisy-chain via Ethernet cables. Connect the PC to the C3 Bus In port; connect the C4 Bus Out port of one drive to the C3 Bus In port of the next drive, and so on.

3. Operation guide for the commissioning software:

► Step 1: Enter the EtherCAT firmware update screen, and select the corresponding network adapter of the PC from the drop-down list.

► Step 2: Click Scan, then verify that the number of slaves and detected drives matches the actual connected devices.

► Step 3: After confirmation, click Connect and check if the status changes to Connected.

► Step 4: Follow the same procedures as the NET port flashing process.

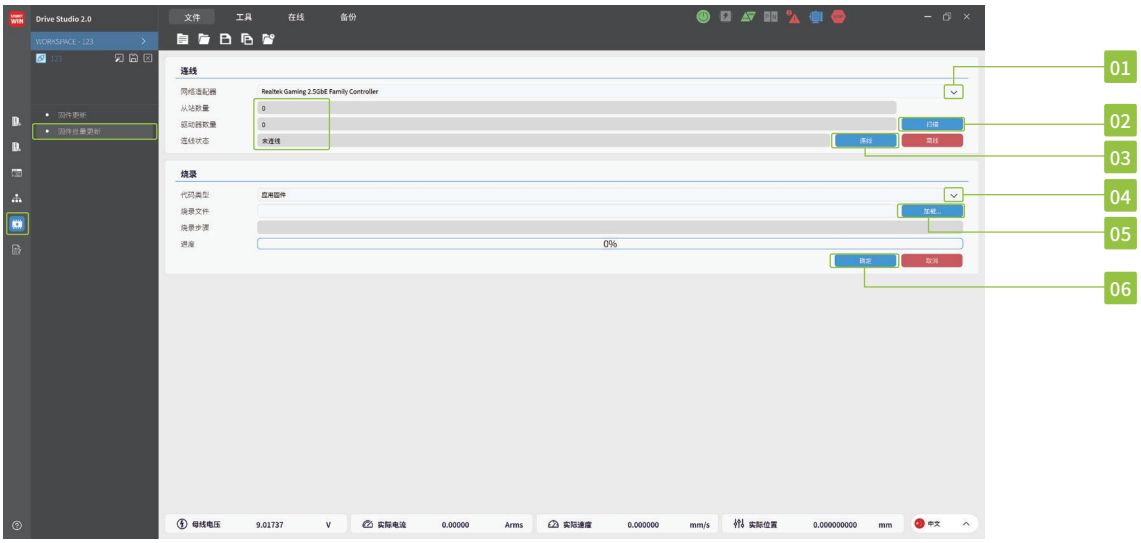


Figure 13-3 Update Firmware via EtherCAT Bus Interface

14. Quick Tuning

14.1 Selecting Motor Parameter

- ▶Step 1: Access the Quick Tuning screen.
- ▶Step 2: Start Auto Tuning.
- ▶Step 3: Double-click to initialize the motor.
- ▶Step 4: Select motor parameters. Refer to section 9.1.2 for setup procedures.
- ▶Step 5: Completed.

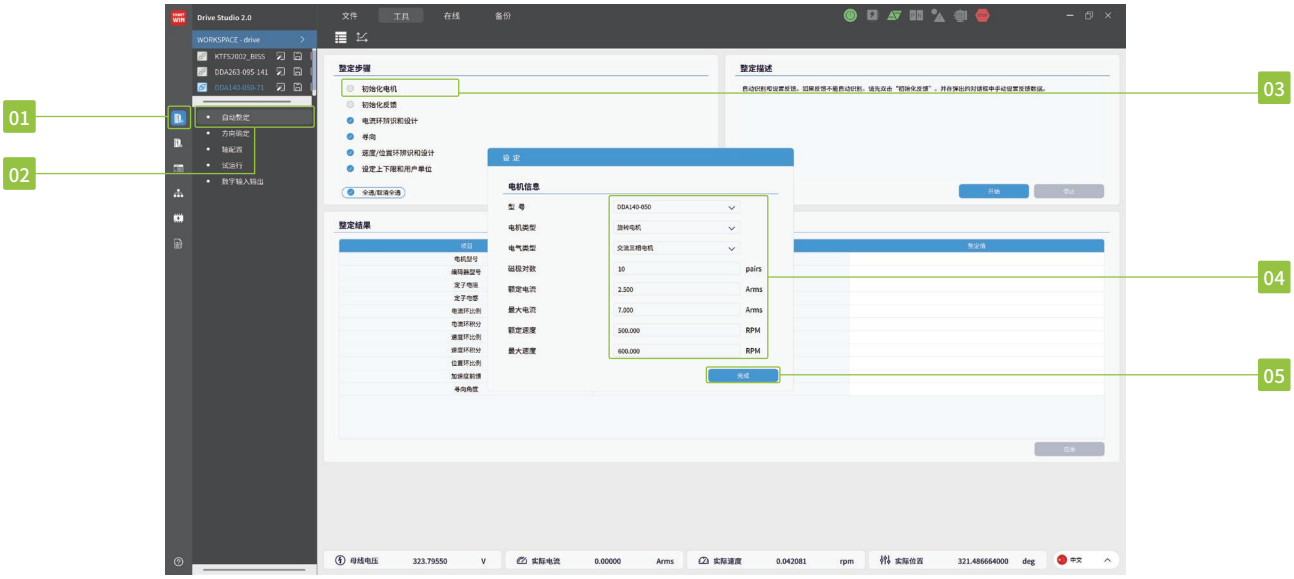


Figure 14-1 Selecting Motor Parameter

14.2 Selecting Encoder Feedback Parameter

- ► Step 1: Double-click to initialize feedback.
- ► Step 2: Select feedback parameters. Refer to sections 9.2.1 and 9.2.2 for settings.
- ► Step 3: Completed.

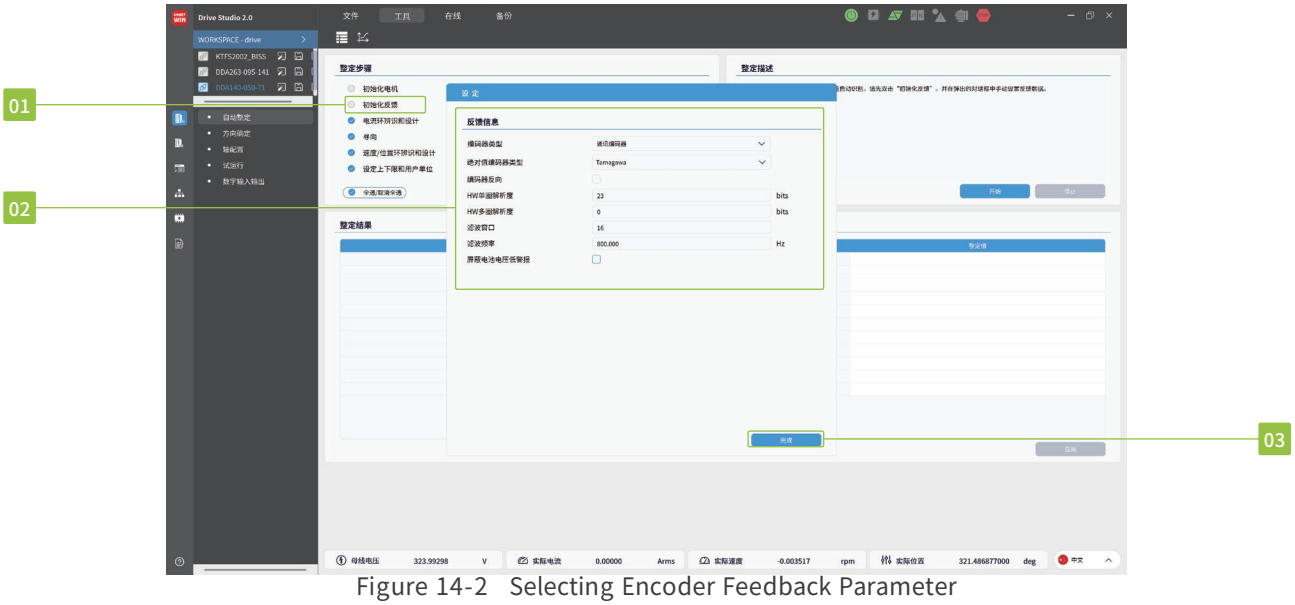


Figure 14-2 Selecting Encoder Feedback Parameter

14.3 Auto Tuning

► Step 1: Click to start Auto Tuning. The system will automatically perform current loop identification, direction detection, speed/position loop identification and other operations.
► Step 2: Confirm the limit values and units are correct to finish tuning. You may modify parameters as required.
► Step 3: Apply the tuned parameters.

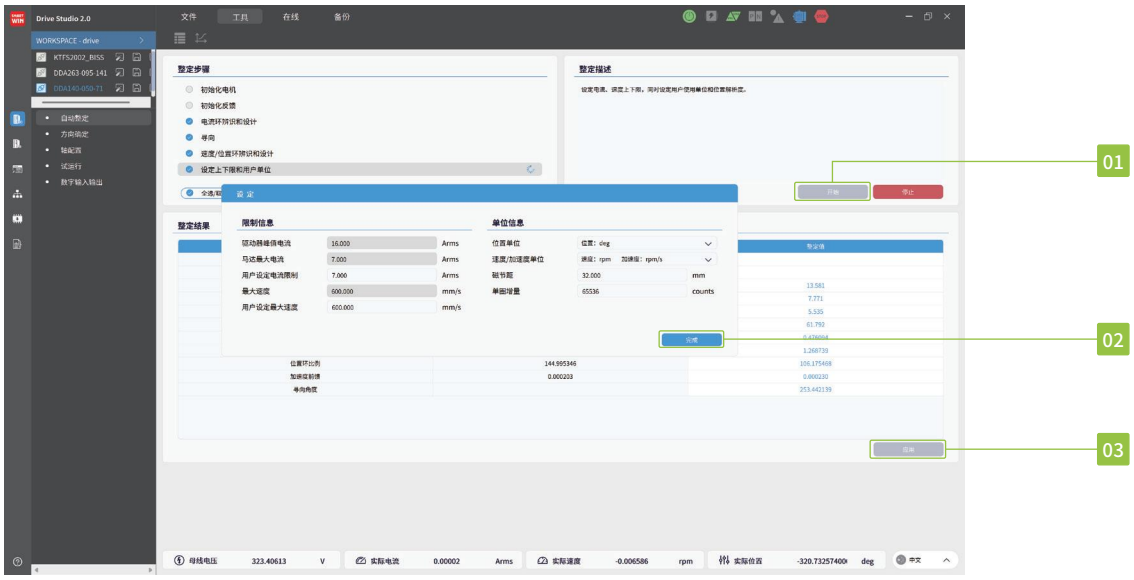


Figure 14-3 Auto Tuning

14.4 Direction Confirmation

► Step 1: Open the Direction Confirmation screen.
► Step 2: Set speed and time, then click Judge Direction.
► Step 3: Click OK.
► Step 4: Observe the motor rotation direction and select the correct option.
► Step 5: Motor direction confirmation completed.

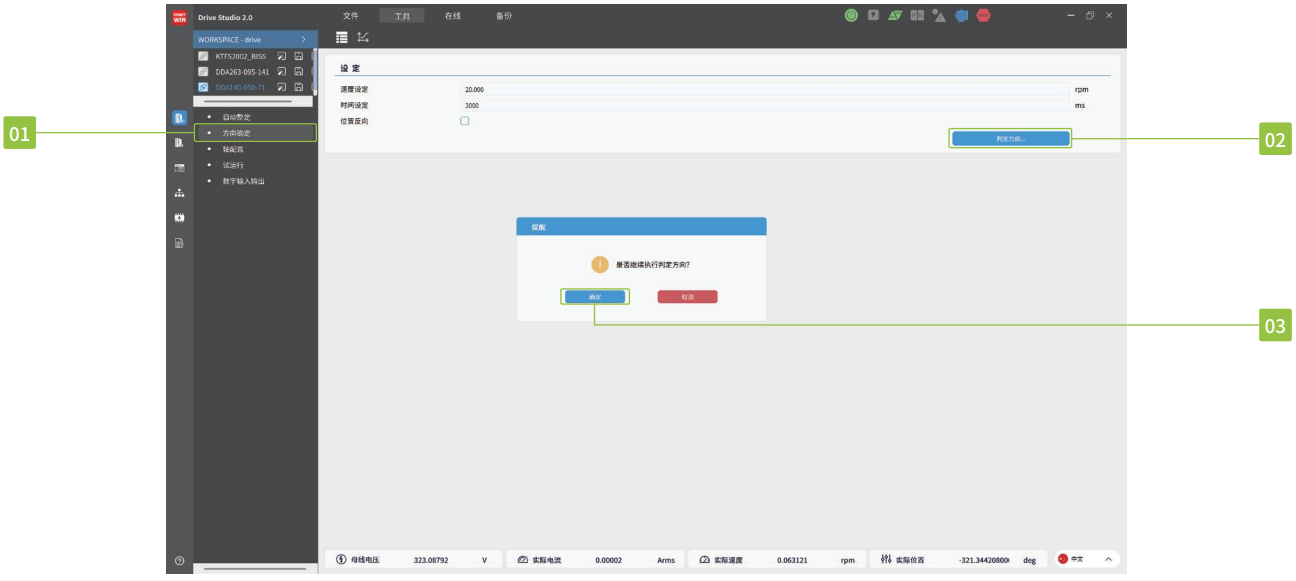


Figure 14-4 Direction Confirmation

14.5 Axis Configuration

- ▶ Step 1: Open the Axis Configuration screen.
- ▶ Step 2: Set parameters as required. Refer to section 9.7 for configuration steps.

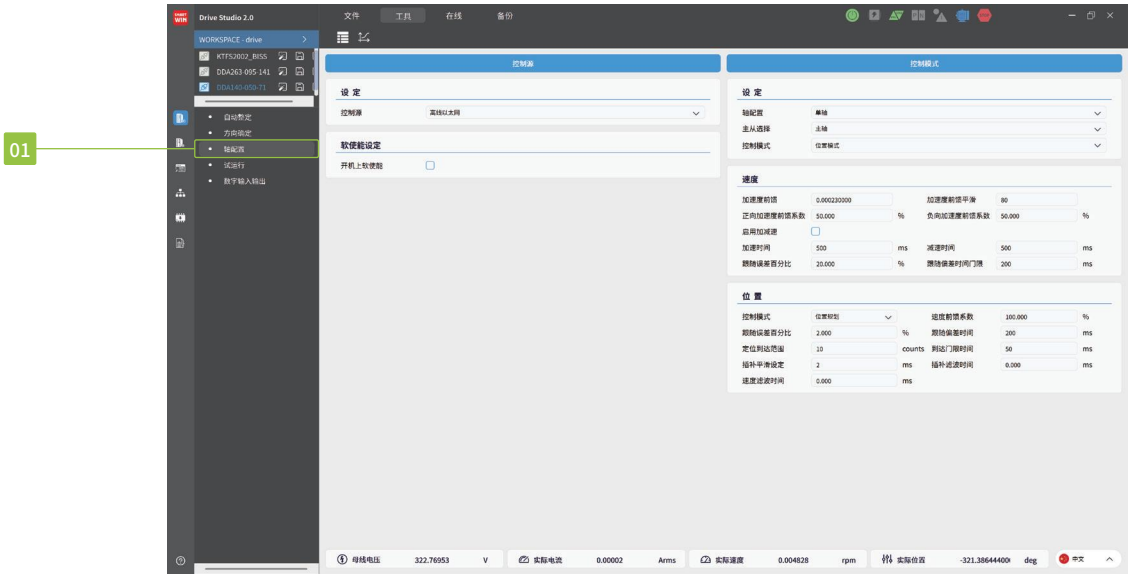


Figure 14-5 Axis Configuration

14.6 Trail Operation

- ▶ Step 1: Open the Trial Operation screen.
- ▶ Step 2: Refer to section 9.8 for trial run debugging procedures.

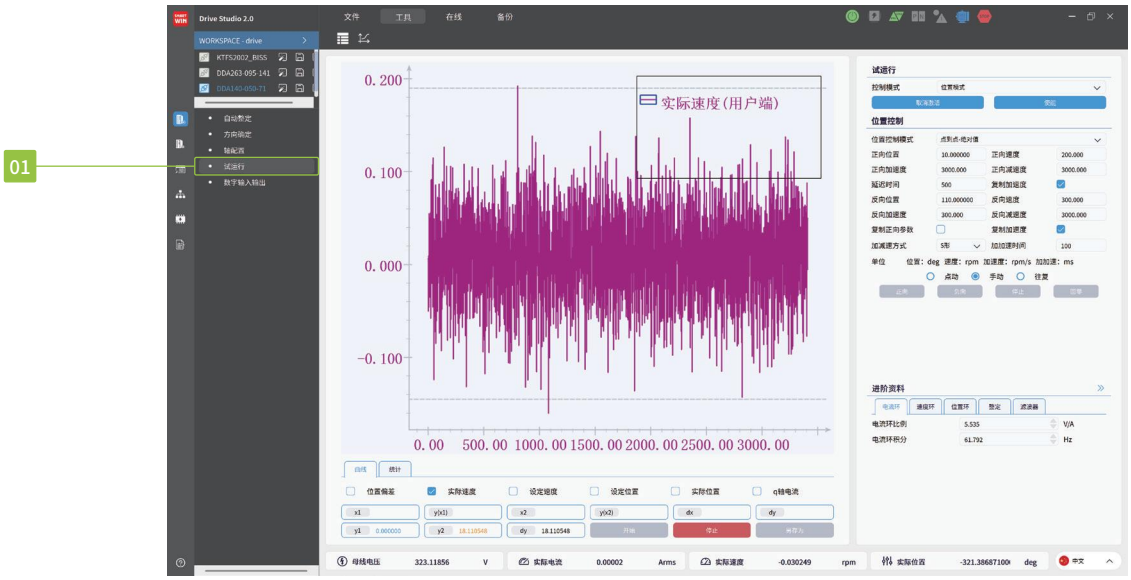


Figure 14-6 Trial Operation

14.7 Digital Input & Output

- ► Step 1: Open the Digital Input/Output screen.
- ► Step 2: Configure digital I/O as required. Refer to section 9.9.1 for setup steps.

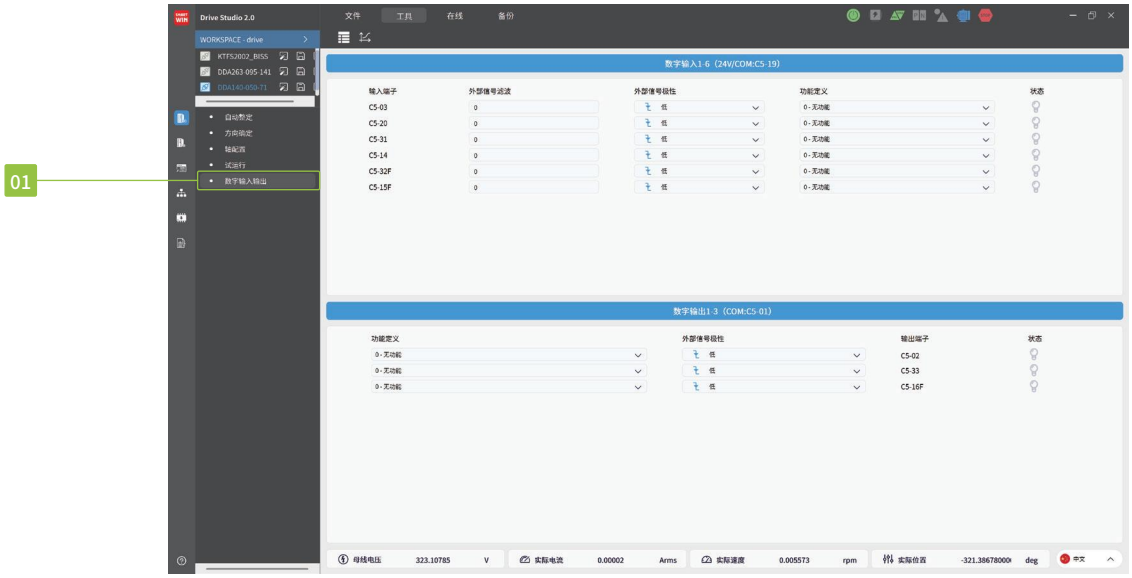


Figure 14-7 Digital Input & Output

15. Troubleshooting

15.1 Faults & Warning

Drive Alarms. When the PC is connected to the drive, you can view all active alarms, including alarm type, occurrence time, description and solutions. Historical alarms record past alerts that have been cleared or generated before drive restart, and the data is stored on the PC.

15.2 Fault Code Instruction

Code	Chinese Description	English Description	Alarm Causes
E0101	RST缺相	LACK_POWER_PHASE	Phase loss detected continuously for 3ms after enable and during operation. This alarm can be masked.
E0103	继电器未吸合	RELAY_FAULT	After enabling, the bus voltage relay is detected not to pull in or be inactive. Please check this alarm. You can set the detection delay or disable this check function.
E0104	母线电压过低	DC_UNDER_VOLTAGE	Bus voltage is detected to stay below the set minimum threshold for 100ms continuously after enable.
E0105	母线电压放电错误	DC_REGEN	The bus voltage continuously exceeds the set discharge voltage by 20V for the duration defined in parameter P11276.
E0106	母线电压过高	DC_OVER_VOLTAGE	The bus voltage exceeds the set upper voltage limit continuously for 4ms.
E0107	无实际电压输出	No output voltage	Output voltage remains at 90% continuously for 100ms, while actual speed and current feedback are below 10%.
E0111	软使能未开启	Software disable	Please check if the soft enable is activated.
E0113	监测电流过大	MONITOR_CURRENT	The current exceeds the value set by P11287 for the duration defined in P11322 continuously.
E0115	过电流	OVER_CURRENT	Hardware overcurrent detected. This alarm can be masked. Note: Once this alarm occurs, disable the drive for 3 to 5 minutes before reuse. Otherwise, other alarms such as E107 may be triggered or the drive may be damaged.
E0116	STO丢失	STO_LOST	Check if STO is inserted
E0118	外部ADC读取错误	ADCEXT_ERROR	Restart the drive or change it
E0120	电机I2T警报	MOTOR_I2T	I2T alarm set for motor (current will fold back after the it is selected in the function options)
E0121	驱动器I2T警报	DRIVE_I2T	I2T alarm Set for motor (current will fold back after the it is selected in the function options)
E0123	驱动器过温	D_OVER_TEMPERATURE	Adjust thr alarm temperature of drive by modifying p11280
E0125	马达过温	M_OVER_TEMPERATURE	Adjust the alarm temperature of motor by modifyingo11316
E0150	编码器离线	ENCODER_DISCONNECT	Check if the encoder line is normal
E0151	脉冲输入离线	PULSEIN_DISCONNECT	Check if the pulse input line is normal or if the alarm can be shileded (user can choose to inspect the alarm after enabling or not)

Code	Chinese Description	English Description	Warning Causes
E0152	编码器读值突变错误	ENCODER_FAULT	If the encoder input has mutation after enabled, set the times of abrupt change inspections or shield it.
E0153	编码器通讯错误	ENCODER_COMM	1.For the communication error of absolute value encoder,shield or set offline inspection time. 2. CD phase error of sincos encoder.
E0154	SinCos编码器处理错	ENCODER_HANDLE	Unmatched AB phase and CD phases of sincos encoder
E0155	绝对值编码器故障	ENCODER_ALM	1. Battery error of Tamagawa encoder 2. AB phase error of Sick or Heidenhain sincos encoder 3.Amplitude error of AB phase or CD phase of sincos 4. AB phase error of ABZ encoder
E0157	编码器Z脉冲与QEP A/B count数不匹配	ENCODER_ZPULSE	Check if Z phase signal is matched with A/B signal
E0158	第二编码器离线	ENCODER_SECDISCONN ECT	Check if the second encoder is in normal connection
E0159	编码器类型设置错误	ENCODER_TYPE	Check if the 1st and 2nd encoders and pulse input channel are correctly and in conflict with each other
E0160	速度偏差	SPEED_TRACK	Improper rigidity adjustment or abnormal encoder feedback. Check if it enters motor or drive current fold back (enter I2T protection).Adjust the threshold percentage and time via screen.
E0161	马达过速	M_OVER_SPEED	Exceed 50% of the set max. motor speed (Note: 10% for 16C05 hardware and previous ones, 50% for the subsequent ones).
E0163	马达堵转	MOTOR_LOCKED	Set motor locking current p11338 and min. locking speed p11331; at the same time, set locking time p11332 (3s by fault). The alarm is not inspected under non-current, speed and position modes.
E0170	位置偏差	POSITION_FOLLOW	Improper adjustment of position loop or speed loop rigidity; adjust the threshold percentage and time via screen. It indicates shielding, if any of the items is set as 0.
E0171	位置控制设定错误	POSITION_SET	Any of speed, acceleration or deceleration cannot be 0.
E0172	设定位置超过上限	POSITION_OVER_RANGE	The upper limit of position can be set.
E0177	位置控制或者插补周期设置错误	POSITION_CYCLE_ERROR	Check if the cycle is set correctly after enabling. Speed will jitter if the setting is incorrect.
E0178	上位机控制器设定速度超过限定	POSITION_SETVELO_LIMIT	The speed command issued by the host exceeds 50% of the user-set maximum speed. This alarm can be masked.
E0200	首先要寻向	ENCODER_NOTALIGN	
E0201	编码器寻向错误	ENCODER_ALIGNFAILURE	1.For searching direction manually, check the error percentage of encoder. The reason why the data is set as p11016 probably lies in: Setting error of motor pole-pairs, encoder resolution or motor magnetic pitch 2.When auto-direction finding is enabled, the automatic direction search may be blocked at one side or motor does not move.

Code	Chinese Description	English Description	Warning Cause
E0202	电机电流环辨识错误	MOTOR_TEST	Check if the soft enabling can be enabled
E0203	回零方式错误	HOMING_METHOD_ERROR	Check if homing mode is supported
E0204	回零超时错误	HOMING_TIMEOUT	Check if the zero returning parameter is correct or proper (for soft gantry, check if the timeout time is proper, 60s by fault)
E0205	回零不被允许	HOMING_NOTPERMIT	Check if zero returning is done before the previous action is ended
E0206	回零失败	HOMING_FAILURE	Check if the zero returning switches are set correctly (positive and negative limits; check if the zero returning switch is under IO reflection)
E0215	总线离线	EtherCAT offline	Failure to receive PDO data from upper computer by setting p11606 setting time continuously after enabling. Check if any interference of network environment. The alarm can be shielded. Improve smoothness after shielding to avoid speed jitter.
E0216	上位机插补周期与驱动器插补周期设置不匹配	ETCONTROLLER_OFFLINE	Check if the interpolation cycle of screen EtherCAT is set correctly. Alarm will be given out if it is unmatched for continuously 3s (higher than 10%). The alarm can be shielded. General configurations: p11610=500us.
E0226	寻向类型错误	ALIGN_TYPEERROR	Do not choose Identify current direction and Identify encoder offset under non-diection search mode
E0227	多轴控制内部总线离线	MULTIAXIS_OFFLINE	Check the internal communication line inside the spindle and if Baud rate p11685 is matched with communication delay time p11880.
E0228	龙门主从位置偏差过大	GANTRY_POSIDELTA	Adjust the flexible and rigid gantries using different methofs. Threshold can be modified by p11350 at the same time
E0229	从轴错误	SLAVE_ERROR	The spindle monitors the alarm of slaveaxis. Please check the reason of the alarm.
E0230	从轴使能超时	SLAVE_ENATIMEOUT	Check if the slave axisruns normally. Slalve axis needs finishing enabling within 16s by default; otherwise, error will be reported
E0231	底层固件需要升级	FIRMWARE_VERSIONERR	Underlying fireware needs upgrading, for some special functions are required.
E0250	资料保存错误	EEPROM_ERROR	It is forbidden to save parameters when catching curve when trial running is activated.
E0256	中断超时	ISR over time	Please contact us.

16. Field Bus Status

16.1 EtherCAT Status LED

The network cable is connected normally. The green LEDs of port C3 and C4 stay on steadily, while the orange LEDs flash rapidly during data transmission.

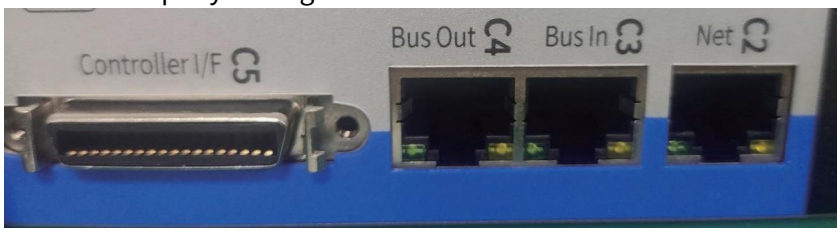


图16-1 EtherCAT Status LED

16.2 Communication Status

Communicate normally with the host computer. Check if the drive is in OP state. You can also view the communication cycle, PDO quantity and packet loss status.

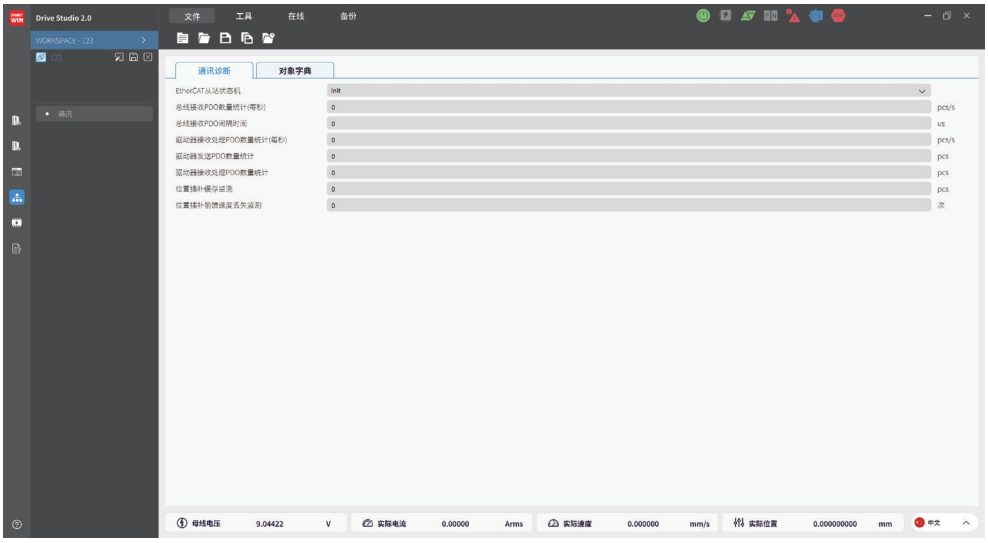


Figure 16-2 Communication Status

16.3 Communication Data Diagnosis

View object dictionary data in real time. Real-time commands can be sent from the host to the drive, or transmitted from the drive to the host, enabling quick troubleshooting of abnormalities.,

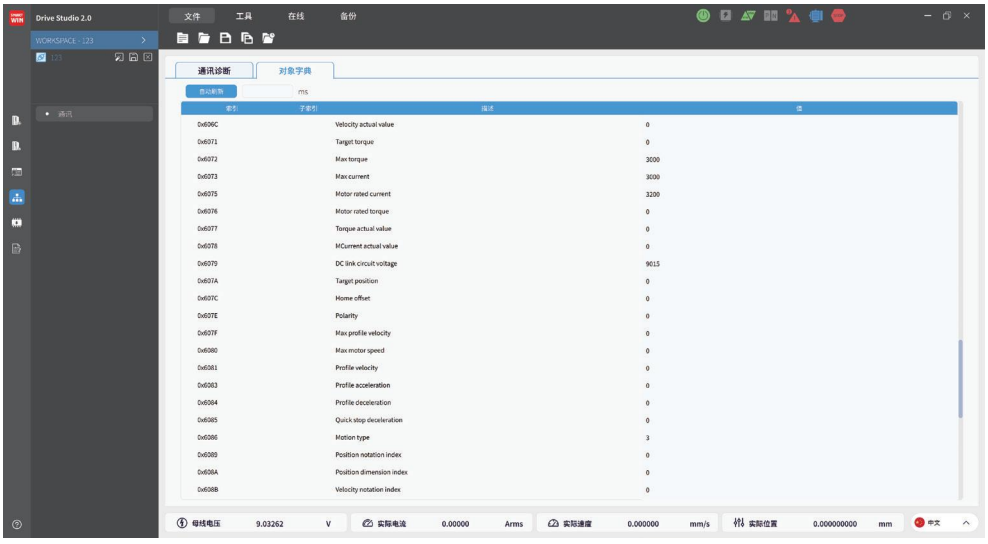


Figure 16-3 Communication Data Diagnosis

17. Brief Introduction to Menu Functions of Debug Software

17.1 Debugging Software Icons Instruction

No.	Icons	Name	Function Instruction
1	文件	File	Operate with Workspace (Click to execute)
2	工具	Tool	Drive parameter list & curve functions (Click to operate)
3	在线	Online	Drive project operation icon (Click to operate)
4	备份	Backup	Backup drive data & database comparison (Click to operate)
5		Soft start	Restart drive without power off (Double-click to operate)
6		Motor Enable	Icon lights up after motor enabled (Non-operable)
7		Left & Right Limit Status	Lights up when limit reached (Non-operable)
8		STO1/STO2	External STO1/STO2 normal (icon turns green, non-operable)
9		Warning	Drive alarm prompt. Hover over icon to view alarm details; double-click to clear all alarms.
10		Motor Soft Enable	Soft enable icon. Lights up when soft enabled (Click to operate)
11		Emergency Stop	Emergency Stop (Click to operate)
12		New Project	New Project (Click to operate)
13		Open Project	Open Project (Click to operate)
14		Reserve Drive Document	Save drive data in project (Click to operate)
15		Backup Project	Backup real-time parameters (Click to operate)
16		Recent Projects	View recently opened project files on PC (Click to operate)
17		Diagnosis Icon	Modify parameters (Click to operate)
18		Curve Iron	Capture curve (Click to operate)
19		Reserve Document in ROM	Save current parameters to drive ROM (Click to operate
20		Load data from ROM	Read parameters from drive ROM to PC (Click to operate)
21		Read Data from RAM	Read parameters from drive RAM to PC (Click to operate)
22		Reset Drive Data	Restore factory settings (Click to operate)
23		Backup Drive Data	Backup drive data (Click to operate)
24		Download Backup data in ROM	Import backed-up parameters to drive and save to ROM (Click to operate)
25		Download Parameter in RAM	Import parameters from opened project file to drive RAM (Click to operate)
26		Database Comparison	Compare current drive parameters with library parameters (Click to operate)

No.	Icons	Name	Function Instruction
27		Quick Tuning	Quick setup wizard for new drives/motors (Click to operate)
28		Expert Tuning	Detailed commissioning & mounting for new drives/motors (Click to operate)
29		Axis Application	Auxiliary function list (View details) (Click to operate)
30		Communication	View EtherCAT communication related parameters (Click to operate)
31		System	Update drive or FPGA firmware (Click to operate)
32		Data Management	Import parameter list for new drives or motors (Click to operate)
33		Helping	View tool version information (Click to operate)
34		Language Setup	Switch between Chinese and English (Click to operate)
35		Bus Voltage	Real-time DC bus voltage display (Non-operable)
36		Actual Current	Real-time actual current display (Non-operable)
37		Actual Speed	Real-time actual speed display (Non-operable)
38		Actual Position	Real-time actual position display (Non-operable)

17.2 File Menu



Figure 17-1 File Menu

1	New Project	Create New Project File
2	Open Project	Open previously saved parameter project files
3	Reserve the drive file of the project	Overwrite the active project file with modified parameters
4	Backup Project	Backup real-time parameters
5	Recent projects	Click to view recently opened project files on PC

17.3 Tool Menu



Figure 17-2 Tool Menu

1	Diagnosis	Modify File Parameter
2	Curve	Capture Curve

17.4 Online Menu



Figure 17-3 Online Menu

1		Reserve drive file in ROM	Save modified real-time parameters to drive ROM
2		Load drive data from ROM	Read parameters from drive ROM to PC
3		Read drive data from RAM	Read parameters from drive RAM to PC
4		Reset drive data	Restore Factory Defaults

17.5 Backup Menu



Figure 17-4 Backup Menu



Figure 17-5 Backup Menu

1		Backup drive data	Backup parameter data
2		Download backup drive data and save to ROM	Import backed-up parameters to drive and save to ROM
3		Download drive data	Import parameters from opened project file to drive RAM
4		Database comparison	Compare current drive parameters with database parameters and check differences

17.6 Helping Menu



Figure 17-6Helping Menu

1		Helping	View debug software version information
2	中文	Language Setup	Switch between Chinese and English

17.7 Introduction to the Logo at the Top Right Corner

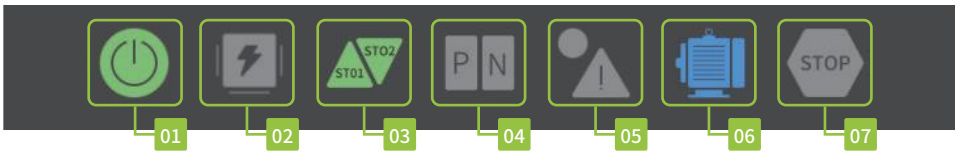


Figure 17-7 Introduction to icons in the upper right corner

1		Restart	Restart drive via software (Power cycle is mandatory after firmware update)
2		Enable Status Display	Enable status: lit in blue
3		STO Status Display	Green light indicates STO active status
4		Left & Right limit status	Yellow light indicates limit triggered status
5		Warning Status Display	Red light for alarm status. Double-click to clear alarms.
6		Soft enable status display	Blue light indicates soft enable active status
7		Stop	

18. Curve Drawing

18.1 Curve Sampling Parameter Settings

► Step 1: Click to select the curve function.

► Step 2: Click the Select window to set the sampling mode and interval, or quickly select the recommended curves.

► Step 3: Click the three-dot icon below the Select list to open the curve selection menu. Check the parameters to be captured and click OK. You may also directly enter the DataID of target curve data.

► Step 4: Click the color icon under the Color list, then pick a color for the curve in the color panel.

► Step 5: Check the curves to be captured in the Active list.

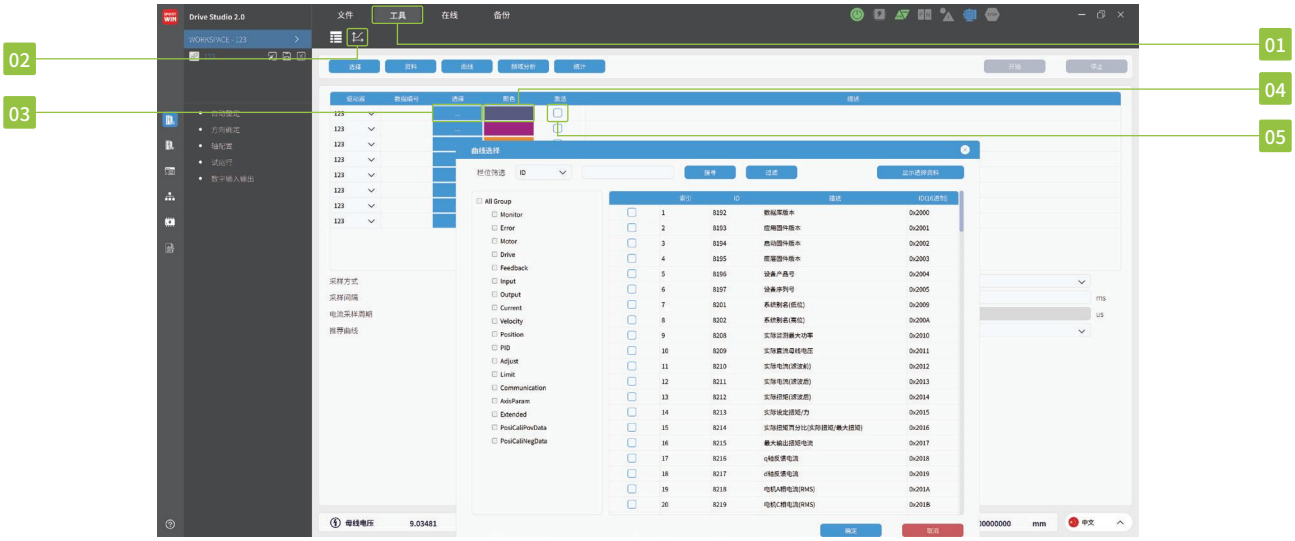


Figure 18-1 Curve Sampling Parameter Configuration

► Step 6: Click Start to begin curve tracking, and click Stop to end it.

► Step 7: View curves in the Curve tab. Click the items at the upper right of the oscilloscope to switch the vertical axis range.

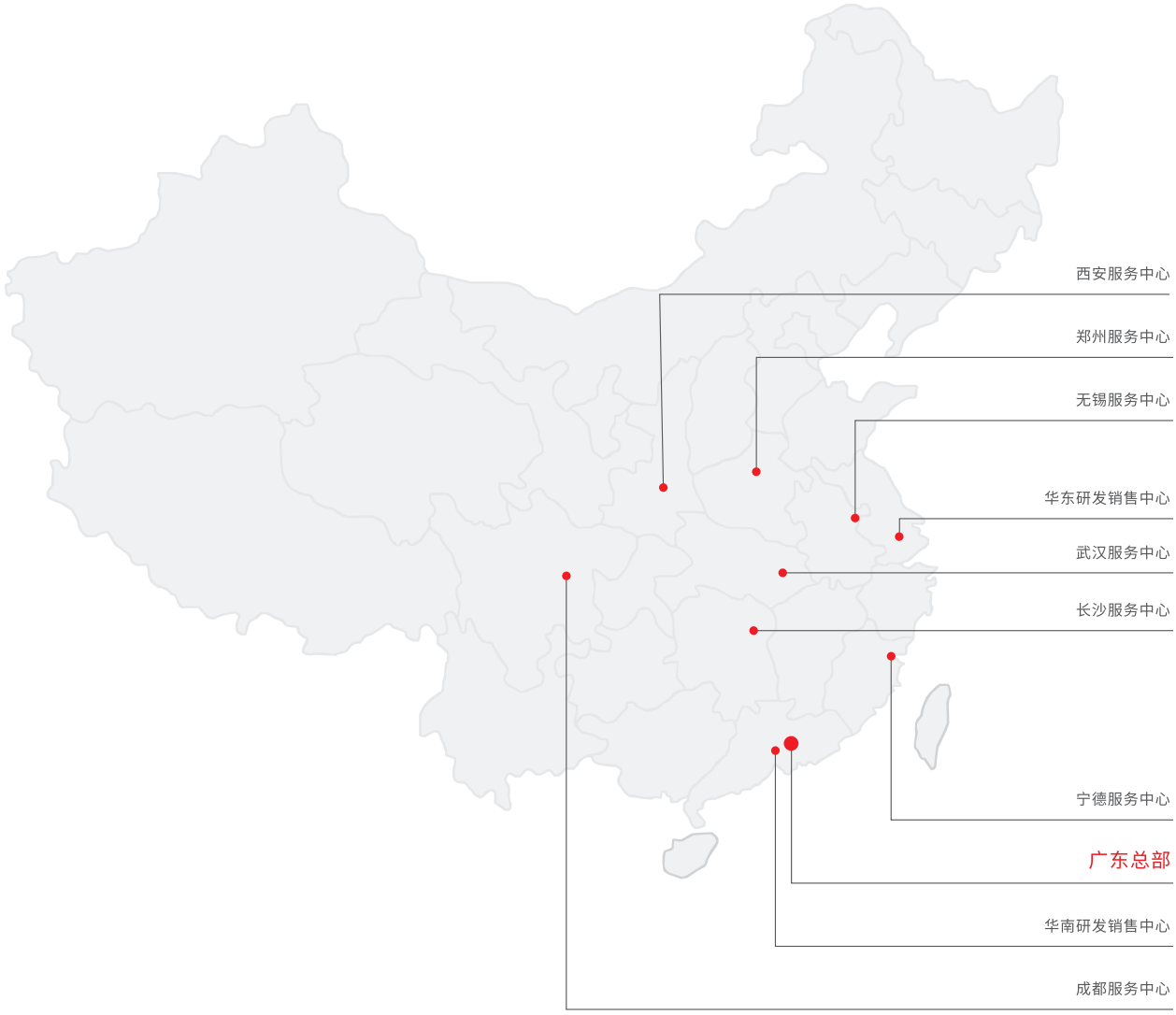
► Step 8: In the Curve tab, hold down the left mouse button to zoom in on local areas; right-click to access other zoom functions.

► Step 9: Save or delete curves in the Data tab.



Figure 18-2 Curve Sampling Parameter Configuration

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